BANGALORE INSTITUTE OF TECHNOLOGY

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Affiliated to

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

BELGAVI – 590018, KARNATAKA, INDIA

DEPARTMENT OF MECHANIAL ENGINEERING

FINITE ELEMENT METHODS

Subject Code: BME701

Semester: VII

Prepared By

DR. E.R. Babu,

ASSISTANT PROFESSOR

PROGRAM OUTCOMES (PO'S)

1. Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of Complex engineering problems.

2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the needs with appropriate consideration for the public health and safety, and the cultural, Societal and environmental considerations.

4. Conduct investigations on complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. Modern tool usage: Create, select, and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of and need for sustainable development.

8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. Individual and team work: Function effectively as an individual and as a member or leader in diverse teams and in multidisciplinary settings.

10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large such as being able to comprehend and write effective reports and design documentation, make effective presentations and give and receive clear instructions.

11. Project Management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

DEPARTMENT VISION

Mechanical Engineering Department Strive to Transform the Students Technically Educated, Broadly Knowledge and Competent Mechanical Engineers.

DEPARTMENT MISSION

1. Create an Environment as a Center of Learning
2. Train the Staff to Ensure the Student to get an Excellent Education at par with Latest Technology
3. Provide an Effective Means of Learning, Advance Knowledge, and Facilitates Personal Commitment to the Educational Success of Students
4. Preparing the Students for a Productive Careers Globally
5. Inspire Hope and Contribute to Society by Providing the Best Practices
6. Prepare the Students for Research and Advanced Problem Solving Technique Needed to Address Complex Mechanical Engineering Problems
7. Educate Students to become highly Successful Alumni who contribute to the Profession in the Global Society.

PROGRAM EDUCATIONAL OBJECTIVIES (PEO's)

MEPEO1: To prepare the students for Successful professional careers with strong fundamental knowledge in Science, Mathematics, English and Engineering Sciences so as to enable them to analyze the Mechanical Engineering related problems leading to leadership, entrepreneurship or pursuing higher education (Preparation).

MEPEO2: To develop ability among the students for acquiring technical knowledge in specialized areas of Mechanical Engineering such as Materials, Design, Manufacturing and Thermal Engineering with a focus on research and innovation and gaining the technical skills in classical software packages (Core competence and professionalism).

MEPEO3: To provide opportunities for the students to work with multidiscipline field of engineering so as to enlarge the ability among the students to understand the different industrial environments (Breadth).

MEPEO4: To promote the students for continuous learning, research and development with strong professional, moral and ethical values and zeal for life-long learning. (Learning environment).

PROGRAM SPECIFIC OUTCOME (PSO's)

PSO - 1: An ability that meets the needs of Industries/ society and the country in general and to synthesize data and technical concepts for application to mechanical engineering software.

PSO - 2 : Ability to accept new challenges for multi-disciplinary projects with a sound foundation in mathematical, scientific and engineering areas necessary to achieve excellence in solving and analyzing engineering problems and to prepare for graduate studies.

PSO - 3: Ability to promote and develop an ambition of life-long learning and to apply them to professional ethics and codes of professional practices.

COURSE OBJECTIVES

CO's	The main objectives of this laboratory course are :
CO1	1. Provide a fundamental understanding of the theory and principles behind the Finite Element Method (FEM).
CO2	1. Develop proficiency in using commercial FEA software for solving a wide range of engineering problems in solid mechanics, heat transfer, and dynamics.
CO3	1. Enable students to model, analyze, and interpret results for complex structures including bars, trusses, beams, plates, and composite materials.
CO4	1. Impart knowledge on advanced FEA topics like thermal analysis, dynamic analysis, contact mechanics, and interoperability between software tools.
CO5	1. Foster an ability to validate FEA results and understand the assumptions and limitations of the method.

COURSE OUTCOMES

	On completion of this Course students will be able to:
CO1	**Model and Analyze 1D Elements:** Construct FEA models for axial members (constant, tapered, stepped cross-sections, and different materials) and truss structures under mechanical loads to determine deformations, stresses, and reactions.
CO2	**Analyze Beam Structures:** Formulate and solve problems related to various types of beams (simply supported, cantilever) subjected to different loading conditions (point load, UDL, varying load) to determine deflections, slopes, bending moments, and shear forces.
CO3	**Perform 2D Stress Concentration Analysis:** Model a 2D planar structure (e.g., plate with a hole) and analyze the stress concentration phenomenon, comparing FEA results with theoretical values.
CO4	**Conduct Thermal Analysis:** Simulate 1D and 2D heat transfer problems involving conduction and convection boundary conditions, including composite sections, to determine temperature distribution and heat flux.
CO5	**Perform Dynamic Analysis:** Determine the natural frequencies and mode shapes of a fixed-fixed beam and analyze its forced vibration response to a given forcing function.
CO6	**Execute Advanced Analyses:** Perform a basic contact analysis to understand the procedure and model/analyze simple components made from composite materials using appropriate element types and material models.

CO TO PO & PSO MAPPING

FINITE ELEMENT ANALYSIS LAB-BME701															
CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
CO1	3	3	2	3									3	2	2
CO2	3	3	2	3									3	2	2
CO3	3	3	3	3	2								3	3	2
CO4	3	3	3	3	2								3	3	2
CO5	3	3	3	3	2								3	3	2
CO6	3	3	3	3	2								3	3	2
AVG	**3**	**3**	**2.66**	**3**	**2**								**3**	**2.66**	**2**

FINITE ELEMENT ANALYSIS LAB

COURSE CONTENT

1) Bars of constant cross section area, tapered cross section area and stepped bar with different materials

2) Trusses – (Minimum 3 exercises of different areas of cross sections of links, different supports such as fixed support, rolling support)

3) Beams – Simply supported, cantilever, beams with point load, UDL, beams with varying load etc. (Minimum 6 exercises)

4) Stress analysis of a rectangular plate with a circular hole

5) Thermal Analysis – 1D & 2D problem with conduction and convection boundary conditions (Heat transfer through composite section) Minimum of 2 exercises

6) Natural frequency of beam with fixed – fixed end condition 7 Response of beam with fixed – fixed end conditions subjected to forcing function

7) Demonstrate the use of graphics standards (IGES, STEP etc) to import the model from modeler to solver.

8) Demonstrate one example of contact analysis to learn the procedure to carry out contact analysis.

9) Demonstrate at least two different type of example to model and analyze bars or plates made from composite material.

LAB TITLE: MODELING AND ANALYSIS LAB

Faculty In charge Name:	Dr. E.R Babu		Academic Year:	2025-2026			
Department:	Mechanical Engineering						
Course Code	Course Title	Core / Elective	Prerequisite	Contact Hours			Total Hrs./ Sessions
				L	T	P	
BME701	FINITE ELEMENT METHODS	Core	Mathematics, FEM	03	-	02	50

COURSE CONTENT

PART – A

Study of a FEA package and modeling and stress analysis of:

1. Bars of constant cross section area, tapered cross section area and stepped bar

2. Trusses – **(Minimum 2 exercises of different types)**

3. Beams – Simply supported, cantilever, beams with point load, UDL, beams with varying load etc

(Minimum 6 exercises different nature).

4. Stress analysis of a rectangular plate with a circular hole.

PART – B

1) Thermal Analysis – 1D & 2D problem with conduction and convection boundary conditions **(Minimum 4 exercises of different types)**

2) Dynamic Analysis to find

a) Fixed – fixed beam for natural frequency determination 41

b) Bar subjected to forcing function

c) Fixed – fixed beam subjected to forcing function

PART – C (only for demo and oral exam)

1) Demonstrate the use of graphics standards (IGES, STEP etc) to import the model from modeler to solver.
2) Demonstrate one example of contact analysis to learn the procedure to carry out contact analysis.
3) Demonstrate at least two different type of example to model and analyze bars or plates made from composite material.

REFERENCE BOOKS:

1. A first course in the Finite element method, Daryl L Logan, Thomason, Third Edition
2. Fundaments of FEM, Hutton – McGraw Hill, 2004
3. Finite Element Analysis, George R. Buchanan, Schaum Series

Scheme for Examination:
One Question from Part A - 32 Marks (08 Write up +24)
One Question from Part B - 32 Marks (08 Write up +24)
Viva-Voce - 16 Marks
Total 80 Marks

LIST OF EXPERIMENTS TO BE COMPLETED FOR MODELING AND ANALYSIS LAB COVERING ALL THE SIX CO'S

Drawing Sheet Number/ Experiment Number	Details of Drawing/ Experiment	CO'S	Matching PO'S
1	Introduction to FEM, Applications, Introduction to ANSYS, Steps involved in solving any problem in Ansys, Capabilities of Ansys etc(Self Learning - To be evaluated for 10 marks by the staff member)	CO1	PO1,PO2,PO3,PO4,PO5
2	Exercise 1- One dimensional problem.	CO1	PO1,PO2,PO3,PO4,PO5
3	Exercise 2- One dimensional problem.	CO1	PO1,PO2,PO3,PO4,PO5
4	Exercise 3- Problem on truss structure.	CO2	PO1,PO2,PO3,PO4,PO5
5	Exercise 4- Problem on truss structure.	CO2	PO1,PO2,PO3,PO4,PO5
6	Exercise 5- Problem on Beams.	CO3	PO1,PO2,PO3,PO4,PO5
7	Exercise 6- Problem on Beams.	CO3	PO1,PO2,PO3,PO4,PO5
8	Exercise 7- Problem on Beams.	CO3	PO1,PO2,PO3,PO4,PO5
9	Exercise 8- Problem on Beams.	CO3	PO1,PO2,PO3,PO4,PO5
10	Exercise 9- Problem on Beams.	CO3	PO1,PO2,PO3,PO4,PO5
11	Exercise10- Problem on Beams.	CO3	PO1,PO2,PO3,PO4,PO5
12	Exercise11- Problem on Plate with a circular hole.	CO4	PO1,PO2,PO3,PO4,PO5
13	Exercise12- Problem on Typical Bracket.	CO4	PO1,PO2,PO3,PO4,PO5
14	Exercise13- Thermal Analysis- Conduction boundary conditions.	CO5	PO1,PO2,PO3,PO4,PO5
15	Exercise14- Thermal Analysis- Conduction boundary conditions.	CO5	PO1,PO2,PO3,PO4,PO5
16	Exercise15- Thermal Analysis- Conduction and convection boundary conditions.	CO5	PO1,PO2,PO3,PO4,PO5
17	Exercise16- Thermal Analysis- Conduction and convection boundary conditions.	CO5	PO1,PO2,PO3,PO4,PO5
18	Exercise17- Dynamic Analysis- Mode frequencies and mode shapes for a cantilever beam.	CO6	PO1,PO2,PO3,PO4,PO5,PO10,PO12

CHAPTER 1

General Introduction to FINITE ELEMENT METHOD Lab

1.1 What Is FEM?

In the field of Engineering Design we come across many complex problems, the Mathematical Formulation of which is tedious and usually not possible by analytical methods. At such instants we resort to the use of Numerical techniques. Here lies the importance of FEM, which is a very powerful tool for getting the Numerical solution of a wide range of engineering problems. The basic concept is that a body or structure may be divided into smaller elements of finite dimensions called as "Finite Elements". The original body or structure is then considered as an assemblage of these elements connected at a finite number of joints called as "Nodes" or "Nodal Points". The properties of the elements are formulated and combined top obtain the properties of the entire body.

The equations of equilibrium for the entire structure or body are then obtained by combining the equilibrium equation of each element such that the continuity is ensured at each node. The necessary boundary conditions are then imposed and the equations of equilibrium are the solved to obtain the required variables such as Stress, Strain, Temperature Distribution or Velocity Flow depending on the application.

Thus instead of solving the problem for the entire structure or body in one operation, in the method attention is mainly devoted to the formulation of properties of the constituent elements. A common procedure is adopted for combining the elements, solution of equations and evaluation of the required variables in all fields. Thus the modular structure of the method is well exploited in various disciplines of Engineering.

1.2 Applications Of Finite Element Method.

With the advent of Computer Aided Design (CAD), the Finite Element Method has been finding wide variety of applications. Some typical applications of FEM are listed below.

PLANE STRESS PROBLEMS:

- Stress Flow around Reinforced Opening:

In steel pressure vessel and aircraft structures, openings have to be introduced. This result is stress concentrations in a localized region. The stress analysis can be carried out by simply discretizing the surface near the edge of the hole by very thin triangular element.

- Dam Subjected To External Pressure.

AXI-SYMMETRIC PROBLEMS:

➢ Application Related To Shell:

A shell is a structure which can be derived from a thin plate by initially forming the middle plane to a singly curved surface. In Finite Element Method, it is assumed that the behavior of a continuously curved surface can be adequately represented by the behavior of a surface built up of small, flat elements. In a shell, the element will be subject both to bending and "in-plane" stresses.

One common example of shell is Cooling Tower. The tower is subjected to unsymmetrical wind loading. The tower is divided into small triangular elements for carrying out stress analysis.

➢ Belleville Spring:

The Belleville Spring, also called Belleville Washer, is a conical disk spring. The load is applied on the periphery of the circle and supported at the bottom. As load is applied in the axial direction, the supporting edge moves out. Only the rectangular cross-section area needs to be modeled. Load-deflection characteristics and stress distribution can be obtained by dividing the area into elements and using a computer program. In the Belleville spring, the load-deflection curve is non-linear. The stiffness depends on geometry. We can find a good approximate solution by an incremental approach.

APPLICATIONS RELATED TO FLUID FLOW:

➢ Aircraft Wing:

FEM can be used to find the lift and drag forces on an aircraft wing in the design stages itself. Given a cross-section of an aero foil and its relative speed with respect to air, we can find the pressure distribution of air around the aero foil. Basic idea here is to discretize the region around the aero foil, thus we can calculate the lift and drag forces on it. The method can also be used to find the optimum cross-section of the aero foil for a particular speed.

➢ Aerodynamics Of A Vehicle :

FEM can be used to compare the Aerodynamics of various proposed vehicle shapes. It is used to calculate the air resistance acting on a particular vehicle shape. Here we discretize region surrounding the vehicle boundary. Thus we can use FEM to get a sleek body for a vehicle.

APPLICATIONS RELATED TO HEAT TRANSFER:

FEM is a very powerful method for Heat Transfer analysis in the design stages.

➢ Engine Piston:

To find out various parameters such as area required for heat transfer, cooling fluid flow, material selection, etc. temperature distribution in a piston must be initially found out. Using FEM the temperature distribution throughout the piston can be found out.

- Fins:

Fin is an extended surface of a structure used to increase the rate of Heat Transfer. For example, fins are used as extension of cylinder head for heat dissipation in engine. Geometry and arrangement of fins is complex. FEM can be used to find Temperature Distribution and Heat Transfer in the fins.

1.3 What is ANSYS?

ANSYS is general-purpose finite element analysis (FEA) software package. Finite Element Analysis is a numerical method of deconstructing a complex system into very small pieces (of user-designated size) called elements. The software implements equations that govern the behavior of these elements and solves them all, creating a comprehensive explanation of how the system acts as a whole. These results then can be presented in tabulated, or graphical forms. This type of analysis is typically used for the design and optimization of a system far too complex to analyze by hand. Systems that may fit into this category are too complex due to their geometry, scale, or governing equations.

ANSYS is the standard FEA teaching tool within the Mechanical Engineering Department at many colleges. ANSYS is also used in Civil and Electrical Engineering, as well as the Physics and Chemistry departments.[1]

Why use it?

ANSYS provides a cost-effective way to explore the performance of products or processes in a virtual environment. This type of product development is termed virtual prototyping.

With virtual prototyping techniques, users can iterate various scenarios to optimize the product long before the manufacturing is started. This enables a reduction in the level of risk, and in the cost of ineffective designs. The multifaceted nature of ANSYS also provides a means to ensure that users are able to see the effect of a design on the whole behavior of the product, be it electromagnetic, thermal, mechanical etc.

1.4 ANSYS in the Professional World:

If you are interested in ANSYS and what it accomplishes in the professional world, here are two links that briefly discuss different aspects of the company. (*note: Their headquarters is based 20 minutes outside of Pittsburgh.) For more information, just use www.google.com

- http://www.shef.ac.uk/mecheng/staff/ay/cae/tutorials/ansys/ansys1.html
- http://www.post-gazette.com/businessnews/20020406topstockbiz7P7.asp

1.5 Generic Steps to Solving any Problem in ANSYS:

Like solving any problem analytically, you need to define (1) your solution domain, (2) the physical model, (3) boundary conditions and (4) the physical properties. You then solve the problem and present the results. In numerical methods, the main difference is an extra step called mesh generation. This is the step that divides the complex model into small elements that become solvable in an otherwise too complex situation. Below describes the processes in terminology slightly more attune to the software.

Build Geometry

Construct a two or three dimensional representation of the object to be modeled and tested using the work plane coordinate system within ANSYS.

Define Material Properties

Now that the part exists, define a library of the necessary materials that compose the object (or project) being modeled. This includes thermal and mechanical properties.

Generate Mesh

At this point ANSYS understands the makeup of the part. Now define how the modeled system should be broken down into finite pieces.

Apply Loads

Once the system is fully designed, the last task is to burden the system with constraints, such as physical loadings or boundary conditions.

Obtain Solution

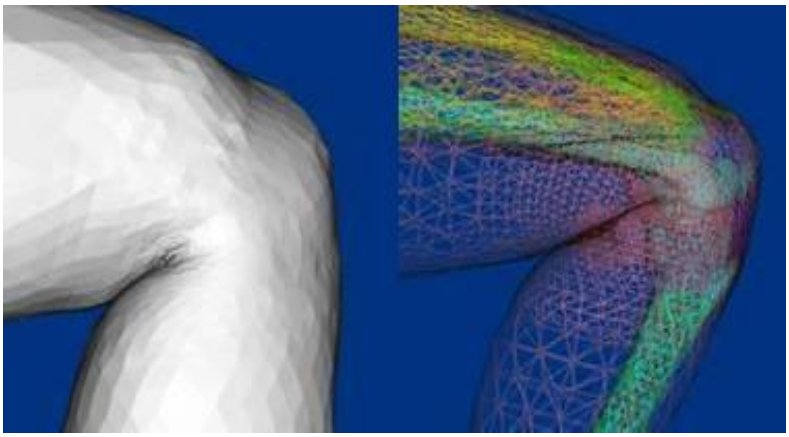
This is actually a step, because ANSYS needs to understand within what state (steady state, transient… etc.) the problem must be solved.

Present the Results

After the solution has been obtained, there are many ways to present ANSYS' results, choose from many options such as tables, graphs, and contour plots.

Specific Capabilities of ANSYS:

Structural:



http://www.cs.unc.edu/~hirota/fem/
Fig: QuickTime movie of the knee joint here.
http://www.cs.unc.edu/~hirota/fem/movies/op2tr.mov

Structural analysis is probably the most common application of the finite element method as it implies bridges and buildings, naval, aeronautical, and mechanical structures such as ship hulls, aircraft bodies, and machine housings, as well as mechanical components such as pistons, machine parts, and tools.

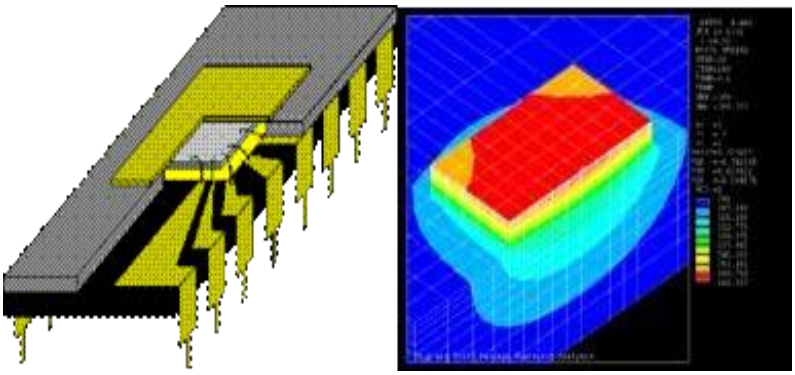
- **Static Analysis** - Used to determine displacements, stresses, etc. under static loading conditions. ANSYS can compute both linear and nonlinear static analyses. Nonlinearities can include plasticity, stress stiffening, large deflection, large strain, hyper elasticity, contact surfaces, and creep.

- **Transient Dynamic Analysis** - Used to determine the response of a structure to arbitrarily time-varying loads. All nonlinearities mentioned under Static Analysis above are allowed.

- **Buckling Analysis** - Used to calculate the buckling loads and determine the buckling mode shape. Both linear (eigenvalue) buckling and nonlinear buckling analyses are possible.

In addition to the above analysis types, several special-purpose features are available such as **Fracture mechanics**, **Composite material analysis**, **Fatigue**, and both **p-Method and Beam analyses.**

Thermal:



ANSYS is capable of both steady state and transient analysis of any solid with thermal boundary conditions.

Steady-state thermal analyses calculate the effects of steady thermal loads on a system or component. Users often perform a steady-state analysis before doing a transient thermal analysis, to help establish initial conditions. A steady-state analysis also can be the last step of a transient thermal analysis; performed after all transient effects have diminished. ANSYS can be used to determine temperatures, thermal gradients, heat flow rates, and heat fluxes in an object that are caused by thermal loads that do not vary over time. Such loads include the following:

- Convection
- Radiation
- Heat flow rates
- Heat fluxes (heat flow per unit area)
- Heat generation rates (heat flow per unit volume)
- Constant temperature boundaries

A steady-state thermal analysis may be either linear, with constant material properties; or nonlinear, with material properties that depend on temperature. The thermal properties of most material vary with temperature. This temperature dependency being appreciable, the analysis becomes nonlinear. Radiation boundary conditions also make the analysis nonlinear. Transient calculations are time dependent and ANSYS can both solve distributions as well as create video for time incremental displays of models.

Coupled Fields:

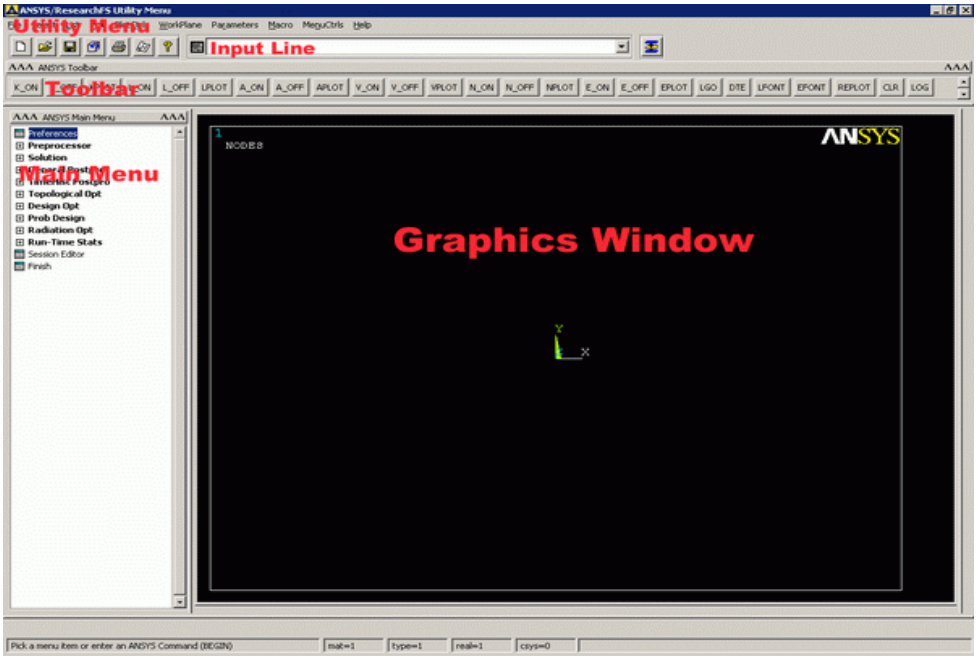
A *coupled-field analysis* is an analysis that takes into account the interaction (coupling) between two or more disciplines (fields) of engineering. A piezoelectric analysis, for example, handles the interaction between the structural and electric fields: it solves for the voltage distribution due to applied displacements, or vice versa. Other examples of coupled-field analysis are thermal-stress analysis, thermal-electric analysis, and fluid-structure analysis.

Some of the applications in which coupled-field analysis may be required are pressure vessels (thermal- stress analysis), fluid flow constrictions (fluid-structure analysis), induction heating (magnetic-thermal analysis), ultrasonic transducers (piezoelectric analysis), magnetic forming (magneto-structural analysis), and micro-electro mechanical systems (MEMS).

1.6 ANSYS Environment.

The ANSYS Environment for ANSYS 7.0 contains 2 windows: the Main Window and an Output Window. Note that this is somewhat different from the previous version of ANSYS which made use of 6 different windows.

Main Window:



Within the Main Window are 5 divisions:

Utility Menu

The Utility Menu contains functions that are available throughout the ANSYS session, such as file controls, selections, graphic controls and parameters.

Input Line

The Input Line shows program prompt messages and allows you to type in commands directly.

Toolbar

The Toolbar contains push buttons that execute commonly used ANSYS commands. More push buttons can be added if desired.

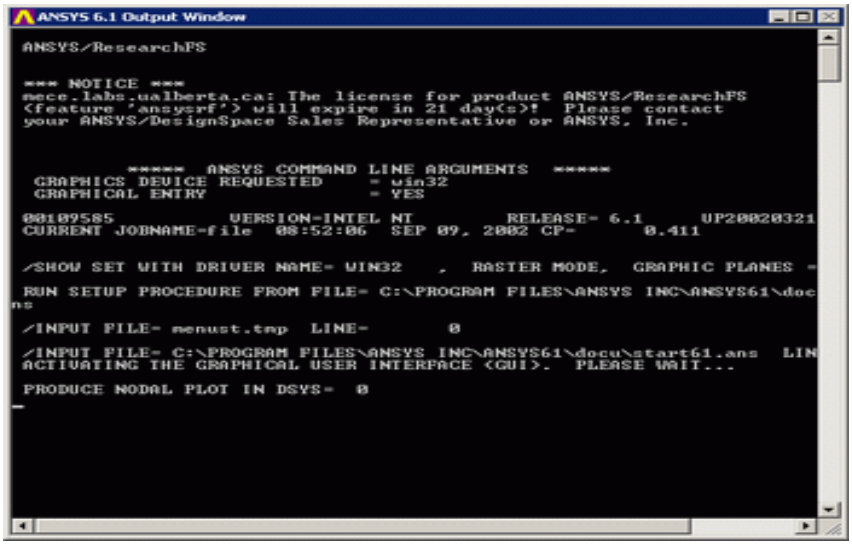
Main Menu

The Main Menu contains the primary ANSYS functions, organized by preprocessor, solution, general postprocessor, design optimizer. It is from this menu that the vast majority of modeling commands are issued. This is where you will note the greatest change between previous versions of ANSYS and version 7.0. However, while the versions appear different, the menu structure has not changed.

Graphics Window

The Graphic Window is where graphics are shown and graphical picking can be made. It is here where you will graphically view the model in its various stages of construction and the ensuing results from the analysis.

Output Window



The Output Window shows text output from the program, such as listing of data etc. It is usually positioned behind the main window and can be put to the front if necessary.

1.7 ANSYS Interface. Graphical Interface vs. Command File Coding

There are two methods to use ANSYS. The first is by means of the graphical user interface or GUI. This method follows the conventions of popular Windows and X-Windows based programs.

The second is by means of command files. The command file approach has a steeper learning curve for many, but it has the advantage that an entire analysis can be described in a small text file, typically in less than 50 lines of commands. This approach enables easy model modifications and minimal file space requirements.

The tutorials in this website are designed to teach both the GUI and the command file approach, however, many of you will find the command file simple and more efficient to use once you have invested a small amount of time into learning the code.

For information and details on the full ANSYS command language, consult:

Help > Table of Contents > Commands Manual.

1.8 FEM Convergence Testing.

Introduction

A fundamental premise of using the finite element procedure is that the body is sub-divided up into small discrete regions known as finite elements. These elements defined by nodes and interpolation functions. Governing equations are written for each element and these elements are *assembled* into a global matrix. Loads and constraints are applied and the solution is then determined.

The Problem

The question that always arises is: *How small do I need to make the elements before I can trust the solution?*

What to do about it...

In general there are no real firm answers on this. It will be necessary to conduct convergence tests! By this we mean that you begin with a mesh discretization and then observe and record the solution. Now repeat the problem with a finer mesh (i.e. more elements) and then compare the results with the previous test. If the results are nearly similar, then the first mesh is probably good enough for that particular geometry, loading and constraints. If the results differ by a large amount however, it will be necessary to try a finer mesh yet.

The Consequences

Finer meshes come with a cost however: more calculation time and large memory requirements (both disk and RAM)! It is desired to find the minimum number of elements that give you a converged solution.

Beam Models

For beam models, we actually only need to define a single element per line unless we are applying a distributed load on a given frame member. When point loads are used, specifying more than one element per line will not change the solution, it will only slow the calculations down. For simple models it is of no concern, but for a larger model, it is desired to minimize the number of elements, and thus calculation time and still obtain the desired accuracy.

General Models

In general however, it is necessary to conduct convergence tests on your finite element model to confirm that a fine enough element discretization has been used. In a solid mechanics problem, this would be done by creating several models with different mesh sizes and comparing the resulting deflections and stresses, for example. In general, the stresses will converge more slowly than the displacement, so it is not sufficient to examine the displacement convergence.

1.9 ANSYS: Saving and Restoring Jobs.

Saving Your Job

It is good practice to save your model at various points during its creation. Very often you will get to a point in the modeling where things have gone well and you like to save it at the point. In that way, if you make some mistakes later on, you will at least be able to come back to this point.

To save your model, select Utility Menu Bar -> File -> Save As Jobname.db. Your model will be saved in a file called jobname.db, where jobname is the name that you specified in the Launcher when you first started ANSYS.

It is a good idea to save your job at different times throughout the building and analysis of the model to backup your work in case of a system crash or other unforeseen problems.

Recalling or Resuming a Previously Saved Job

Frequently you want to start up ANSYS and recall and continue a previous job. There are two methods to do this:

- Using the Launcher...
  - In the ANSYS Launcher, select Interactive... and specify the previously defined job name.
  - Then when you get ANSYS started, select Utility Menu -> File -> Resume Jobname.db .
  - This will restore as much of your database (geometry, loads, solution, etc) that you previously saved.
- Or, start ANSYS and select Utility Menu -> File -> Resume from... and select your job from the list that appears.

1.10 ANSYS Files.

Introduction

A large number of files are created when you run ANSYS. If you started ANSYS without specifying a jobname, the name of all the files created will be FILE.* where the * represents various extensions described below. If you specified a jobname, say Frame, and then the created files will all have the file prefix, Frame again with various extensions:

frame.db

Database file (binary). This file stores the geometry, boundary conditions and any solutions.

frame.dbb

Backup of the database file (binary).

frame.err

Error file (text). Listing of all error and warning messages.

frame.out

Output of all ANSYS operations (text). This is what normally scrolls in the output window during an ANSYS session.

frame.log

Logfile or listing of ANSYS commands (text). Listing of all equivalent ANSYS command line commands used during the current session.

etc...

Depending on the operations carried out, other files may have been written. These files may contain results, etc.

What to save?

When you want to clean up your directory, or move things from the /scratch directory, what files do you need to save?

- If you will always be using the GUI, then you only require the .db file. This file stores the geometry, boundary conditions and any solutions. Once the ANSYS has started, and the jobname has been specified, you need only activate the resume command to proceed from where you last left off (see Saving and Restoring Jobs).

- If you plan on using ANSYS command files, then you need only store your command file and/or the log file. This file contains a complete listing of the ANSYS commands used to get you model to its current point. That file may be rerun as is, or edited and rerun as desired (Command File Creation and Execution).

If you plan to use the command mode of operation, starting with an existing log file, rename it first so that it does not get over-written or added to, from another ANSYS run.

1.11 Printing and Plotting ANSYS Results to a File.

Printing Text Results to a File

ANSYS produces lists and tables of many types of results that are normally displayed on the screen. However, it is often desired to save the results to a file to be later analyzed or included in a report.

- Stresses: instead of using 'Plot Results' to plot the stresses, choose 'List Results'. Select 'Elem Table Data', and choose what you want to list from the menu. You can pick multiple items. When the list appears on the screen in its own window, Select 'File'/'Save As...' and give a file name to store the results.
- Any other solutions can be done in the same way. For example select 'Nodal Solution' from the 'List Results' menu, to get displacements.
- Preprocessing and Solution data can be listed and saved from the 'List' menu in the 'Utility Menu bar'. Save the resulting list in the same way described above.

ANALYSIS OF ONE DIMENSIONAL PROBLEMS

CHAPTER TWO

ONE DIMENSIONAL ELEMENT (Bars)

2.1 Introduction.

Elements having spread in a single direction are known as one-dimensional elements. Such elements have a particular dimension quite longer compared to its other dimension. Such elements are pipes, shafts, truss members, cables, small diameter chimneys, thin towers etc.

For one dimensional problem, the stress, strain, displacement and loading depend only on the variable x. The degree of freedom is a variable that describes the behavior of a node in an element. In a one dimensional problem, every node is permitted to displace only in the ± x direction. Thus each node has only one degree of freedom.

2.2 Element used in one dimensional problem - LINK1 2-D Spar

LINK1 Element Description

LINK1 can be used in a variety of engineering applications. Depending upon the application, you can think of the element as a truss, a link, a spring, etc. The 2-D spar element is a uniaxial tension-compression element with two degrees of freedom at each node: translations in the nodal x and y directions. As in a pin-jointed structure, no bending of the element is considered.

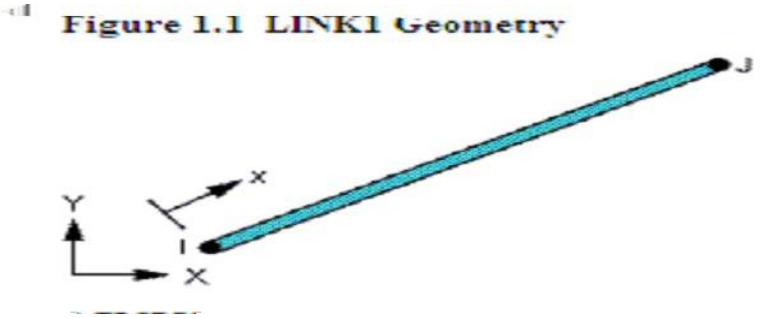


Fig: LINK1 Geometry

LINK1 Input Data

Figure "LINK1 Geometry" shows the geometry, node locations, and the coordinate system for this element. The element is defined by two nodes, the cross-sectional area, an initial strain, and the material properties. The element x-axis is oriented along the length of the element from node I toward node J.

"LINK1 Input Summary" summarizes the element input. Element Input gives a general description of element input.

LINK1 Input Summary

Nodes

I, J

Degrees of Freedom

UX, UY

Real Constants

AREA - Cross-sectional area

ISTRN - Initial strain

Material Properties

EX, ALPX (or CTEX *or* THSX), DENS, DAMP

Surface Loads

None

Body Loads

Temperatures --

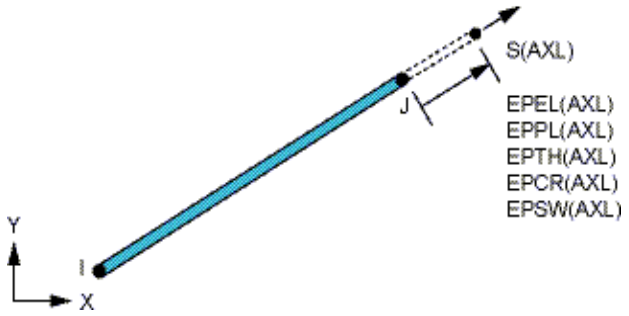
T(I), T(J)

LINK1 Output Data

The solution output associated with the element is in two forms:
Nodal displacements included in the overall nodal solution
Additional element output as shown in Table "LINK1 Element Output Definitions".

Figure "LINK1 Stress Output" illustrates several items. A general description of solution output is given in Solution Output. **LINK1 Stress Output**

Figure 1.2 LINK1 Stress Output



The Element Output Definitions table uses the following notation:

A colon (:) in the Name column indicates the item can be accessed by the Component Name method [**ETABLE**, **ESOL**]. The O column indicates the availability of the items in the file Jobname.OUT. The R column indicates the availability of the items in the results file.

In either the O or R columns, Y indicates that the item is *always* available, a number refers to a table footnote that describes when the item is *conditionally* available, and a - indicates that the item is *not* available.

2.3 LINK1 Element Output Definitions

Name	Definition	O	R
EL	Element Number	Y	Y
NODES	Element node numbers (I and J)	Y	Y
MAT	Material number for the element	Y	Y
VOLU:	Element volume	-	Y
XC, YC	Location where results are reported	Y	2
TEMP	Temperature at nodes I and J	Y	Y
FLUEN	Fluence at nodes I and J	Y	Y
MFORX	Member force in the element coordinate system X direction	Y	Y
SAXL	Axial stress in the element	Y	Y
EPELAXL	Axial elastic strain in the element	Y	Y

Name	Definition	O	R
EPTHAXL	Axial thermal strain in the element	Y	Y
EPINAXL	Axial initial strain in the element	Y	Y
SEPL	Equivalent stress from the stress-strain curve	1	1
SRAT	Ratio of trial stress to the stress on yield surface	1	1
EPEQ	Equivalent plastic strain	1	1
HPRES	Hydrostatic pressure	1	1
EPPLAXL	Axial plastic strain	1	1
EPCRAXL	Axial creep strain	1	1
EPSWAXL	Axial swelling strain	1	1

Only if the element has a nonlinear material

Available only at centroid as a ***GET** item.

The Item and Sequence Number... table lists output available through the **ETABLE** command using the Sequence Number method.

Output Quantity Name

Output quantity as defined in the Element Output Definitions table.

Item

Predetermined Item label for **ETABLE** command **E**
Sequence number for single-valued or constant element data **I,J**
Sequence number for data at nodes I and J

2.4 LINK1 Item and Sequence Numbers

Output Quantity Name	ETABLE and ESOL Command Input			
	Item	E	I	J
SAXL	LS	1	-	-
EPELAXL	LEPEL	1	-	-
EPTHAXL	LEPTH	1	-	-
EPSWAXL	LEPTH	2	-	-
EPINAXL	LEPTH	3	-	-
EPPLAXL	LEPPL	1	-	-
EPCRAXL	LEPCR	1	-	-
SEPL	NLIN	1	-	-
SRAT	NLIN	2	-	-
HPRES	NLIN	3	-	-
EPEQ	NLIN	4	-	-
MFORX	SMISC	1	-	-
FLUEN	NMISC	-	1	2
TEMP	LBFE	-	1	2

LINK1 Assumptions and Restrictions

The spar element assumes a straight bar, axially loaded at its ends, of uniform properties from end to end. The length of the spar must be greater than zero, so nodes I and J must not be coincident.

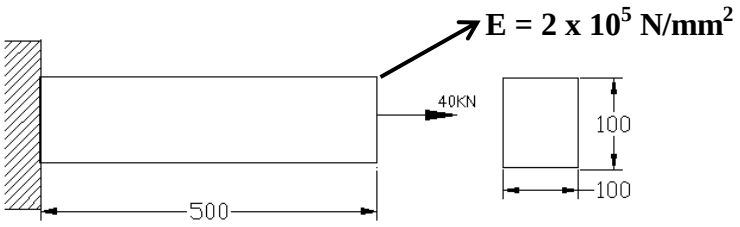
The spar must lie in an X-Y plane and must have an area greater than zero. The temperature is assumed to vary linearly along the length of the spar.

The displacement shape function implies a uniform stress in the spar.

The initial strain is also used in calculating the stress stiffness matrix, if any, for the first cumulative iteration.

Bars of Constant Cross-section Area

1. A bar is subjected to an axial load of 40 KN and has a length of 500 mm. Find the total extension of the bar and stress due to tension. Also find the reaction forces at the supports.take poisons ratio as 0.27.



Procedure:

Step 1: Ansys Utility Menu

File – clear and start new – do not read file – ok – yes.

Preprocessing: Defining the problem

Step 2: Ansys Main Menu – Preferences
select – STRUCTURAL - ok

Step 3: Preprocessor

Element type – Add/Edit/Delete – Add – Link – 2D spar 1 – ok – close.
Real constants – Add – ok – real constant set no – 1 – c/s area – 100*100 – ok.
Material Properties – material models – Structural – Linear – Elastic – Isotropic – EX – 2e5 – PRXY – 0.27 – ok – close.

Step 4: Preprocessor

Modeling – Create – Nodes – In Active CS – Apply (first node is created) – X,Y,Z location in CS – 500 (X value w.r.t first node) – ok (second node is created).
Create –Elements – Element Attributes – Material – 1 - real constant set no – 1 – ok
Auto numbered– Thru Nodes – pick 1 & 2 – ok (element created through nodes).

Assigning loads:

Step 5: Preprocessor
Loads – Define loads – apply – Structural – Displacement – on Nodes- pick node 1 – apply – DOFs to be constrained – All DOF – ok.
Loads – Define loads – apply – Structural – Force/Moment – on Nodes- pick node 2 – apply – direction of For/Mom – FX – Force/Moment value – 40000N (+ve value) – ok.

Solution Phase:

Step 6: Solution

Solve – current LS – ok (Solution is done is displayed) – close.

Post processing: Viewing the results

Step 7: General Post Processor

Element table – Define table – Add –User label for item – Type **STRESS** – ' Results data item' – By Sequence num – LS, – **LS,1** – ok.

Step 8: General Post Processor

Plot Results – contour plot – nodel solution – DOF solution – X-component of Displacement – ok.

Plot Results – contour plot – Line Element Results – Elem table item at node I – LS,1 – Elem table item at node J – LS,1 – ok (Line Stress diagram will be displayed).

OR

Plot Results – contour plot – Elem Table – item to be plotted – select Stress - ok(Line Stress diagram will be displayed).

List Results – reaction solution – items to be listed – All items – ok (reaction forces will be displayed with the node numbers).

List Results – Element Table Data – item to be listed – Select STRESS – ok.(stresses in each element will be displayed)

Step 9: Ansys Utility Menu

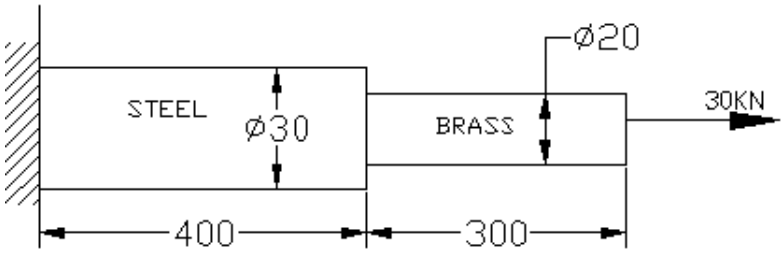
PlotCtrls – style – size and shape – Display of elements – ON – ok.

Step 10: Ansys Utility Menu

PlotCtrls - Animate – Deformed shape – def+undeformed-ok

Stepped Bars with different Cross-Section Area

2. Find the total extension of the bar subjected to a tensile force of 30 KN. Take E of steel = 2 x 10^5 N/ mm^2 and that of brass 1 x 10^5 N/ mm^2 . Determine the following 1. Nodal solutions 2. Reactions at the supports 3. Stresses in each element.



Procedure:

Step 1: Ansys Utility Menu
File – clear and start new – do not read file – ok – yes.

Preprocessing: Defining the problem

Step 2: Ansys Main Menu – Preferences
select – STRUCTURAL - ok

Step 3: Preprocessor

Element type – Add/Edit/Delete – Add – Link – 2D spar 1 – ok – close.
Real constants – Add – ok – real constant set no – 1 – c/s area – (π*30**2/4) – apply - real constant set no – 2 – c/s area – (π*20**2/4) – ok – close.
Material Properties – material models – Material model number 1 - Structural – Linear – Elastic – Isotropic – EX1 – 2e5 – PRXY – 0.25 – ok
Material Main Menu - Material – New model – Define material ID – 2 – ok – Material model number 2 - Structural – Linear – Elastic – Isotropic – EX2 – 1e5 – ok – close.

Step 4: Preprocessor

Modeling – Create – Nodes – In Active CS – Apply (first node is created) – X,Y,Z location in CS – 400 (X value w.r.t first node) – Apply (second node is created) - X,Y,Z location in CS – 700 (X value w.r.t first node) - ok(Third node is created).
Create –Elements – Element Attributes – Material – 1 - real constant set no – 1 – ok
Auto numbered– Thru Nodes – pick 1 & 2 – ok (first element created through nodes).
Create –Elements – Element Attributes – Material – 2 - real constant set no – 2 – ok
Auto numbered– Thru Nodes – pick 2 & 3 – ok (second element created through nodes).

Assigning loads:

Step 5: Preprocessor

Loads – Define loads – apply – Structural – Displacement – on Nodes- pick node 1 – apply – DOFs to be constrained – All DOF – ok.

Loads – Define loads – apply – Structural – Force/Moment – on Nodes- pick node 3 – apply – direction of For/Mom – FX – Force/Moment value – 30000N (+ve value) – ok.

Solution Phase:

Step 6: Solution

Solve – current LS – ok (Solution is done is displayed) – close.

Post processing: Viewing the results

Step 7: General Post Processor

Element table – Define table – Add –User label for item – Type **STRESS** – ' Results data item' – By Sequence num – LS, – **LS,1** – ok.

Step 8: General Post Processor

Plot Results – contour plot – nodel solution – DOF solution – X-component of Displacement – ok.

Plot Results – contour plot – Line Element Results – Elem table item at node I – LS,1 – Elem table item at node J – LS,1 – ok (Line Stress diagram will be displayed).

OR

Plot Results – contour plot – Elem Table – item to be plotted – select Stress - ok(Line Stress diagram will be displayed).

List Results – reaction solution – items to be listed – All items – ok (reaction forces will be displayed with the node numbers).

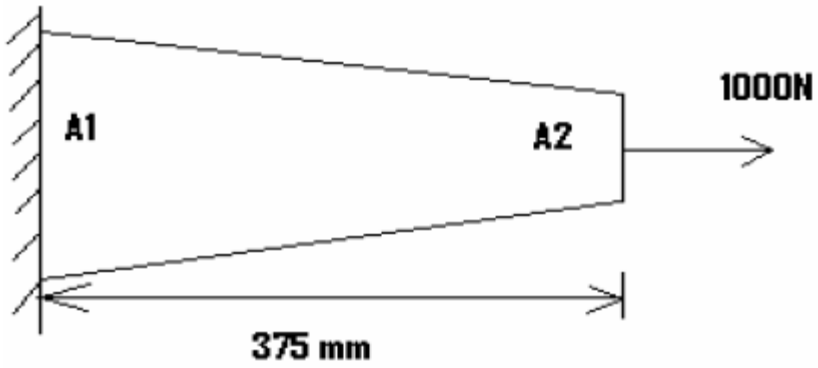
List Results – Element Table Data – item to be listed – Select STRESS – ok.(stresses in each element will be displayed)

Step 9: Ansys Utility Menu : PlotCtrls – style – size and shape – Display of elements – ON – ok.

Step 10: Ansys Utility Menu : PlotCtrls - Animate – Deformed shape – def+undeformed-ok

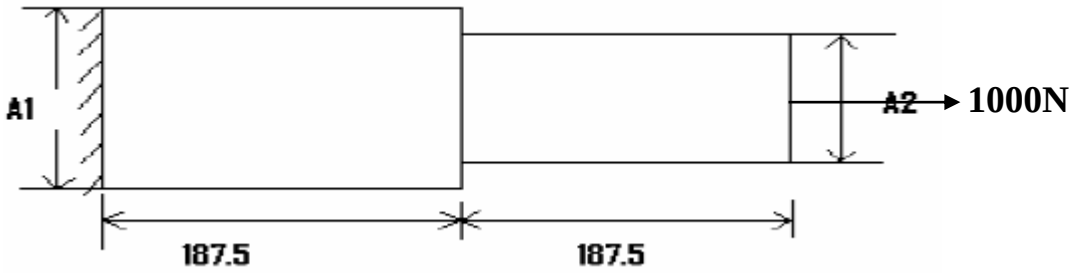
Tapered Cross-Section Area

3. For the tapered bar shown in the figure determine the displacement, stress and reaction in the bar.
   Given $A_1 = 1000\ mm^2$ and $A_2 = 500\ mm^2$, $E = 2x10^5\ N/mm^2$



Note: The Tapered bar is modified into **2 elements** as shown below with modified area of cross section.

$(1000+500)/2 = 750\ mm^2$
$A_1 = (1000+750)/2 = 875\ mm^2$
$A_2 = (750+500)/2 = 625\ mm^2$



Procedure:

Step 1: Ansys Utility Menu
File – clear and start new – do not read file – ok – yes.

Preprocessing: Defining the problem

Step 2: Ansys Main Menu – Preferences
select – STRUCTURAL - ok

Step 3: Preprocessor

Element type – Add/Edit/Delete – Add – Link – 2D spar 1 – ok – close.
Real constants – Add – ok – real constant set no – 1 – c/s area – **875 mm** – apply - real constant set no – 2 – c/s area – **625 mm**– ok – close.
Material Properties – material models – Material model number 1 - Structural – Linear – Elastic – Isotropic – EX – 2e5 – PRXY – 0.27 – ok – close.

Step 4: Preprocessor

Modeling – Create – Nodes – In Active CS – Apply (first node is created) – X,Y,Z location in CS – 187.5 (X value w.r.t first node) – Apply (second node is created) - X,Y,Z location in CS – 375 (X value w.r.t first node) - ok(Third node is created).
Create –Elements – Element Attributes – Material – 1 - real constant set no – 1 – ok
Auto numbered– Thru Nodes – pick 1 & 2 – ok (first element created through nodes).
Create –Elements – Element Attributes – Material – 1 - real constant set no – 2 – ok
Auto numbered– Thru Nodes – pick 2 & 3 – ok (second element created through nodes).

Assigning Loads:

Step 5: Preprocessor

Loads – Define loads – apply – Structural – Displacement – on Nodes- pick node 1 – apply – DOFs to be constrained – All DOF – ok.
Loads – Define loads – apply – Structural – Force/Moment – on Nodes- pick node 3 – apply – direction of For/Mom – FX – Force/Moment value – 1000N (+ve value) – ok.

Solution Phase:

Step 6: Solution

Solve – current LS – ok (Solution is done is displayed) – close.

Post Processing: Viewing the result

Step 7: General Post Processor

Element table – Define table – Add –User label for item – Type **STRESS** – ' Results data item' – By Sequence num – LS, – **LS,1** – ok.

Step 8: General Post Processor

Plot Results – contour plot – nodel solution – DOF solution – X-component of Displacement – ok.

Plot Results – contour plot – Line Element Results – Elem table item at node I – LS,1 – Elem table item at node J – LS,1 – ok (Line Stress diagram will be displayed).

OR

Plot Results – contour plot – Elem Table – item to be plotted – select Stress - ok(Line Stress diagram will be displayed).

List Results – reaction solution – items to be listed – All items – ok (reaction forces will be displayed with the node numbers).

List Results – Element Table Data – item to be listed – Select STRESS – ok.(stresses in each element will be displayed)

Step 9: Ansys Utility Menu : PlotCtrls – style – size and shape – Display of elements – ON – ok.

Step 10: Ansys Utility Menu : PlotCtrls - Animate – Deformed shape – def+undeformed-ok.

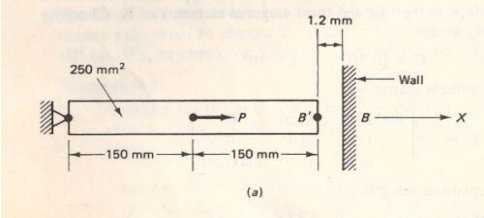
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PRACTICE PROBLEMS

One Dimensional Problems

BARS OF CONSTANT CROSS SECTIONAL AREA

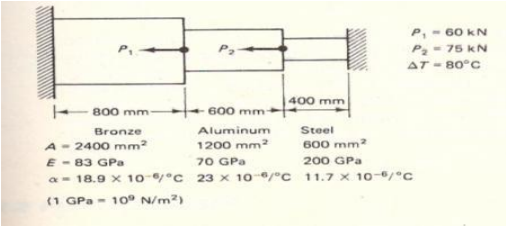
A load of P=60x10^3 is applied as shown. Determine the nodal displacement, stresses and forces in each member and support reactions taking E=20 x 10^3 N/mm^2



STEPPED BARS

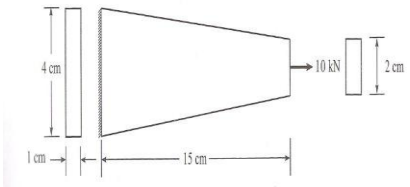
The structure is subjected to an increase in temperature DT=80 deg C.

a) Determine the nodal displacements.
b) Stresses and forces in each material.
c) Reaction forces.

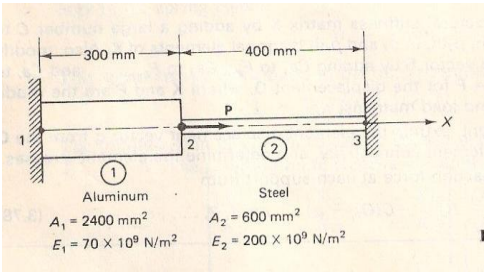


BARS OF TAPERED CROSS SECTIONAL AREA

For the tapered bar as shown below a) Determine the nodal displacements)Stresses and forces in each material and c)Reaction forces considering three finite elements.



Consider the bar shown as below. Determine the following when an axial load of P=200000 N is applied. Nodal displacements. Forces and Stresses in each material, Reaction forces.



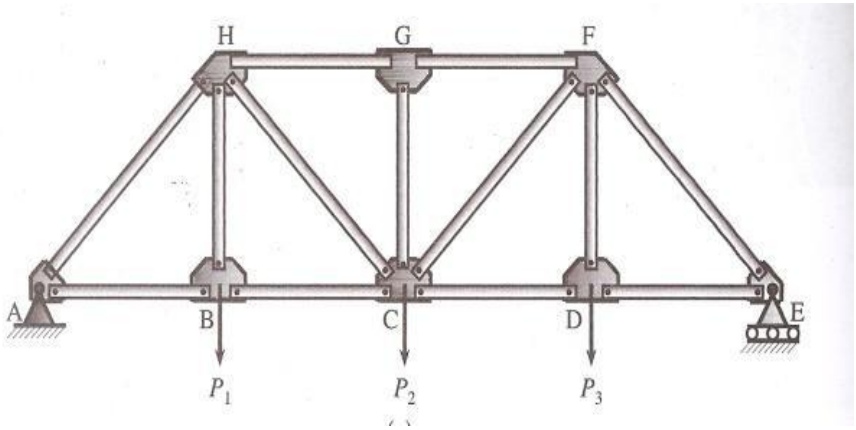
ANALYSIS OF TRUSSES

CHAPTER 3

Analysis of Trusses

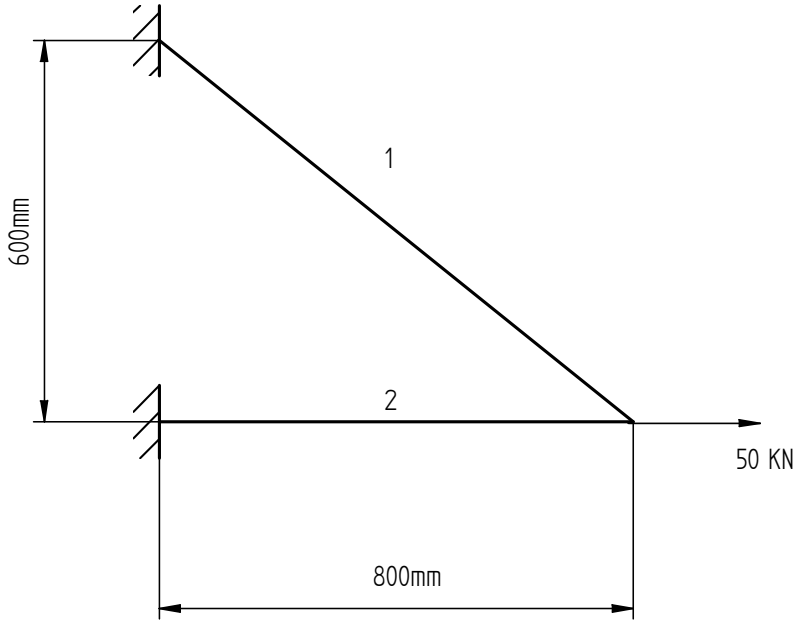
3.1 Introduction

Truss is an engineering structure with each member in either direct tension or compression. Hence forces developed within the members of the truss are purely axial i.e.no bending is considered. Each member is connected to the adjoining member by means of welded connections or by bolts, rivets, pins. A truss element is two dimensional i.e. it can displace both in x and y directions. Thus we have two degrees of freedom at each node. Analysis of trusses is of vital importance as they find wide applications in bridges, roofs of buildings, cranes, transmission towers etc. Typical truss structure is as shown below.



Two Bar/ Multi bar element trusses

1. Consider the two bar truss shown in fig for the given data, Find, i).Nodal Displacement, ii) Stress in the element iii) Reaction Forces and Forces in each member, Take **A_1= 250mm^2 , E_1= 2x10^5 N/mm^2 and A_2=800mm^2 , E_2=2.5x10^5 N/mm2**



Procedure:

Step 1: Ansys Utility Menu
File – clear and start new – do not read file – ok – yes.

Preprocessing: Defining the Problem

Step 2: Ansys Main Menu – Preferences
select – STRUCTURAL - ok

Step 3: Preprocessor

Element type – Add/Edit/Delete – Add – Link – 2D spar 1 – ok – close.
Real constants – Add – ok – real constant set no – 1 – c/s area – **250mm^2** – apply - real constant set no – 2 – c/s area – **800mm^2**– ok – close.
Material Properties – material models – Material model number 1 - Structural – Linear – Elastic – Isotropic – EX1 – **2e5 N/mm^2** – PRXY – 0.25 – ok
Material Main Menu - Material – New model – Define material ID – 2 – ok – Material model number 2 - Structural – Linear – Elastic – Isotropic – EX2 – **2.5e5 N/mm^2** – ok – close.

Step 4: Preprocessor

Modeling – Create – Nodes – In Active CS – Apply (first node is created) – X,Y,Z location in CS – 800 (X value w.r.t first node) – Apply (second node is created) - X,Y,Z location in CS – 0,600 (X,Y value w.r.t first node) - ok(Third node is created).
Create –Elements – Element Attributes – Material – 1 - real constant set no – 1 – ok
Auto numbered– Thru Nodes – pick 2 & 3 – ok (first element created through nodes).
Create –Elements – Element Attributes – Material – 2 - real constant set no – 2 – ok
Auto numbered– Thru Nodes – pick 1 & 2 – ok (second element created through nodes).

Assigning Loads:

Step 5: Preprocessor
Loads – Define loads – apply – Structural – Displacement – on Nodes- pick node 1 & 3 – apply – DOFs to be constrained – All DOF – Displacement Value – 0 - ok.
Loads – Define loads – apply – Structural – Force/Moment – on Nodes- pick node 2 – ok – direction of For/Mom – FX – Force/Moment value – 50000N (+ve value) – ok.

Solution Phase:

Step 6: Solution
Solve – current LS – ok (Solution is done is displayed) – close.

Post Processing: Viewing the result

Step 7: General Post Processor
Element table – Define table – Add –User label for item – Type **STRESS** – ' Results data item' – By Sequence num – LS, – **LS,1** – ok.

Step 8: General Post Processor
Plot Results – contour plot – nodel solution – DOF solution – X-component of Displacement – ok.
Plot Results – contour plot – Element Results – Elem table item at node I – LS,1 – Elem table item at node J – LS,1 – ok (Line Stress diagram will be displayed).

OR

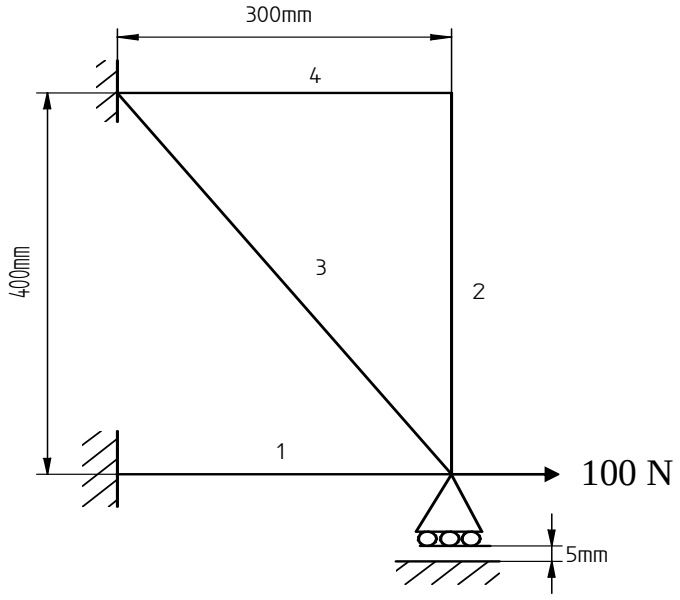
Plot Results – contour plot – Elem Table – item to be plotted – select Stress - ok(Line Stress diagram will be displayed).

List Results – reaction solution – items to be listed – All items – ok (reaction forces will be displayed with the node numbers).
List Results – Element Table Data – item to be listed – Select STRESS – ok.(stresses in each element will be displayed)

Step 9: Ansys Utility Menu : PlotCtrls – style – size and shape – Display of elements – ON – ok.

Step 10: Ansys Utility Menu : PlotCtrls - Animate – Deformed shape – def+undeformed-ok

2. For the following truss members, find out (i) Nodal displacement (ii) Reaction forces (iii) stress in each member (iv) Axial elastic strain, Take **A=100mm^2** and **E=3x10^5 Mpa**



Procedure:

Step 1: Ansys Utility Menu
File – clear and start new – do not read file – ok – yes.

Preprocessing: Defining the problem

Step 2: Ansys Main Menu – Preferences
select – STRUCTURAL - ok

Step 3: Preprocessor

Element type – Add/Edit/Delete – Add – Link – 2D spar 1 – ok – close.
Real constants – Add – ok – real constant set no – 1 – c/s area – **100mm^2**– ok.
Material Properties – material models – Structural – Linear – Elastic – Isotropic – EX – **3e5 Mpa(N/mm^2)**– PRXY – 0.27 – ok – close.

Step 4: Preprocessor

Modeling – Create – Nodes – In Active CS – Apply (first node is created) – X,Y,Z location in CS – 300 (X value w.r.t first node) – Apply (second node is created) - X,Y,Z location in CS – 300,400 (X,Y value w.r.t first node) – Apply (third node is created) - X,Y,Z location in CS – 0,400 (X,Y value w.r.t first node) – ok (fourth node is created).
Create –Elements – Element Attributes – Material – 1 - real constant set no – 1 – ok
Auto numbered– Thru Nodes – pick 1 & 2 – ok (first element created through nodes).

Auto numbered– Thru Nodes – pick 2 & 3 – ok (second element created through nodes).
Auto numbered– Thru Nodes – pick 3 & 4 – ok (third element created through nodes).
Auto numbered– Thru Nodes – pick 4 & 2 – ok (fourth element created through nodes).

Assigning Loads:

Step 5: Preprocessor

Loads – Define loads – apply – Structural – Displacement – on Nodes- pick node 1 & 4– apply – DOFs to be constrained – All DOF – Displacement value – 0 - apply – pick node 2 – ok - DOFs to be constrained – UY - Displacement value – 5 - ok.

Loads – Define loads – apply – Structural – Force/Moment – on Nodes- pick node 2 – ok – direction of For/Mom – FX – Force/Moment value – 100N (+ve value) – ok.

Solution Phase:

Step 6: Solution

Solve – current LS – ok (Solution is done is displayed) – close.

Post Processing: Viewing the result

Step 7: General Post Processor

Element table – Define table – Add –User label for item – Type **STRESS** – ' Results data item' – By Sequence num – LS, – **LS,1** – apply - User label for item – Type **STRAIN** – ' Results data item' – By Sequence num – LEPEL, – **LEPEL,1** – ok – close.

Step 8: General Post Processor

Plot Results – contour plot – nodel solution – DOF solution – X-component of Displacement – ok.

Plot Results – contour plot – Line Element Results – Elem table item at node I – LS,1 – Elem table item at node J – LS,1 – ok (Line Stress diagram will be displayed).

Plot Results – contour plot – Line Element Results – Elem table item at node I – LEPEL,1 – Elem table item at node J – LEPEL,1 – ok (Line Strain diagram will be displayed).

OR

Plot Results – contour plot – Elem Table – item to be plotted – select Stress - ok(Line Stress diagram will be displayed).

Plot Results – contour plot – Elem Table – item to be plotted – select Stress - ok(Line Strain diagram will be displayed).

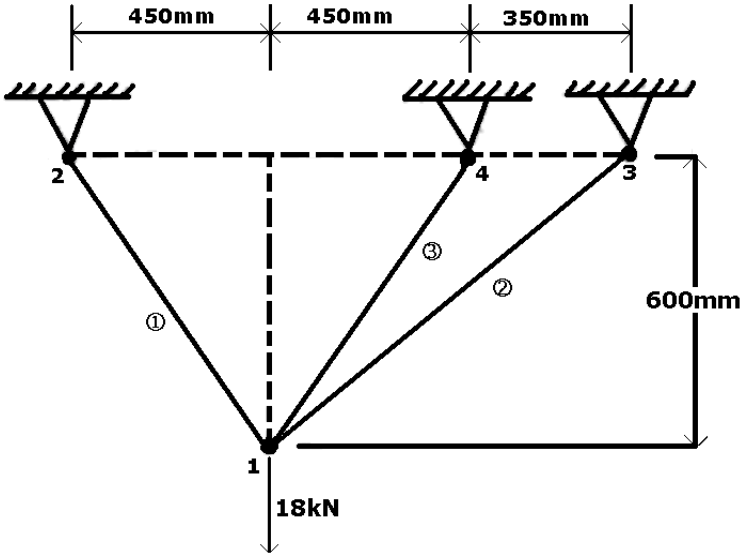
List Results – reaction solution – items to be listed – All items – ok (reaction forces will be displayed with the node numbers).

List Results – Element Table Data – item to be listed – Select STRESS & STRAIN – ok. (stresses & strain in each element will be displayed)

Step 9: Ansys Utility Menu : PlotCtrls – style – size and shape – Display of elements – ON – ok.

Step 10: Ansys Utility Menu : PlotCtrls - Animate – Deformed shape – def+undeformed-ok.

3. For the two bar truss shown in figure, determine the reaction forces, nodal displacement and element Stress. take **E=2.1x 10^5 N/mm2, A= 1000mm2.**



Procedure:

Step 1: Ansys Utility Menu
File – clear and start new – do not read file – ok – yes.

Preprocessing: Defining the problem

Step 2: Ansys Main Menu – Preferences
select – STRUCTURAL - ok

Step 3: Preprocessor

Element type – Add/Edit/Delete – Add – Link – 2D spar 1 – ok – close.
Real constants – Add – ok – real constant set no – 1 – c/s area – **1000mm^2**– ok.
Material Properties – material models – Structural – Linear – Elastic – Isotropic – EX – **2.1e5 N/mm^2**– PRXY – 0.27 – ok – close.

Step 4: Preprocessor

Modeling – Create – Nodes – In Active CS – Apply (first node is created) – X,Y,Z location in CS – -450,600 (X,Y value w.r.t first node) – Apply (second node is created) - X,Y,Z location in CS – 450,600 (X,Y value w.r.t first node) – Apply (third node is created) - X,Y,Z location in CS – 800,400 (X,Y value w.r.t first node) – ok (fourth node is created).

Create –Elements – Element Attributes – Material – 1 - real constant set no – 1 – ok
Auto numbered– Thru Nodes – pick 1 & 2 – ok (first element created through nodes).

Auto numbered– Thru Nodes – pick 1 & 4 – ok (second element created through nodes).
Auto numbered– Thru Nodes – pick 1 & 3 – ok (third element created through nodes).

Assigning Loads:

Step 5: Preprocessor

Loads – Define loads – apply – Structural – Displacement – on Nodes- pick node 2,4 & 4– ok – DOFs to be constrained – UX,UY – Displacement value – 0 - ok.
Loads – Define loads – apply – Structural – Force/Moment – on Nodes - pick node 1 – ok – direction of For/Mom – FY – Force/Moment value – (**-18000N**) (-ve value) – ok.

Solution Phase:

Step 6: Solution

Solve – current LS – ok (Solution is done is displayed) – close.

Post Processing: Viewing the result

Step 7: General Post Processor

Element table – Define table – Add –User label for item – Type **STRESS** – ' Results data item' – By Sequence num – LS, – **LS,1** – ok.

Step 8: General Post Processor

Plot Results – contour plot – nodel solution – DOF solution –Displacement vector sum – ok.
Plot Results – contour plot – Line Element Results – Elem table item at node I – LS,1 – Elem table item at node J – LS,1 – ok (Line Stress diagram will be displayed).

OR

Plot Results – contour plot – Elem Table – item to be plotted – select Stress - ok(Line Stress diagram will be displayed).

List Results – reaction solution – items to be listed – All items – ok (reaction forces will be displayed with the node numbers).
List Results – Element Table Data – item to be listed – Select STRESS – ok. (stresses in each element will be displayed)

Step 9: Ansys Utility Menu : PlotCtrls – style – size and shape – Display of elements – ON – ok.

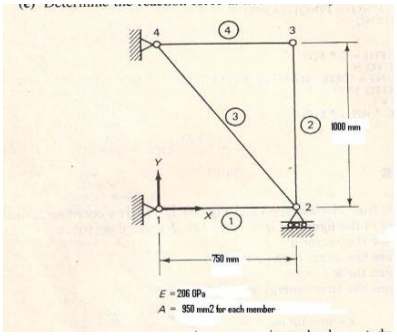
Step 10: Ansys Utility Menu : PlotCtrls - Animate – Deformed shape – def+undeformed-ok.

******

PRACTICE PROBLEMS

Analysis of Trusses

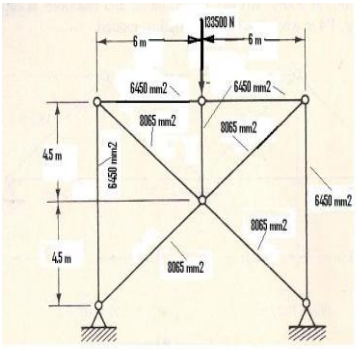
For the truss a horizontal load of P = 18000 N is applied in the x direction at node 2.



Determine

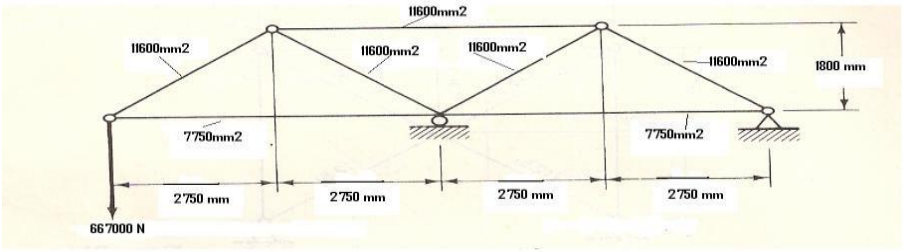
a) Nodal displacements.
b) Reaction forces.
c) Forces in each member.
d) Stresses in each member.

Consider the truss loaded as shown. Cross sectional area in square mm are shown in parenthesis. Consider symmetry and model only one half of the truss shown. Taking E = 200 GPa Determine the following.

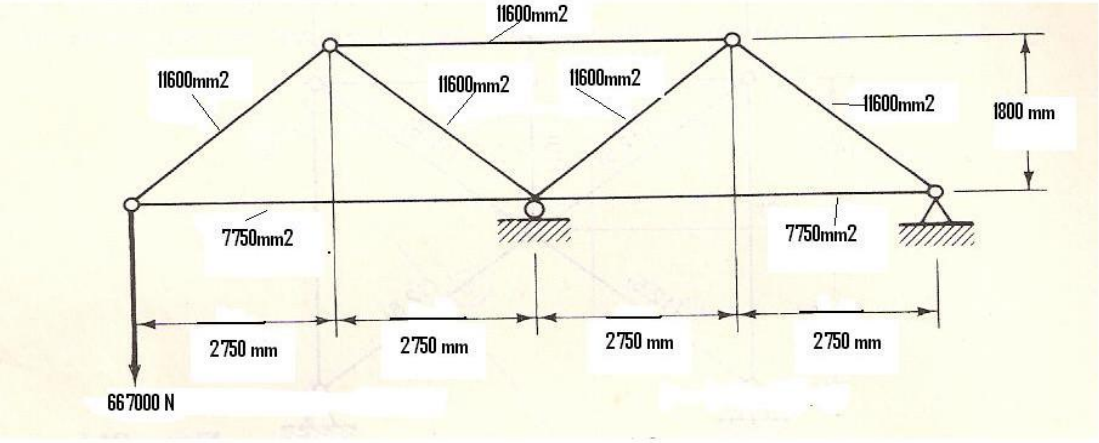


For the truss shown below determine the following. Taking E 206 GPa determine the following

a)Nodal displacements b)Reaction forces c)Forces in each member d)Stresses in each member.



SELF LEARNING EXERCISE: For the truss shown below determine the following. Taking E 206 GPa determine the following a) Nodal displacements) Reaction forces) Forces in each member) Stresses in each member.



ANALYSIS OF BEAMS

CHAPTER 4

ANALYSIS OF BEAMS

4.1 Introduction

A beam is a long, slender structural member generally subjected to transverse loading and produces significant bending deformation. Long horizontal members used in buildings and bridges, and shafts supported in bearings are some examples of beams.

Beams with cross sections that are symmetrical with respect to plane of loading are considered here. A general horizontal beam as shown below shows the cross-section and bending stress distribution. For small deflections, we recall from the elementary beam theory that

$\sigma = MY/I \qquad \varepsilon = \sigma/E \qquad d^2v/dx^2 = M/EI$

Where σ is the normal stress, ε is the normal strain, M is the bending moment at the section, v is the deflection of the centroidal axis at x, and I is the moment of Inertia of the section about the neutral axis.

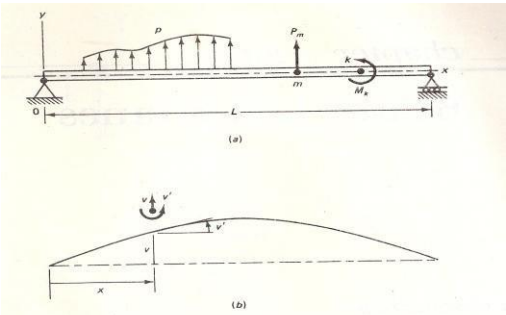


Figure 8.1 (*a*) Beam loading and (*b*) deformation of the neutral axis.

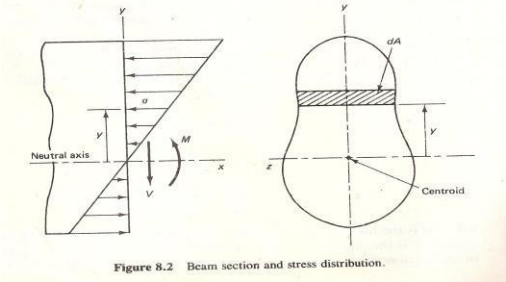


Figure 8.2 Beam section and stress distribution.

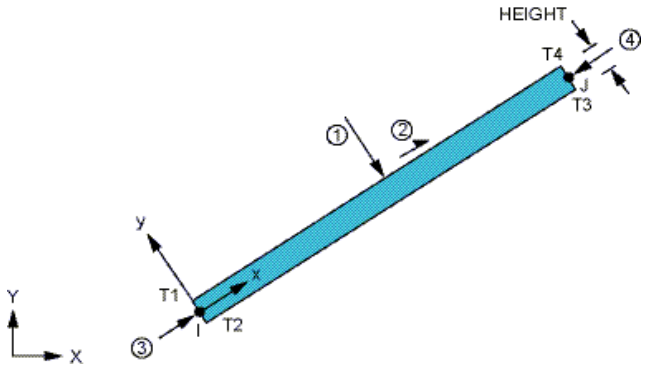
Element used in this chapter is BEAM3 2-D Elastic Beam

4.2 BEAM3 Element Description

BEAM3 is a uniaxial element with tension, compression, and bending capabilities. The element has three degrees of freedom at each node: translations in the nodal x and y directions and rotation about the nodal z-axis.

BEAM3 Geometry

Figure 3.1 BEAM3 Geometry



BEAM3 Input Data

Figure 3.1: "BEAM3 Geometry" shows the geometry, node locations, and the coordinate system for this element. The element is defined by two nodes, the cross-sectional area, the area moment of inertia, the height, and the material properties. The initial strain in the element (ISTRN) is given by Δ/L, where Δ is the difference between the element length, L (as defined by the I and J node locations), and the zero strain length.

"BEAM3 Input Summary" summarizes the element input. Element Input contains a general description of element input.

BEAM3 Input Summary

Nodes

I, J

Degrees of Freedom

UX, UY, ROTZ

Real Constants

AREA - Cross-sectional area

IZZ - Area moment of inertia

HEIGHT - Total beam height

SHEARZ - Shear deflection constant

ISTRN - Initial strain

ADDMAS - Added mass per unit length

Note: SHEARZ goes with the IZZ. If SHEARZ = 0, there is no shear deflection in the element Y direction.

Material Properties

EX, ALPX (or CTEX *or* THSX), DENS, GXY, DAMP

Surface Loads

Pressure --

face 1 (I-J) (-Y normal direction)

face 2 (I-J) (+X tangential direction)

face 3 (I) (+X axial direction)

face 4 (J) (-X axial direction) (use a negative value for loading in the opposite direction)

Body Loads

Temperatures --

T1, T2, T3, T4

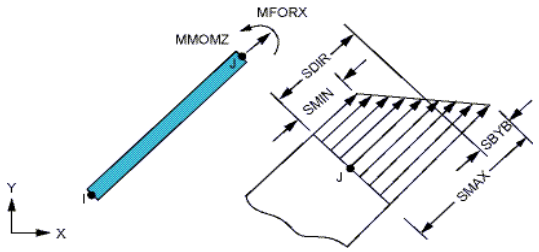
BEAM3 Output Data

The solution output associated with the element is in two forms:

- Nodal displacements included in the overall nodal solution
- Additional element output as shown in Table 3.1: "BEAM3 Element Output Definitions".

Figure 3.2: "BEAM3 Stress Output" illustrates several items. Solution Output gives a general description of solution output. Figure 3.2 BEAM3 Stress Output

Figure 3.2 BEAM3 Stress Output



The Element Output Definitions table uses the following notation:

A colon (:) in the Name column indicates the item can be accessed by the Component Name method [**ETABLE**, **ESOL**]. The O column indicates the availability of the items in the file Jobname.OUT. The R column indicates the availability of the items in the results file.

4.3 BEAM3 Element Output Definitions

Name	Definition	O	R
EL	Element Number	Y	Y
NODES	Element nodes - I, J	Y	Y
MAT	Element material number	Y	Y
VOLU:	Element volume	N	Y
XC, YC	Location where results are reported	Y	3
TEMP	Temperatures T1, T2, T3, T4	Y	Y
PRES	Pressure P1 at nodes I,J; OFFST1 at I,J; P2 at I,J; OFFST2 at I, J; P3 at I; P4 at J	Y	Y
SDIR	Axial direct stress	1	1
SBYT	Bending stress on the element +Y side of the beam	1	1
SBYB	Bending stress on the element -Y side of the beam	1	1
SMAX	Maximum stress (direct stress + bending stress)	1	1

Name	Definition	O	R
SMIN	Minimum stress (direct stress - bending stress)	1	1
EPELDIR	Axial elastic strain at the end	1	1
EPELBYT	Bending elastic strain on the element +Y side of the beam	1	1
EPELBYB	Bending elastic strain on the element -Y side of the beam	1	1
EPTHDIR	Axial thermal strain at the end	1	1
EPTHBYT	Bending thermal strain on the element +Y side of the beam	1	1
EPTHBYB	Bending thermal strain on the element -Y side of the beam	1	1
EPINAXL	Initial axial strain in the element	1	1
MFOR(X, Y)	Member forces in the element coordinate system X and Y direction	2	Y
MMOMZ	Member moment in the element coordinate system Z direction	2	Y

The following tables list output available through the **ETABLE** command using the Sequence Number method.

Name

Output quantity as defined in the Table 3.1: "BEAM3 Element Output Definitions"

Item

Predetermined Item label for **ETABLE** command **E**
Sequence number for single-valued or constant element data **I,J**
Sequence number for data at nodes I and J **IL*N***
Sequence number for data at Intermediate Location.

4.4 BEAM3 Item and Sequence Numbers (KEYOPT (9) = 0)

Output Quantity Name	ETABLE and ESOL Command Input			
	Item	E	I	J
SDIR	LS	-	1	4
SBYT	LS	-	2	5
SBYB	LS	-	3	6
EPELDIR	LEPEL	-	1	4
EPELBYT	LEPEL	-	2	5
EPELBYB	LEPEL	-	3	6
EPTHDIR	LEPTH	-	1	4
EPTHBYT	LEPTH	-	2	5
EPTHBYB	LEPTH	-	3	6
EPINAXL	LEPTH	7	-	-
SMAX	NMISC	-	1	3
SMIN	NMISC	-	2	4
MFORX	SMISC	-	1	7
MFORY	SMISC	-	2	8
MMOMZ	SMISC	-	6	12
P1	SMISC	-	13	14
OFFST1	SMISC	-	15	16
P2	SMISC	-	17	18
OFFST2	SMISC	-	19	20

Output Quantity Name	ETABLE and ESOL Command Input			
	Item	E	I	J
P3	SMISC	-	21	-
P4	SMISC	-	-	22

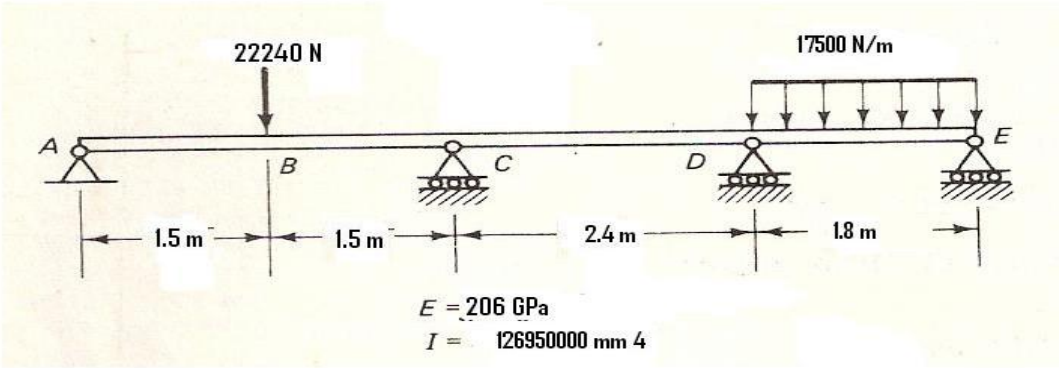
BEAM3 Assumptions and Restrictions

- The beam element must lie in an X-Y plane and must not have a zero length or area.
- The beam element can have any cross-sectional shape for which the moment of inertia can be computed. However, the stresses are determined as if the distance from the neutral axis to the extreme fiber is one-half of the height.
- The element height is used only in the bending and thermal stress calculations.
- The applied thermal gradient is assumed linear across the height and along the length.
- The moment of inertia may be zero if large deflections are not used.

SOLVED PROBLEM - BEAM

1. A three span beam is as shown. Determine the following taking(Assume square cross section).

a) Determine the deflection curve of the beam.
b) Evaluate the reaction at the supports.
c) Draw the B.M and S.F diagram.
d) List bending moment and bending stress Values at different key locations on the beam hence identify the position of maximum bending stress using Ansys.
e) Verify maximum bending stress hence obtained using theoretical procedures.



ANSYS SOLUTION PROCEDURE

Preprocessing: Defining the problem

Preferences-structural-Ok

Preprocessor - element type - add/edit/delete – add – beam - 2delastic3 – OK - close

Real constant – add/edit/delete – add – OK

Area - 0 .039 m^2

Izz - 0.00012695m^4

Height - 0.1975 m (calculate using Izz = bxd^3 /12, assume square cross section, b = d, find d which is the height of the beam section)

Ok – Close

Material properties – material models – structural – linear – elastic – isotropic – Ex 206e9

Prxy 0.3 OK – material – exit

Modelling – create – keypoints – inactiveCS

Keypoint 1st

Location in active CS 0,0,0

2nd

1.5,0,0

3rd

3,0,0

4th

5.4,0,0

5th

7.2,0,0

Create – lines – lines – straight lines – click on keypoints 1 & 2

Click on keypoints 2 & 3

Click on keypoints 3 & 4

Click on keypoints 4 & 5

OK

Meshing – sizecontrols – Manual size – lines – picked lines – pick all the four lines – OK Number of element divisions 5 OK

Meshing – mesh – lines – pick all.

Plot controls – numbering keypoint numbers ON

Element numbers

OK

Plot – multiplot

Solution Phase: Assigning loads and solving

Loads – define loads – apply – structural – displacement - on keypoint

Pick keypoint 1 Ok Select Ux & Uy OK.

On keypoint – click at keypoints 3, 4 and 5 Ok - select Uy only – OK

Force/moment – on keypoints – click at keypoint 2 Ok - select Fy -22240

Ok.

Pressure - on beams – click on element 4 – Ok

Pressure value at node I 17500 OK

Solution – solve – current LS – Ok

Post processing : Viewing the results

General postprocessor – element table – define table – add

User label for item BG1

By sequence number – select SMISC type SMISC, 2 below - Apply

User label for item BG2

By sequence number – select SMISC type SMISC, 8 below - Apply

User label for item BG3

By sequence number – select SMISC type SMISC, 6 below - Apply

User label for item BG4

By sequence number – select SMISC type SMISC, 12 below - Apply

User label for item BG5

By sequence number – select NMISC type NMISC, 1 below - Apply

User label for item BG6

By sequence number – select NMISC type NMISC, 3 below - Ok-Close

(To draw deformation curve)

Plot results – nodal solution – dof solution - Uy – Ok

(To list nodal reactions)

List results – reaction solution – all items – Ok

```
PRINT REACTION SOLUTIONS PER NODE
 ***** POST1 TOTAL REACTION SOLUTION LISTING *****
 LOAD STEP=     1   SUBSTEP=     1
  TIME=    1.0000      LOAD CASE=   0
 THE FOLLOWING X,Y,Z SOLUTIONS ARE IN GLOBAL COORDINATES
   NODE       FX          FY          MZ
      1   0.0000       8886.5
      3                15677.
      4                14050.
      5                15126.
TOTAL VALUES
VALUE     0.0000       53740.      0.0000
```

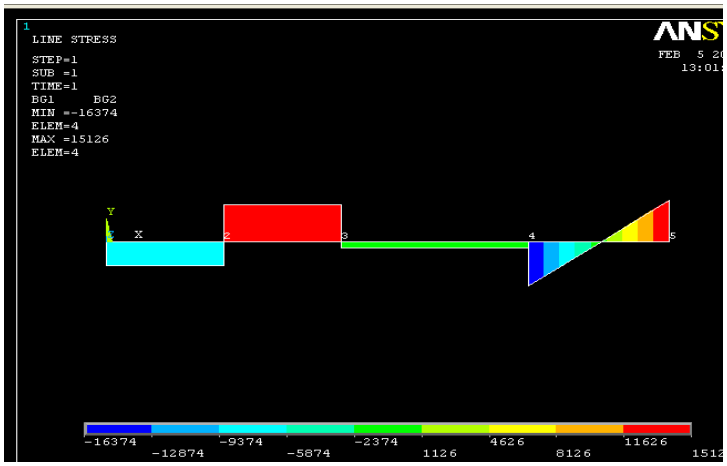
File- close

(To draw SF Diagram)

Plot results – Contour plot – line element Result

Element table item at node I BG1

Element table item at node J BG2 Ok

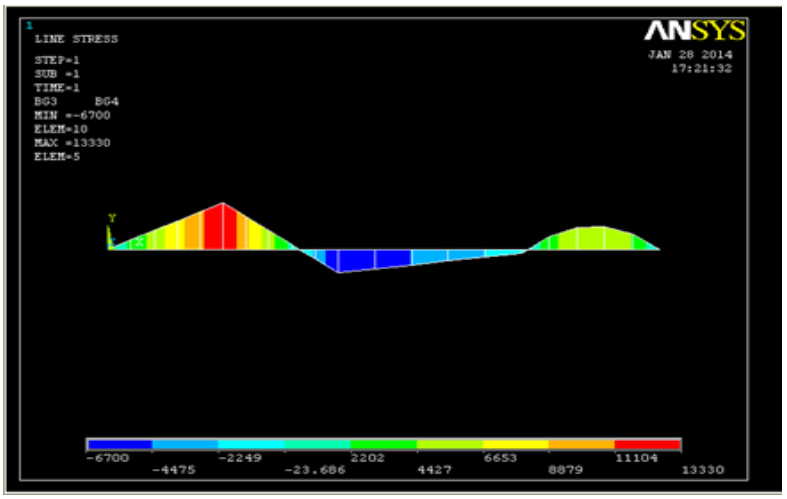


(To draw BM Diagram)

Line element Result

Element table item at node I BG3

Element table item at node J BG4 Ok

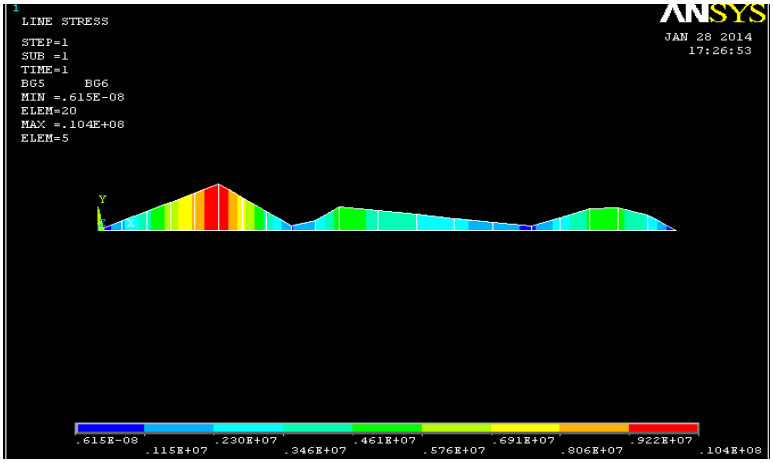


(To draw Bending Stress Diagram)

Line element Result

Element table item at node I BG5

Element table item at node J BG6 Ok



Utility menu – List results – Element table data – BG3 – Ok

(BM at i th node)

```
PRINT ELEMENT TABLE ITEMS PER ELEMENT

 ***** POST1 ELEMENT TABLE LISTING *****

   STAT     CURRENT
   ELEM       BG3
      1  0.68212E-12
      2   13330.
      3  -6700.4
      4  -1123.1
```

File – close

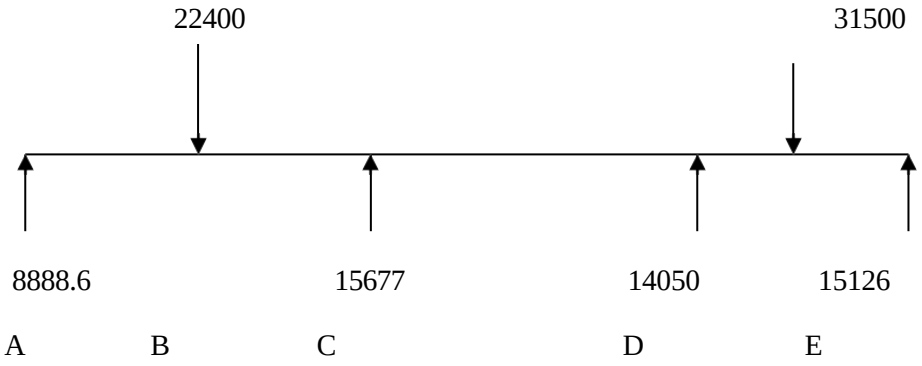
(To list Bending stress values on different elements)

Utility menu – List results – Element table data – BG5, BG6 – Ok

(To View Animation)

Plot controls – Animate – over result

Theoretical verification of Maximum bending stress



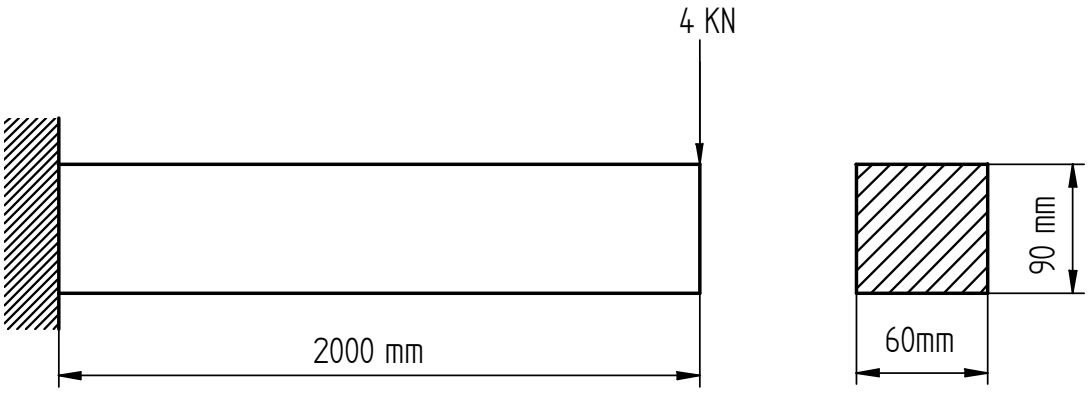
We know from the BM and BS diagram max BM and hence maximum bending stress is occurring at the application of concentrated load i.e., point B

Maximum Bending moment (at B) = 8888.6 x 1.5 = 13328.7 N.m

Maximum Bending Stress (at **B)** = My/I = $\frac{13328.7 \times 0.1975}{0.00012695 \times 2}$ = 10367933.2 N/m^2 = 10.36 Mpa

10.36 Mpa is the max stress shown by BS diagram occurring at the point B on the beam. Hence verified.

2 For a **cantilever beam** as shown in fig. Determine the deflection, maximum stress due to bending, reaction at supports, bending moment and shear force. Take **E=2x10^5 N/mm2.**



Procedure:

Step 1: Ansys Utility Menu
File – clear and start new – do not read file – ok – yes.

Preprocessing: Defining the problem

Step 2: Ansys Main Menu – Preferences
select – STRUCTURAL - ok

Step 3: Preprocessor

Element type – Add/Edit/Delete – Add – **Beam** – **2D elastic 3** – ok – close.

Real constants – Add/Edit/Delete - Add – ok – real constant set no – 1 – c/s area – 60*90 - moment of inertia – 60*90**3/12 – total beam height – 90 – ok.
Material Properties – material models – Structural – Linear – Elastic – Isotropic – EX – 2e5 **N/mm2** – PRXY – 0.27 – ok – close.

Step 4: Preprocessor

Modeling – Create – Nodes – In Active CS – Apply (first node is created) – X,Y,Z location in CS – 2000 (X value w.r.t first node) – ok (second node is created).
Create –Elements – Element Attributes – Element type – BEAM1 – Material – 1 - real constant set no – 1 – ok
Auto numbered– Thru Nodes – pick 1 & 2 – ok (element created through nodes).

Assigning Loads:

Step 5: Preprocessor

Loads – Define loads – apply – Structural – Displacement – on Nodes- pick node 1 – apply – DOFs to be constrained – All DOF – ok.
Loads – Define loads – apply – Structural – Force/Moment – on Nodes- pick node 2 – apply – direction of For/Mom – FY – Force/Moment value – (-4000N) (-ve value) – ok.

Solution Phase:

Step 6: Solution

Solve – current LS – ok (Solution is done is displayed) – close.

Post Processing: Viewing the result

Step 7: General Post Processor

Plot Results – Deformed Shape – def+undeformed – ok
Element table – Define table – Add –User label for item – Type **SFD i** – ' Results data item' – By Sequence num – SMISC, – **SMISC,2** – apply - User label for item – Type **SFD j** – ' Results data item' – By Sequence num – SMISC, – **SMISC,8** – apply - User label for item – Type **BMD i** – ' Results data item' – By Sequence num – SMISC, – **SMISC,6** – apply - User label for item – Type **BMD j** – ' Results data item' – By Sequence num – SMISC, – **SMISC,12** – ok – close.

NOTE: For Shear Force Diagram(SFD) use the combination SMISC, 2 & SMISC, 8, for Bending Moment Diagram(BMD) use the combination SMISC,6 & SMISC,12.

Step 8: General Post Processor

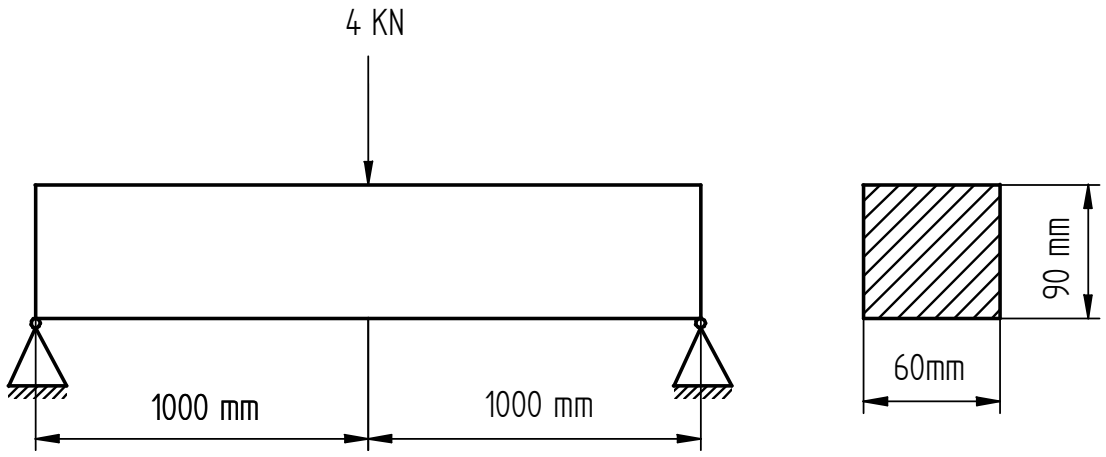
Plot Results – contour plot – nodel solution – DOF solution –Displacement vector sum – ok.
Plot Results – contour plot – Line Element Results – Elem table item at node I – (SFD i) SMISC,2 – Elem table item at node J – (SFD j) SMISC,8 – ok (Shear force diagram will be displayed).
Plot Results – contour plot – Line Element Results – Elem table item at node I – (BMD i) SMISC,6 – Elem table item at node J – (BMD j) SMISC,12 – ok (bending moment diagram will be displayed).

List Results – reaction solution – items to be listed – All items – ok (reaction forces will be displayed with the node numbers).
List Results – Nodal loads – items to be listed – All items – ok (Nodal loads will be displayed with the node numbers).

Step 9: Ansys Utility Menu : PlotCtrls – style – size and shape – Display of elements – ON – ok.

Step 10: Ansys Utility Menu : PlotCtrls - Animate – Deformed shape – def+undeformed-ok

3 For a **simply supported beam** with central loading condition. Determine the deflection, maximum stress due to bending, reaction at supports, bending moment and shear force. Take E=210 Gpa.



Procedure:

Step 1: Ansys Utility Menu
File – clear and start new – do not read file – ok – yes.

Preprocessing: Defining the problem

Step 2: Ansys Main Menu – Preferences
select – STRUCTURAL - ok

Step 3: Preprocessor

Element type – Add/Edit/Delete – Add – **Beam** – **2D elastic 3** – ok – close.

Real constants – Add/Edit/Delete - Add – ok – real constant set no – 1 – c/s area – 60*90 - moment of inertia – 60*90**3/12 – total beam height – 90 – ok.
Material Properties – material models – Structural – Linear – Elastic – Isotropic – EX – 210e3 **N/mm2** – PRXY – 0.27 – ok – close.

Step 4: Preprocessor

Modeling – Create – Nodes – In Active CS – Apply (first node is created) – X,Y,Z location in CS – 1000 (X value w.r.t first node) – apply (second node is created) - X,Y,Z location in CS – 2000 (X value w.r.t first node) – ok (third node is created).
Create –Elements – Element Attributes – Element type – BEAM1 – Material – 1 - real constant set no – 1 – ok
Auto numbered– Thru Nodes – pick 1 & 2 – apply (element created through nodes) – pick 2 & 3 – ok (element created through nodes).

Assigning Loads:

Step 5: Preprocessor

Loads – Define loads – apply – Structural – Displacement – on Nodes- pick node 1 & 3 – ok – DOFs to be constrained – UY(Simply Supported Beam) – ok.

Loads – Define loads – apply – Structural – Force/Moment – on Nodes - pick node 2 – ok – direction of For/Mom – FY – Force/Moment value – (-4000N) (-ve value) – ok.

Solution Phase:

Step 6: Solution

Solve – current LS – ok (Solution is done is displayed) – close.

Post Processing: Viewing the result

Step 7: General Post Processor

Plot Results – Deformed Shape – def+undeformed – ok

Element table – Define table – Add –User label for item – Type **SFD i** – ' Results data item' – By Sequence num – SMISC, – **SMISC,2** – apply - User label for item – Type **SFD j** – ' Results data item' – By Sequence num – SMISC, – **SMISC,8** – apply - User label for item – Type **BMD i** – ' Results data item' – By Sequence num – SMISC, – **SMISC,6** – apply - User label for item – Type **BMD j** – ' Results data item' – By Sequence num – SMISC, – **SMISC,12** – ok – close.

NOTE: For Shear Force Diagram(SFD) use the combination SMISC, 2 & SMISC, 8, for Bending Moment Diagram(BMD) use the combination SMISC,6 & SMISC,12.

Step 8: General Post Processor

Plot Results – contour plot – nodel solution – DOF solution –Displacement vector sum – ok.

Plot Results – contour plot – Line Element Results – Elem table item at node I – (SFD i) SMISC,2 – Elem table item at node J – (SFD j) SMISC,8 – ok (Shear force diagram will be displayed).

Plot Results – contour plot – Line Element Results – Elem table item at node I – (BMD i) SMISC,6 – Elem table item at node J – (BMD j) SMISC,12 – ok (bending moment diagram will be displayed).

List Results – reaction solution – items to be listed – All items – ok (reaction forces will be displayed with the node numbers).

List Results – Nodal loads – items to be listed – All items – ok (Nodal loads will be displayed with the node numbers).

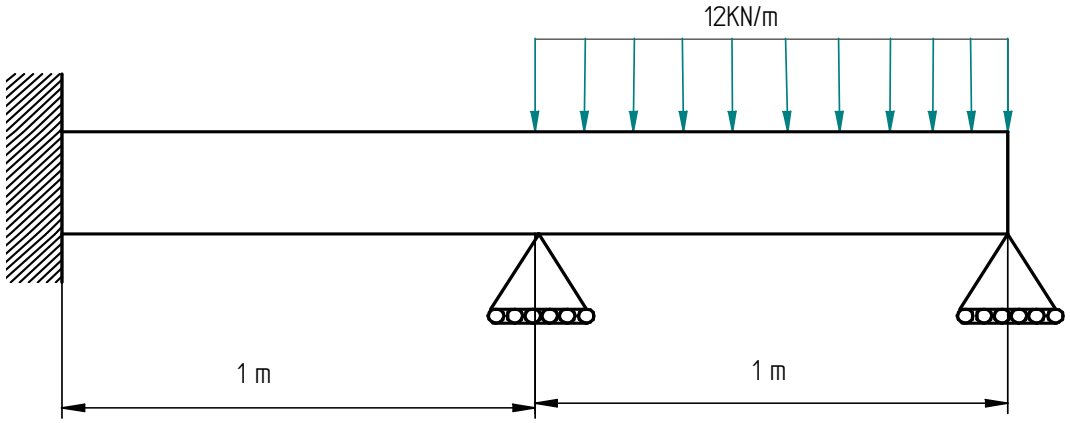
Step 9: Ansys Utility Menu : PlotCtrls – style – size and shape – Display of elements – ON – ok.

Step 10: Ansys Utility Menu : PlotCtrls - Animate – Deformed shape – def+undeformed-ok

4 For a beam is shown in fig. Determine the deflection and maximum stress due to bending, bending moment and shear force. Take **E=200GPa** and **I = 4x10^6 mm^4**

E=200GPa
=200 x10^3 N/mm^2

UDL=12KN/m
=12 x 1000 / 1000
=12N/mm



Convert **m** to **mm**

Assume sqr cross section b = d.
Moment of inertia I = b x d^3 / 12
Cross sectional area A = b x d

Procedure:

Step 1: Ansys Utility Menu
File – clear and start new – do not read file – ok – yes.

Preprocessing: Defining the problem

Step 2: Ansys Main Menu – Preferences
select – STRUCTURAL - ok

Step 3: Preprocessor

Element type – Add/Edit/Delete – Add – **Beam** – **2D elastic 3** – ok – close.

Real constants – Add/Edit/Delete - Add – ok – real constant set no – 1 – c/s area – **83.23*83.23** - moment of inertia – **4x10^6 mm^4** – total beam height – **83.23** – ok.
Material Properties – material models – Structural – Linear – Elastic – Isotropic – EX – 200e3 **N/mm2** – PRXY – 0.27 – ok – close.

Step 4: Preprocessor

Modeling – Create – Nodes – In Active CS – Apply (first node is created) – X,Y,Z location in CS – 1000 (X value w.r.t first node) – apply (second node is created) - X,Y,Z location in CS – 2000 (X value w.r.t first node) – ok (third node is created).
Create –Elements – Element Attributes – Element type – BEAM1 – Material – 1 - real constant set no – 1 – ok
Auto numbered– Thru Nodes – pick 1 & 2 – apply (element created through nodes) – pick 2 & 3 – ok (element created through nodes).

Assigning Loads:

Step 5: Preprocessor

Loads – Define loads – apply – Structural – Displacement – on Nodes - pick node 1 – ok – DOFs to be constrained – ALL DOF – displacement value - 0 – apply - pick node 2 & 3 – ok – DOFs to be constrained – UY (Roller supports in X direction) - displacement value – 0 – ok.
Loads – Define loads – apply – Structural – pressure – on beams - pick beam b/w node 2 & 3 – ok – pressure value at node I – 12 N/mm (+ve value) – pressure value at node J – 12 N/mm (+ve value) – ok.

Solution Phase:

Step 6: Solution

Solve – current LS – ok (Solution is done is displayed) – close.

Post Processing: Viewing the result

Step 7: General Post Processor

Plot Results – Deformed Shape – def+undeformed – ok
Element table – Define table – Add –User label for item – Type **SFD i** – ' Results data item' – By Sequence num – SMISC, – **SMISC,2** – apply - User label for item – Type **SFD j** – ' Results data item' – By Sequence num – SMISC, – **SMISC,8** – apply - User label for item – Type **BMD i** – ' Results data item' – By Sequence num – SMISC, – **SMISC,6** – apply - User label for item – Type **BMD j** – ' Results data item' – By Sequence num – SMISC, – **SMISC,12** – apply - User label for item – Type **MBS i -** ' Results data item' – By Sequence num – NMISC, – **NMISC,1** – apply - User label for item – Type **MBS j -** ' Results data item' – By Sequence num – NMISC, – **NMISC,3** – ok - close

NOTE: For Shear Force Diagram(SFD) use the combination SMISC, 2 & SMISC, 8, For Bending Moment Diagram(BMD) use the combination SMISC,6 & SMISC,12, For maximum stress due to bending(MBS) use the combination NMISC, 1 & NMISC, 3

Step 8: General Post Processor

Plot Results – contour plot – nodel solution – DOF solution –Displacement vector sum – ok.
Plot Results – contour plot – Line Element Results – Elem table item at node I – (SFD i) SMISC,2 – Elem table item at node J – (SFD j) SMISC,8 – ok (Shear force diagram will be displayed).

Plot Results – contour plot – Line Element Results – Elem table item at node I – (BMD i) SMISC,6 – Elem table item at node J – (BMD j) SMISC,12 – ok (bending moment diagram will be displayed).

Plot Results – contour plot – Line Element Results – Elem table item at node I – (MBS i) NMISC,1 – Elem table item at node J – (MBS j) NMISC,3 – ok (maximum stress diagram will be displayed).

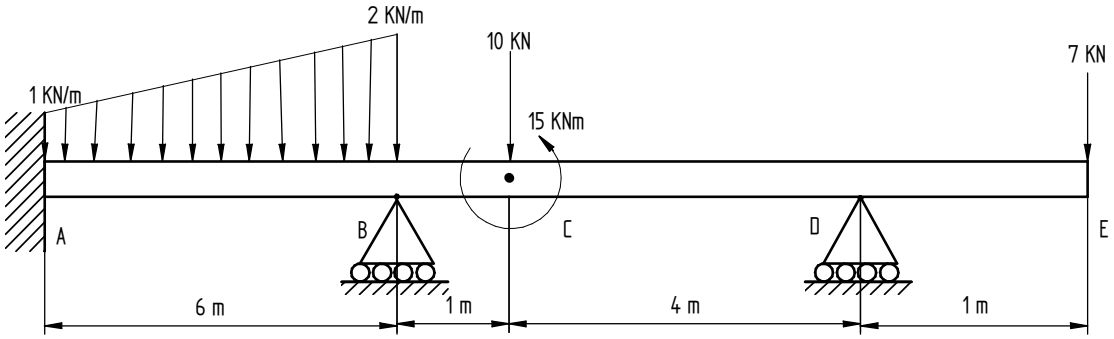
List Results – reaction solution – items to be listed – All items – ok (reaction forces will be displayed with the node numbers).

List Results – Nodal loads – items to be listed – All items – ok (Nodal loads will be displayed with the node numbers).

Step 9: Ansys Utility Menu : PlotCtrls – style – size and shape – Display of elements – ON – ok.

Step 10: Ansys Utility Menu : PlotCtrls - Animate – Deformed shape – def+undeformed-ok

5 A two span continuous beam with an overhang on one side is shown in fig. The beam has a square cross section of side 15cm for the span AB and 20cm for the remaining span. Modulus of elasticity is $2x10^6$ N/cm^2



Procedure:

Step 1: Ansys Utility Menu
File – clear and start new – do not read file – ok – yes.

Preprocessing: Defining the problem

Step 2: Ansys Main Menu – Preferences
select – STRUCTURAL - ok

Step 3: Preprocessor

Element type – Add/Edit/Delete – Add – **Beam** – **2D elastic 3** – ok – close.

Real constants – Add/Edit/Delete - Add – ok – real constant set no – 1 – c/s area – **0.15*0.15 -** moment of inertia – **0.15*0.15**3/12**– total beam height – **0.15** – apply - real constant set no – 2 – c/s area – **0.2*0.2 -** moment of inertia – **0.2*0.2**3/12**– total beam height – **0.2** – ok.
Material Properties – material models – Structural – Linear – Elastic – Isotropic – EX – 210e9 **N/m^2** – PRXY – 0.27 – ok – close.

Step 4: Preprocessor

Modeling – Create – Nodes – In Active CS – Apply (A node is created) – X,Y,Z location in CS – 6mt (X value w.r.t first node) – apply (B node is created) - X,Y,Z location in CS – 7mt (X value w.r.t first node) – apply (C node is created) - X,Y,Z location in CS – 11mt (X value w.r.t first node) – apply (D node is created) - X,Y,Z location in CS – 12mt (X value w.r.t first node) – ok (E node is created).
Create –Elements – Element Attributes – Element type – BEAM1 – Material – 1 - real constant set no – 1 – ok.

Auto numbered– Thru Nodes – pick 1 & 2 – ok (element created through nodes).
Create –Elements – Element Attributes – Element type – BEAM1 – Material – 1 - real constant set no – 2 – ok.
Auto numbered– Thru Nodes – pick 2 & 3 – apply (element created through nodes) – pick 3 & 4 – apply (element created through nodes) – pick 4 & 5 – ok (element created through nodes).

Assigning Loads:

Step 5: Preprocessor

Loads – Define loads – apply – Structural – Displacement – on Nodes - pick node 1 – apply - DOFs to be constrained – ALL DOF – displacement value - 0 – apply – pick node 2 & 4 – ok – DOFs to be constrained – UY(Roller supports in X direction) – displacement value - 0 – ok.
Loads – Define loads – apply – Structural – pressure – on beams - pick beam b/w node 1 & 2 – ok – pressure value at node I – 1000 N/m (+ve value) – pressure value at node J – 2000 N/m (+ve value) – ok.(Uniformly varying load)
Loads – Define loads – apply – Structural – Force/Moment – on Nodes - pick node 3 – ok – direction of force/moment - MZ - Force/Moment value – 15000N/m(+ve value anti clockwise direction) – ok.

Solution Phase:

Step 6: Solution
Solve – current LS – ok (Solution is done is displayed) – close.

Post Processing: Viewing the result

Step 7: General Post Processor

Plot Results – Deformed Shape – def+undeformed – ok
Element table – Define table – Add –User label for item – Type **SFD i** – ‘ Results data item’ – By Sequence num – SMISC, – **SMISC,2** – apply - User label for item – Type **SFD j** – ‘ Results data item’ – By Sequence num – SMISC, – **SMISC,8** – apply - User label for item – Type **BMD i** – ‘ Results data item’ – By Sequence num – SMISC, – **SMISC,6** – apply - User label for item – Type **BMD j** – ‘ Results data item’ – By Sequence num – SMISC, – **SMISC,12** – apply - User label for item – Type **MBS i -** ‘ Results data item’ – By Sequence num – NMISC, – **NMISC,1** – apply - User label for item – Type **MBS j -** ‘ Results data item’ – By Sequence num – NMISC, – **NMISC,3** – ok - close.

NOTE: For Shear Force Diagram(SFD) use the combination SMISC, 2 & SMISC, 8, for Bending Moment Diagram(BMD) use the combination SMISC,6 & SMISC,12**,** for maximum stress due to bending(MBS) use the combination NMISC, 1 & NMISC, 3.

Step 8: General Post Processor

Plot Results – contour plot – nodel solution – DOF solution –Displacement vector sum – ok.

Plot Results – contour plot – Line Element Results – Elem table item at node I – (SFD i) SMISC,2 – Elem table item at node J – (SFD j) SMISC,8 – ok (Shear force diagram will be displayed).

Plot Results – contour plot – Line Element Results – Elem table item at node I – (BMD i) SMISC,6 – Elem table item at node J – (BMD j) SMISC,12 – ok (bending moment diagram will be displayed).

Plot Results – contour plot – Line Element Results – Elem table item at node I – (MBS i) NMISC,1 – Elem table item at node J – (MBS j) NMISC,3 – ok (maximum stress diagram will be displayed).

List Results – reaction solution – items to be listed – All items – ok (reaction forces will be displayed with the node numbers).

List Results – Nodal loads – items to be listed – All items – ok (Nodal loads will be displayed with the node numbers).

Step 9: Ansys Utility Menu : PlotCtrls – style – size and shape – Display of elements – ON – ok.

Step 10: Ansys Utility Menu : PlotCtrls - Animate – Deformed shape – def+undeformed-ok

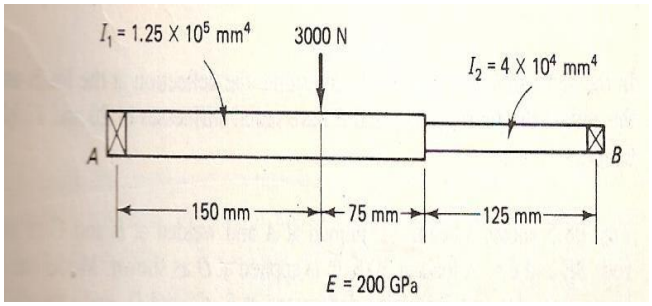
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PRACTICE PROBLEMS ON ANALYSIS OF BEAMS

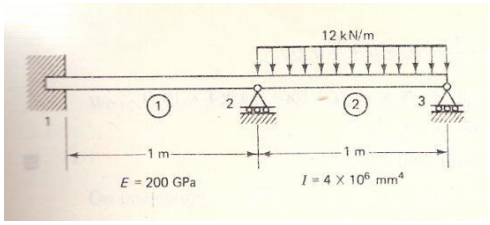
Exercise 1: Consider the shaft to be simply supported at bearings A and B.

Determine the following taking E=200GPa

a) Determine the deflection curve of the beam and max deflection.
b) Evaluate the reaction at the supports.
c) Draw the B.M and S.F diagram.
d) Determine the maximum bending stress value and its location.



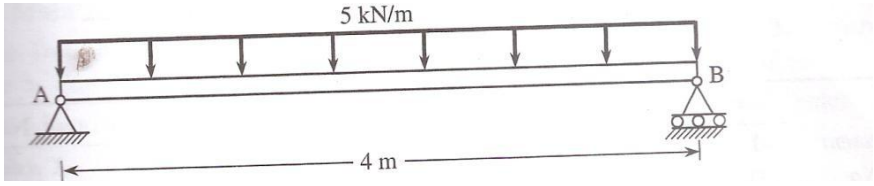
Exercise 2: For the beam and the loading shown determine the following taking E= 200 Gpa ,I = 4 x 10 ^{6} mm^4 .(Assume square cross section)



a) Determine the deflection curve of the beam.
b) Evaluate the reaction at the supports.
c) Draw the B.M and S.F diagram.
d) Determine the maximum bending stress value and its location.

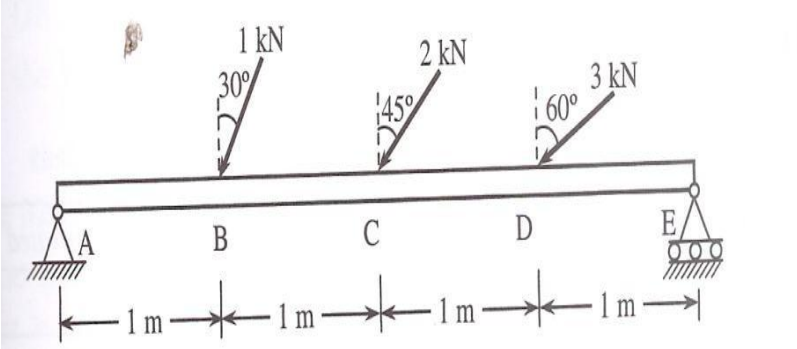
Exercise 3: For the simply supported beam with the UDL as shown determine the following taking cross section of the beam as breadth=150 mm depth = 200 mm and modulus of elasticity E= 2x10^5 N/mm^2

a) Determine the deflection curve of the beam.
b) Evaluate the reaction at the supports.
c) Draw the B.M and S.F diagram.
d) Determine the maximum bending stress value and its location.



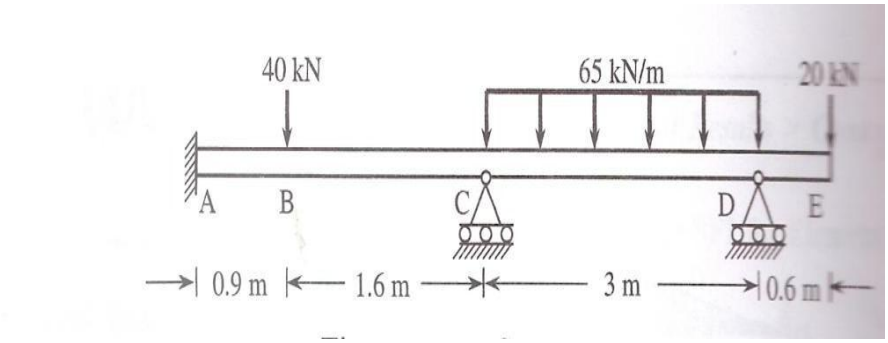
Exercise 4: For the simply supported beam carrying oblique loads as shown determine the following taking cross section of the beam as breadth=120 mm depth = 100 mm and modulus of elasticity E= $2x10^5$ N/mm^2

a) Determine the deflection curve of the beam.
b) Evaluate the reaction at the supports.
c) Draw the B.M and S.F diagram.
d) Determine the maximum bending stress value and its location.

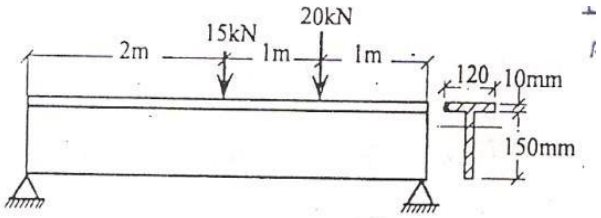


Exercise 5: For the continuous beam with overhang as shown determine the following taking cross section of the beam as breadth=300 mm depth = 450 mm and modulus of elasticity E= $2x10^4$ N/mm^2

a) Determine the deflection curve of the beam.
b) Evaluate the reaction at the supports.
c) Draw the B.M and S.F diagram.
d) Determine the maximum bending stress value and its location.



Self-Learning Exercise 1 (Beams) A beam carries a 'T' cross section with flange of 120 x 10 mm and a web of 150 x 10 mm sizes. The beam is 4m long and simply supported at the ends. The beam carries two point loads of 15 KN and 20 KN at 2m and 3m from LH end. Take E = 200 Gpa.



a) Determine the deflection curve of the beam.
b) Evaluate the reaction at the supports.
c) Draw the B.M and S.F diagram.
d) List bending moment and bending stress Values for different elements .
e) Tabulate the Ansys results with theoretical ones with respect to the above.

(Hint: Find Ixx for the c/s as 7089167 mm^4 and y = 49.44 from top fiber)

Self-Learning Exercise 2 (Beams) An I section with top flange 90 x 10 mm, web 150 x 10 mm and bottom flange 180 x 10 is used as a simply supported beam for a span length of 5 m and carries a udl of 12 KN/m . Take E 210 Gpa.

a) Determine the deflection curve of the beam.
b) Evaluate the reaction at the supports.
c) Draw the B.M and S.F diagram.
d) List bending moment and bending stress Values for different elements.
e) Tabulate the Ansys results. Take atleast five elements.

(Hint: Obtain Ixx as 18800714 mm^4 and y as 67.86 from the bottom fiber)

PLANE STRESS AND PLANE STRAIN ANALYSIS

CHAPTER 5

PLANE STRESS AND PLANE STRAIN ANALYSIS

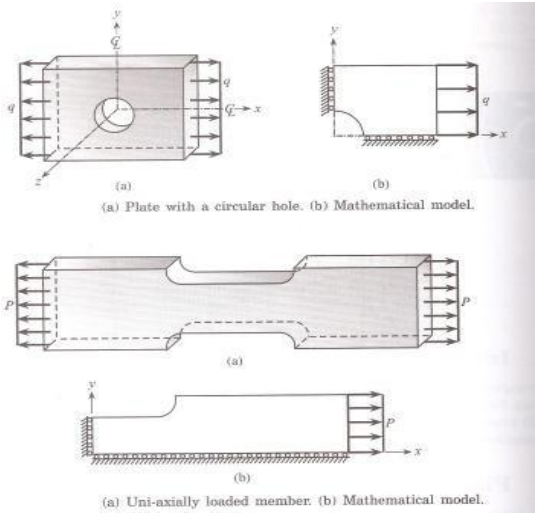
5.1 Introduction

Many engineering problems can be idealized into 2D formulations as plane stress or plane strain analysis. Finite element formulation for 2D problems and derivation of finite element equations for 2D structural elements such as constant strain triangular element and quadrilateral element are considered**.**

Plane Stress

It is a state of stress in which normal stress σ_z and the shear stresses ξ_{xz} and ξ_{yz} directed perpendicular to the normal stress are assumed to be zero. In general, thin member subjected to in-plane loading can be considered as a plane stress distribution.

A thin plate with a circular hole as shown subjected to in-plane loading and uni-axially loaded member with fillet shown can be idealized and analyzed as a plane stress problem.



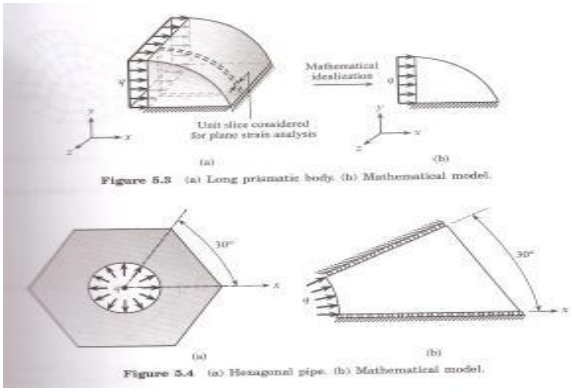
(a) Plate with a circular hole. (b) Mathematical model.

(a) Uni-axially loaded member. (b) Mathematical model.

Plane Strain

It is a state of strain in which normal strain ε_z and the shear strains γxz and γyz are assumed to be zero. Assumptions of plane strain are realistic for cylindrical or prismatic bodies with constant cross-sectional area subjected to transverse loading along x and /or y direction and do not vary in z direction. A unit slice along longitudinal axis of the body can be idealized and analyzed as a plain strain problem.

A long prismatic body subjected to constant uniform load along its length as shown and a long hexagonal pipe subjected to internal pressure as shown can be idealized and analyzed as a plain strain problem



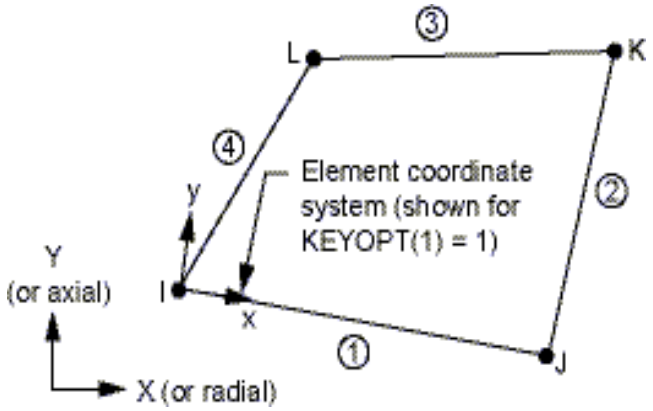
Elements used in this analysis can be either PLANE42 or PLANE82.

5.2 PLANE42 Element Description

PLANE42 is used for 2-D modeling of solid structures. The element can be used either as a plane element (plane stress or plane strain) or as an axisymmetric element. The element is defined by four nodes having two degrees of freedom at each node: translations in the nodal x and y directions. The element has plasticity, creep, swelling, stress stiffening, large deflection, and large strain capabilities.

The element input data includes four nodes as shown below, a thickness (for the plane stress option only) and the orthotropic material properties. Orthotropic material directions correspond to the element coordinate directions.

Plane 42 Geometry

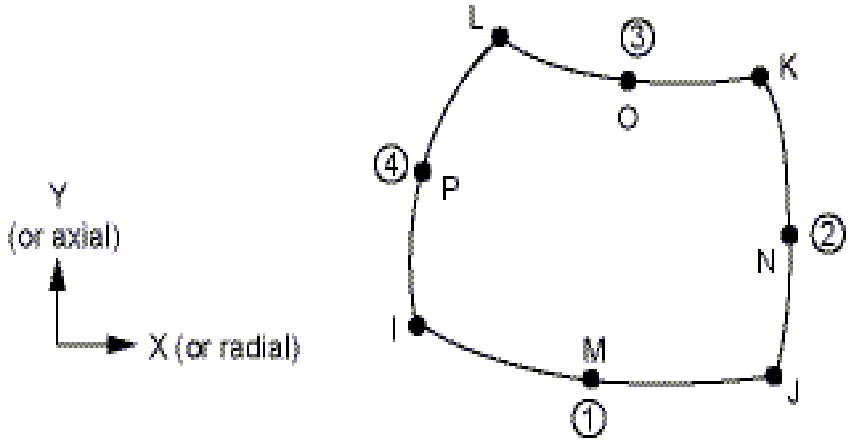


5.3 PLANE82 Element Description

PLANE82 is a higher order version of the 2-D, four-node element (PLANE42). It provides more accurate results for mixed (quadrilateral-triangular) automatic meshes and can tolerate irregular shapes without as much loss of accuracy. The 8-node elements have compatible displacement shapes and are well suited to model curved boundaries.

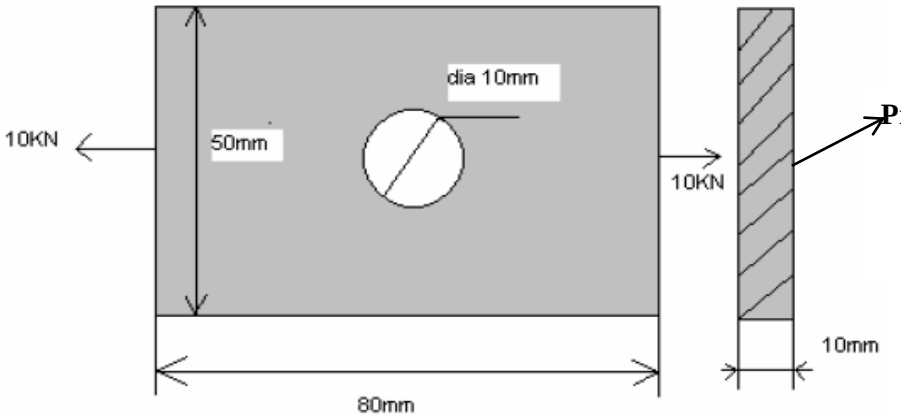
The 8-node element is defined by eight nodes as shown below having two degrees of freedom at each node: translations in the nodal x and y directions. The element may be used as a plane element or as an axisymmetric element. The element has plasticity, creep, swelling, stress stiffening, large deflection, and large strain capabilities.

Plane 82 Geometry



SOLVED PROBLEM 1 – PLATE WITH A HOLE

1. Determine the maximum stress for a rectangular plate of 50mm x 80 mm with hole of 10 mm diameter in the center is loaded in axial tension. Thickness of the Plate is 10mm. Take E = 200 GPa



Pressure(P) = Force / Area
= 10000 / 50 x 10
= 20 N/mm^2

Analytical solution

$$\sigma_{nom} = F / (w - a)\, t = 10000 / (50 - 10)10 = 25 \text{ N/mm}^2$$

$$\text{Stress concentration factor } K_\sigma = \sigma_{max} / \sigma_{nom}$$

$$2.5 = \sigma_{max} / 25$$

$$\sigma_{max} = 2.5 \times 25 = 62.5 \text{ N/mm}^2$$

Procedure:

Step 1: Ansys Utility Menu
File – clear and start new – do not read file – ok – yes.

Preprocessing: Defining the problem

Step 2: Ansys Main Menu – Preferences
select – STRUCTURAL - ok

Step 3: Preprocessor

Element type – Add/Edit/Delete – Add – **solid** – **Quad 4node 42** – ok – **PLANE42** – options – **Element Behavior K3 – plane strs w/thk** – ok - close.

Real constants – Add/Edit/Delete - Add - **PLANE42 –** ok **–** Thickness – 10mm - ok – close.
Material Properties – material models – Structural – Linear – Elastic – Isotropic – EX – 200e3 **N/mm2** – PRXY – 0.27 – ok – close.

Step 4: Preprocessor

Modeling – Create – Areas – Rectangle – By 2 Corners – WP X=0, WP Y=0, width=80mm, Height=50mm – ok.
Create – Area – Circle – solid circle – X, Y, radius – 40mm, 25mm, 5mm – ok.
Modeling - Operate – Booleans – Subtract – Areas – pick area which is not to be deleted (rectangle) – apply – pick area which is to be deleted (circle) – ok.
Meshing – Mesh Tool – Mesh Areas – Quad – Free – Mesh – pick all – ok.
Mesh Tool – Refine – pick all – Level of refinement – 3 – ok.

Assigning Loads:

Step 5: Preprocessor

Loads – Define loads – apply – Structural – Displacement – on Nodes – select box – drag the left side of the area – apply – DOFs to be constrained – ALL DOF – ok.
Loads – Define loads – apply – Structural – Pressure – on Nodes – select box – drag the right side of the area – ok – Load Pressure Value – (-20 N/mm^2) (-ve value) – ok.

OR

Loads – Define loads – apply – Structural – Force/Moment – on Nodes – select box – drag the right side of the area – apply – direction of For/Mom – FX – Force/Moment value – 10000N (+ve value) – ok.

Solution Phase:

Step 6: Solution

Solve – current LS – ok (Solution is done is displayed) – close.

Post Processing: Viewing the result

Step 7: General Post Processor

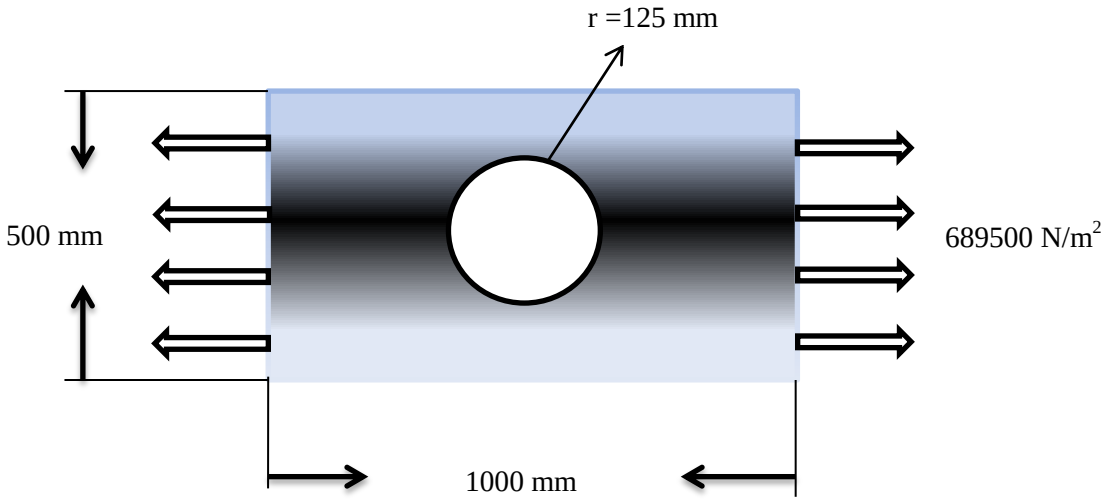
Plot Results – Deformed Shape – def+undeformed – ok
Plot results – contour plot – Element solu – Stress – Von Mises Stress – ok (the stress distribution diagram will be displayed).
Plot results – contour plot – nodal solution – DOF solution – displacement vector sum – ok.

Step 9: Ansys Utility Menu : PlotCtrls – Animate – Deformed shape – def+undeformed-ok

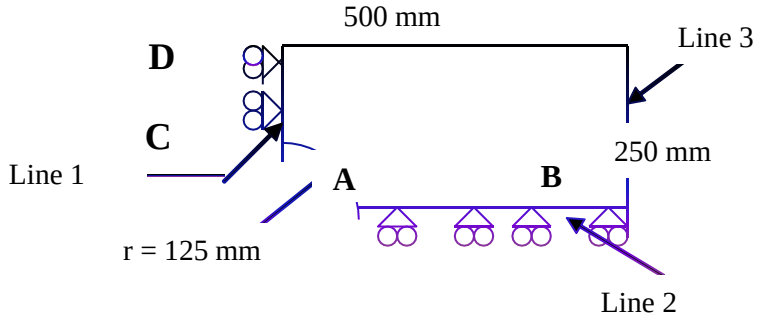
2. In the plate with a hole under plane stress as shown, plot the deformed shape and the stresses on the plate when it is subjected to uniform tensile stress of 689500 N/m^2. The thickness of the plate is 0.25 mm , young's modulus, $E=2x10^5$ N/mm^2 Poisson's ratio, $\sqrt{}$ =0.3, Also find

a) Maximum Deformation and maximum stress(Von Mises stress) on the plate.

b) Plot the variation of Von Mises stress along AB, AC, CD.



CONSIDERING THE QUARTER SYMMETRY WE HAVE



Step 1: Ansys Utility Menu

File – clear and start new – do not read file – ok – yes.

Step 2: Ansys Main Menu – Preferences

select – STRUCTURAL - ok

Step 3: Preprocessor

Element type – Add/Edit/Delete – Add – **solid** – **8 node82** – ok – **PLANE82** – options – **Element Behavior K3 – plane strs w/thk** – ok - close.

Real constants – Add/Edit/Delete - Add – **PLANE82 –** ok **–** Thickness – 0.25mm - ok – close.

Material Properties – material models – Structural – Linear – Elastic – Isotropic – EX – 200e3 **N/mm2** – PRXY – 0.3 – ok – close.

Step 4: Preprocessor

Modeling – Create – Areas – Rectangle – By Dimension –
X1,X2 X coordinates – 0,500
Y1,Y2 Y coordinates – 0,250 – ok.
Create – Area – Circle – solid circle – By Dimensions – RAD1 outer radius – 125mm - ok.
Modeling - Operate – Booleans – Subtract – Areas – pick area which is not to be deleted (rectangle) – apply – pick area which is to be deleted (circle) – ok.
Meshing – Meshing – Mesh tool – smart size – slider to 5 – in the mesh session choose areas & free - Click on Mesh – Pick all

Step 5: Preprocessor

Loads – Define loads – apply – Structural – Displacement –symmetry B C– on lines – select line 1(CD) and line 2(AB) – ok.
Loads – Define loads – apply – Structural – Pressure – on lines – select line 3**(right side line parallel to line 1,CD)** – ok – Load Pressure Value – (-0.6895 N/mm^2) (-ve value) – ok.

Step 6: Solution

Solve – current LS – ok (Solution is done is displayed) – close.

Step 7: General Post Processor

Plot Results – Deformed Shape – def+undeformed – ok
Plot results – contour plot – Element solu – Stress – Von Mises Stress – ok (the stress distribution diagram will be displayed).
Plot results – contour plot – nodal solution – DOF solution – displacement vector sum – ok.

To Plot the variation of Von Mises stress along AB, AC, CD

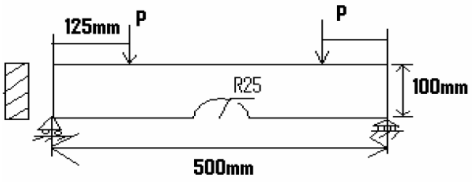
General Postproce - Path operations - Define path - By nodes , Select nodes from C to A Map onto Path - select Stress – von Mises SEQV – ok - Plot Path Item - SEQV – ok

Step 9: Ansys Utility Menu – PlotCtrls – Style – Symmetry Expansion – Periodic/Cyclic Symmetry – select type of cyclic symmetry – 1/4 Dihedral Sym – ok. (full plate will be displayed)

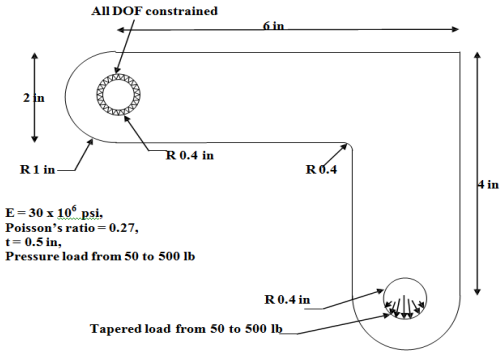
Step 10: Ansys Utility Menu : PlotCtrls – Animate – Deformed shape – def+undeformed-ok

PRACTICE PROBLEMS
On Plane Stress And Plane Strain Analysis

1. Calculate the stresses and displacement for the plate shown below. Let the load be P = 100N applied at equal distance from both ends and E = $3x10^7$ N/mm2. t=5mm



2. The corner angle bracket is shown below. The upper left hand pin-hole is constrained around its entire circumference and a tapered pressure load is applied to the bottom of lower right hand pin-hole. Compute Maximum displacement, Von-Mises stress.



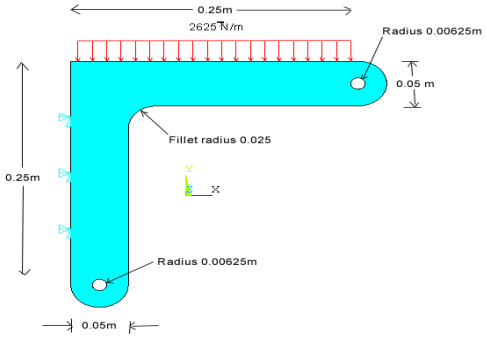
3. The corner angle bracket is shown below , We will model the bracket as a solid 8 node plane stress element. By a plane stress element we are assuming that there are no stresses in the thickness direction of the bracket.

   Geometry: The thickness of the bracket is 3.125 mm

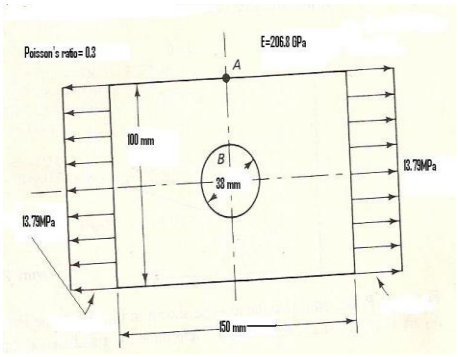
   Material: Assume the structure is made of steel with modulus of elasticity E=200 GPa.

   Boundary conditions: The bracket is fixed at its left edge.

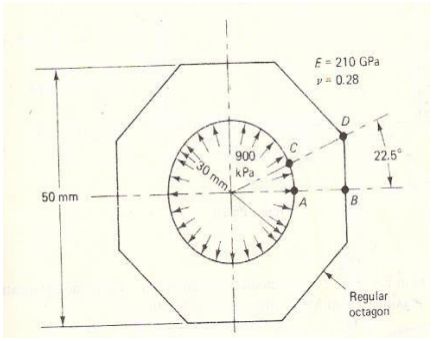
   Loading: The bracket is loaded uniformly along its top surface. The load is 2625 N/meter.



4. In the plate with a hole under plane stress, plot the deformed shape and the stresses on the plate when it is subjected to a tensile stresses of 13.73Mpa as shown. Also obtain a) Maximum Deflection and Maximum Von Mises stresses on the plate. Take thickness of the plate equal to 10 mm. b) Draw also the variation of Von Mises stresses along AB. Use Plane 82(8-node structural solid element) and model only one quarter to perform the analysis.



SELF LEARNING EXERCISE 1: In the pipe with a hole under plane strain, plot the deformed shape and the stresses. Also find a) Maximum Deflection and Maximum stress. b) Plot the stress variation along AB. Use Plane 82(8-node structural solid element) and model only one quarter to perform the analysis.



SELF LEARNING EXERCISE 2: For the culvert/dam made of concrete with the loading as shown plot the deformed shape and the stresses. Plane strain option must be selected for element behavior. Also find a) Maximum Deflection and Maximum stress. Use Plane 82 (8-node structural solid element) and model only one half to perform the analysis .Take E=40Gpa and poison's ratio, √=0.15

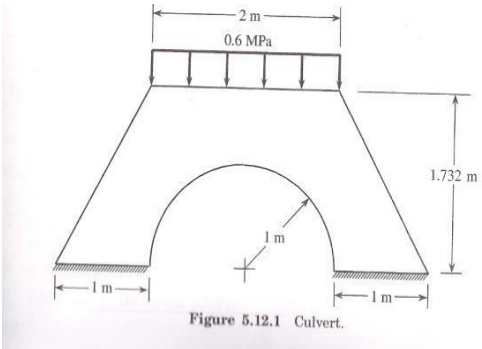


Figure 5.12.1 Culvert.

Thermal Analysis

CHAPTER 6

THERMAL ANALYSIS

6.1 Introduction

Analyzing Thermal Phenomena

A *thermal analysis* calculates the temperature distribution and related thermal quantities in a system or component. Typical thermal quantities of interest are:

- The temperature distributions The amount of heat lost or gained Thermal gradients
- Thermal fluxes.
- Thermal simulations play an important role in the design of many engineering applications, including internal combustion engines, turbines, heat exchangers, piping systems, and electronic components. In many cases, engineers follow a thermal analysis with a stress analysis to calculate *thermal stresses* (that is, stresses caused by thermal expansions or contractions).

How ANSYS Treats Thermal Modeling

Only the ANSYS/Multiphasic, ANSYS/Mechanical, ANSYS/Thermal, and ANSYS/FLOTRAN programs support thermal analyses.

The basis for thermal analysis in ANSYS is a heat balance equation obtained from the principle of conservation of energy. (For details, consult the *ANSYS Theory Reference*.) The finite element solution you perform via ANSYS calculates nodal temperatures, then uses the nodal temperatures to obtain other thermal quantities.

The ANSYS program handles all three primary modes of heat transfer: conduction, convection, and radiation.

Convection

An ANSYS user specifies convection as a surface load on conducting solid elements or shell elements. You specify the convection film coefficient and the bulk fluid temperature at a surface; ANSYS then calculates the appropriate heat transfer across that surface. If the film coefficient depends upon temperature, you specify a table of temperatures along with the corresponding values of film coefficient at each temperature.

Types of Thermal Analysis

ANSYS supports two types of thermal analysis:

1. A steady-state thermal analysis determines the temperature distribution and other thermal quantities under steady-state loading conditions. A steady-state loading condition is a situation where heat storage effects varying over a period of time can be ignored.

2. A transient thermal analysis determines the temperature distribution and other thermal quantities under conditions that vary over a period of time.

Coupled-Field Analyses

Some types of coupled-field analyses, such as thermal-structural and magnetic- thermal analyses, can represent thermal effects coupled with other phenomena. A coupled-field analysis can use matrix-coupled ANSYS elements, or sequential load- vector coupling between separate simulations of each phenomenon. For more information on coupled-field analysis, see the *ANSYS Coupled-Field Analysis Guide*.

Definition of Steady-State Analysis

The ANSYS/Multiphysics, ANSYS/Mechanical, ANSYS/FLOTRAN, and ANSYS/Thermal products support steady-state thermal analysis. A steady-state thermal analysis calculates the effects of steady thermal loads on a system or component. Engineer/analysts often perform a steady-state analysis before doing a transient thermal analysis, to help establish initial conditions. A steady-state analysis also can be the last step of a transient thermal analy sis, performed after all transient effects have diminished.

You can use steady-state thermal analysis to determine temperatures, thermal gradients, heat flow rates, and heat fluxes in an object that are caused by thermal loads that do not vary over time. Such loads include the following:

- Convections Radiation
- Heat flow rates
- Heat fluxes (heat flow per unit area)
- Heat generation rates (heat flow per unit volume) Constant temperature boundaries.

A steady-state thermal analysis may be either linear, with constant material properties; or nonlinear, with material properties that depend on temperature. The thermal properties of most material do vary with temperature, so the analysis usually is nonlinear. Including radiation effects also makes the analysis nonlinear.

6.2 Available Elements for Thermal Analysis

The ANSYS and ANSYS/Thermal programs include about 40 elements (described below) to help you perform steady-state thermal analyses.

For detailed information about the elements, consult the *ANSYS Elements Reference*. That manual organizes element descriptions in numeric order, starting with element LINK1.

Element names are shown in uppercase. All elements apply to both steady-state and transient thermal analyses. The usage notes for each element indicate whether other types of analyses can also use the element.

Two-Dimensional Solid Elements

PLANE35



Dimensions: 2-D
Shape or characteristic: Triangle, six nodes
Degrees of freedom: Temperature (at each node)

PLANE55



Dimensions: 2-D
Shape or characteristic: Quadrilateral, four nodes
Degrees of freedom: Temperature (at each node)

PLANE75



Dimensions: 2-D
Shape or characteristic: Harmonic, four nodes
Degrees of freedom: Temperature (at each node)

PLANE77



Dimensions: 2-D

Shape or characteristic: Quadrilateral, eight nodes

Degrees of freedom: Temperature (at each node) Usage notes: Useful for modeling curved boundaries

PLANE78



Dimensions: 2-D

Shape or characteristic: Harmonic, eight nodes

Degrees of freedom: Temperature (at each node)

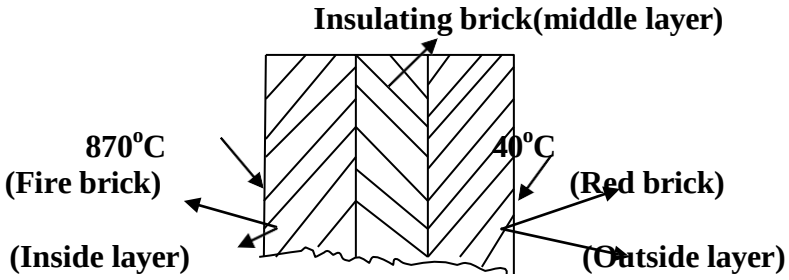
ONE DIMENSIONAL CONDUCTION BOUNDARY CONDITION

1. A furnace wall measuring 5x2 mts comprises three layers. 13.5 cm thick inside layer of fire brick(Thermal conductivity k = 1.2w/m^{o}c) , 7.5 cm thick middle layer of insulating brick(Thermal conductivity k = 0.14w/m^{o}c) and 11.5 cm thick outside layer of red brick(Thermal conductivity k = 0.85w/m^{o}c).. Furnace operates at 870° C and it is anticipated that the outside of this composite wall can be maintained at 40° C by the circulation of air.

Find the rate of heat loss from the furnace and the wall interface temperatures using

a) Theoretical methods

b) Using Ansys Plane77 element (Thermal Solid 8node 77 elements)



Q=Heat flow from inside of the furnace to outside of the composite wall = Heat loss

$$= \frac{(T_1 - T_4)}{\frac{L_1}{K_1 A} + \frac{L_2}{K_2 A} + \frac{L_3}{K_3 A}}$$

$$= \frac{(870 - 40)}{\frac{0.135}{1.2x10} + \frac{0.075}{0.14x10} + \frac{0.115}{0.85x10}} = 10593.3\ W$$

Thermal flux ,q = Q/A = 10593.3/10 = 1059.33 W/m^2

To find wall interface temperatures

Since heat flowing through each layer is same

$$10593.3 = \frac{870 - T_2}{\frac{0.135}{1.2x10}} \qquad T_2 = 750.82°C$$

$$10593.3 = \frac{750.82 - T_3}{\frac{0.075}{0.14x10}} \qquad T_3 = 183.3°C$$

Step 1: Ansys Utility Menu

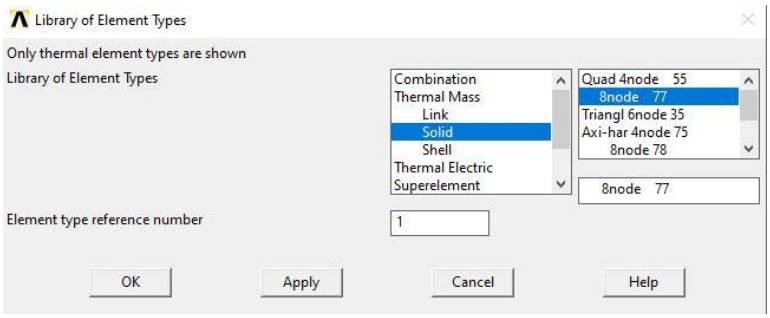
File – clear and start new – do not read file – ok – yes.

Step 2: Ansys Main Menu – Preferences

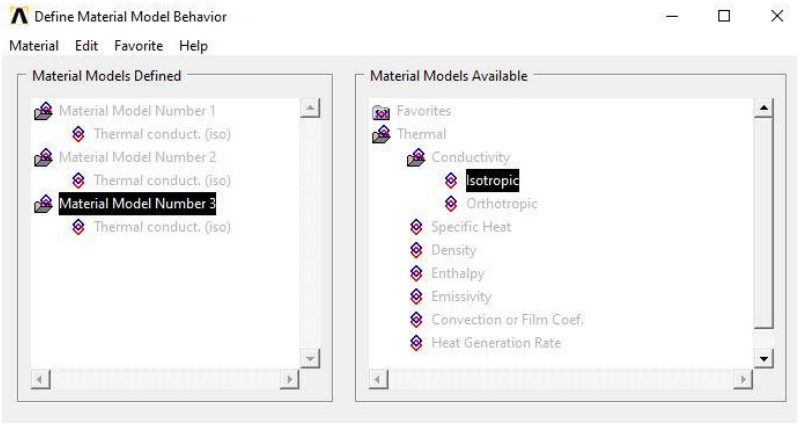
select – THERMAL - ok

Step 3: Preprocessor

Element type – Add/Edit/Delete – Add – solid – 8 Node77 – ok - close.

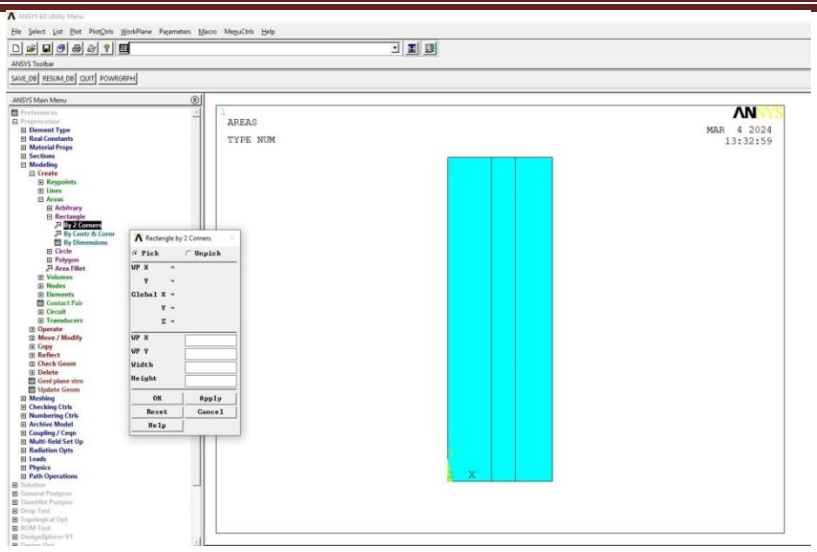


Material Properties – material models – material model number1 – Thermal – Conductivity – Isotropic – **Kxx** – 1.2 – ok – material – new model –model ID – 2 – ok – material model number2 - Thermal – Conductivity – Isotropic – **Kxx** – 0.14 – ok - material – new model –model ID – 3 – ok – material model number3 - Thermal – Conductivity – Isotropic – **Kxx** – 0.85 – ok – close.



Step 4: Preprocessor

Modeling – Create – Areas – rectangle – By two corners – X-0 - Y-0 - Width - 0.135– height – 1 – Apply(Area of fire brick created) - X-0.135 - Y-0 - Width – 0.21– height – 1 – Apply(Area of Insulating brick created) - X-0.21 - Y-0 - Width – 0.325– height – 1 – ok(Area of Red brick created).



Modeling - Operate – Booleans **-** Glue **-** areas – Pick All

Meshing – Mesh Attributes – Picked areas – Pick the left rectangle – material Number – 1 –ok
Mesh Attributes – Picked areas – Pick the middle rectangle – material Number – 2 –ok
Mesh Attributes – Picked areas – Pick the right rectangle – material Number – 3 –ok
Meshing – Size controls – Manual size – Lines – All lines – No of element Divisions – 10 –Ok
Mesh – Areas – Free – Pick All –Ok

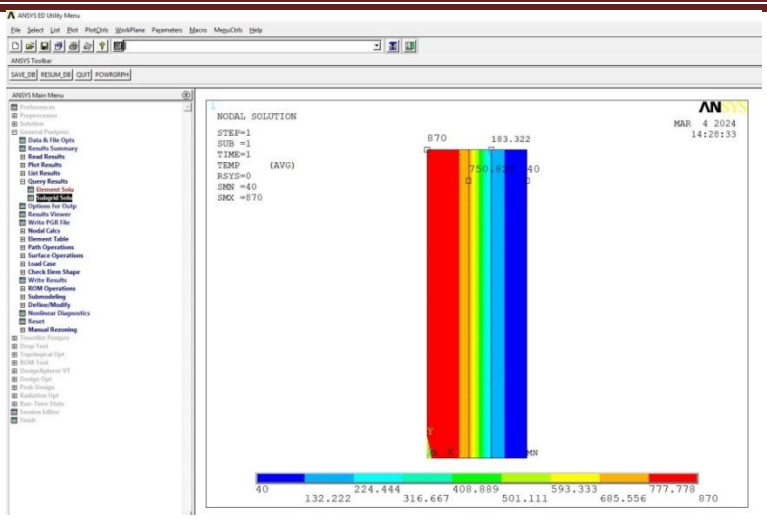
Step 5: Preprocessor

Loads – Define loads – Apply – Thermal – Temperature – on lines – select the left vertical edge – Ok- Temp – Assign 870 – Ok
Define loads – Apply – Thermal – Temperature – on lines – select the right vertical edge – Ok- Temp – Assign 40 – Ok

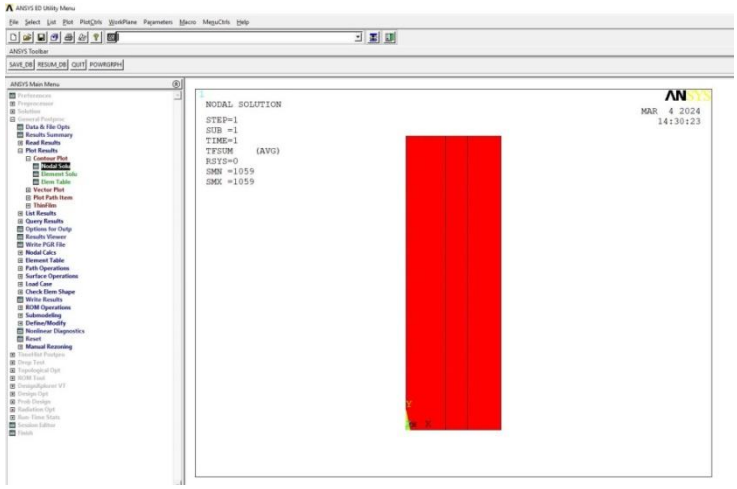
Step 6: Solution – Solve – Current LS – Ok – Close

Step 7: General Postproc

General Postproc – Plot results – Contour Plot – Nodal solution – DOF solution – Nodal temperature – Ok
General Postproc – Query results – subgrid solution – select nodes **-** ok



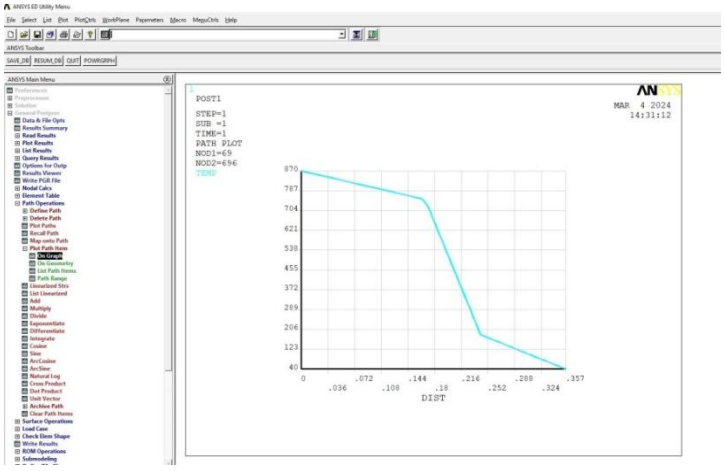
General Postproc – Plot results – Contour Plot – Nodal solution – Thermal Flux – Thermal flux Vector sum – Ok



General Postproce - Path operations - Define path - By nodes - Select nodes

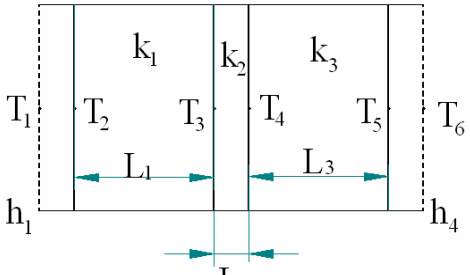
Map onto Path – user label for items type TEMP - select DOF solution – Temperature TEMP – ok

- Plot Path Item – on graph – Select TEMP – ok



TWO DIMENSIONAL CONDUCTION AND CONVECTION BOUNDARY CONDITIONS

2. A furnace wall is made of inside silica brick (k = 1.5 W/mK) and outside magnesia brick (k = 4.9 W/mK), each 10 cm thick. The inner and outer surfaces are exposed to fluids at temperatures of 820^0 C and 110^0 C respectively. The contact resistance is $0.001m^2$ K/W. The heat transfer coefficient for inner and outside surfaces is equal to 35 W/m^2 K. Find the heat flow through the wall per unit area per unit time and temperature distribution across the wall. **(**March/April 2001)



k_1 = 1.5 W/mK
k_3 = 4.9 W/mK
$L_1 = L_3$ = 100mm=0.1m
$T_1 = 820^0C$
$T_6 = 110^0C$
$R_c = 0.001m^2K/W.$
$h_1 = h_4 = 35\ W/m^2K.$

$$R_c = \frac{L_2}{K_2 A}$$

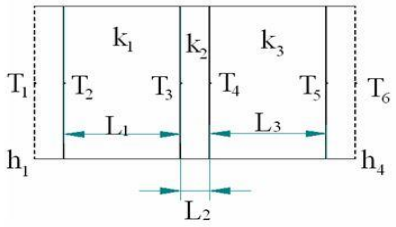
Take L_2=1mm=0.001m
K_2 = 1 W/mK
A = $1m^2$

COMPOSITE WALL1 – Theoretical solution:

$$Q = \frac{(T_1 - T_6)}{\frac{1}{A}\left(\frac{1}{h_1} + \frac{L_1}{K_1} + \frac{L_2}{K_2} + \frac{L_3}{K_3} + \frac{1}{h_4}\right)} = 4889.21 W/m^2$$

$$Q = h_1 A(T_1 - T_2) \Rightarrow T_2 = 680.31^0 C$$

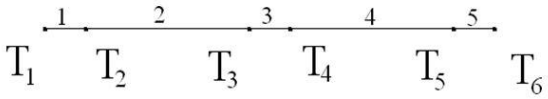
$$O = \frac{k_1 A(T_2 - T_3)}{L_1} \Rightarrow T_3 = 354.36^0 C$$



$$O = \frac{k_2 A(T_3 - T_4)}{L_2} \Rightarrow T_4 = 349.47^0 C$$

$$O = \frac{k_3 A(T_4 - T_5)}{L_3} \Rightarrow T_5 = 249.69^0 C$$

COMPOSITE WALL1- ANSYS analysis



For FE model,

Element no.	Element Type
2,3,4	LINK 32
1,5	LINK 34

Step 1: Ansys Utility Menu

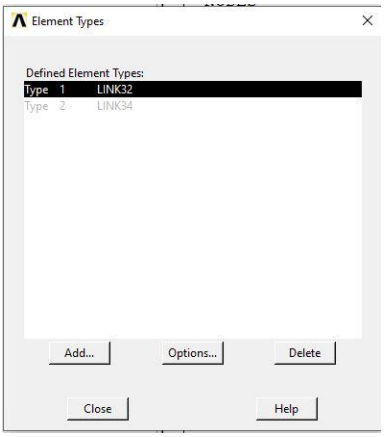
File – clear and start new – do not read file – ok – yes.

Step 2: Ansys Main Menu – Preferences

select – THERMAL - ok

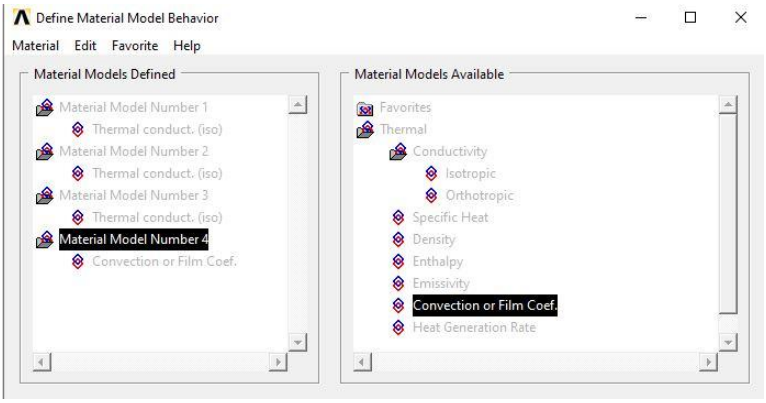
Step 3: Preprocessor

Element type – Add/Edit/Delete – Add – Link – 2D Conduction 32 – ref number – 1 - apply – Link – Convection 34 – ref number – 2 – ok - close.



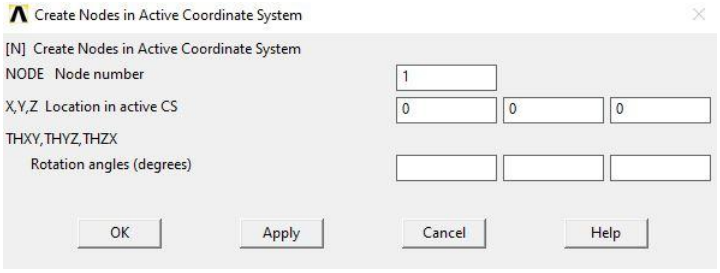
Real constants – Add/Edit/Delete - Add – Type 1 Link32 - ok – real constant set no – 1 – c/s area – 1 – ok – Type 2 Link34 – ok - real constant set no – 2 – c/s area – 2 – convection surface area – 1 – ok – close.

Material Properties – material models – material model number1 – Thermal – Conductivity – Isotropic – **Kxx** – 1.5 – ok – material – new model –model ID – 2 – ok – material model number2 - Thermal – Conductivity – Isotropic – **Kxx** – 1 – ok - material – new model –model ID – 3 – ok – material model number3 - Thermal – Conductivity – Isotropic – **Kxx** – 4.9 – ok - material – new model –model ID – 4 – ok – material model number4 - Thermal – Convection or Film Coef – **HF** – 35 – ok – close.



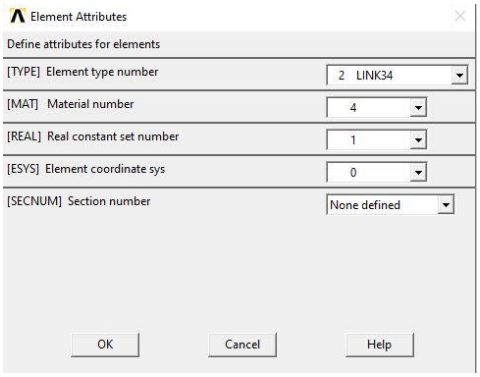
Step 4: Preprocessor

Modeling – Create – Nodes – In Active CS – Apply (first node is created)



– X,Y,Z location in CS – 0.001 (X value w.r.t first node) – Apply (second node is created) - X,Y,Z location in CS – 0.101 (X value w.r.t first node) – Apply (Third node is created) - X,Y,Z location in CS – 0.102 (X value w.r.t first node) – Apply (Fourth node is created) - X,Y,Z location in CS – 0.202 (X value w.r.t first node) – Apply (Fifth node is created) - X,Y,Z location in CS – 0.203 (X value w.r.t first node) – ok (Sixth node is created).

Create –Elements – Element Attributes – Element Type number – 2 Link34 – Material number – 4 - real constant set no – 2 – ok.



Auto numbered– Thru Nodes – pick 1 & 2 – ok (element created through nodes).
Create – Elements – Element Attributes – Element Type number – 1 Link32 – Material number – 1 - real constant set no – 1 – ok.
Auto numbered– Thru Nodes – pick 2 & 3 – ok (element created through nodes).
Create –Elements – Element Attributes – Element Type number – 1 Link32 – Material number – 2 - real constant set no – 1 – ok.
Auto numbered– Thru Nodes – pick 3 & 4 – ok (element created through nodes).
Create –Elements – Element Attributes – Element Type number – 1 Link32 – Material number – 3 - real constant set no – 1 – ok.
Auto numbered– Thru Nodes – pick 4 & 5 – ok (element created through nodes).
Create –Elements – Element Attributes – Element Type number – 2 Link34 – Material number – 4 - real constant set no – 2 – ok.
Auto numbered– Thru Nodes – pick 5 & 6 – ok (element created through nodes).

Step 5: Ansys Utility Menu – plotctrls – Numbering – Elem/numbering – Element numbers – ok.

Step 6: Preprocessor

Loads – Define loads – apply – Thermal – Temperature – on Nodes- pick node 1 – ok – DOFs to be constrained – TEMP – load TEMP value – 820 – Apply - pick node 6 – ok – DOFs to be constrained – TEMP – load TEMP value – 110 - ok.

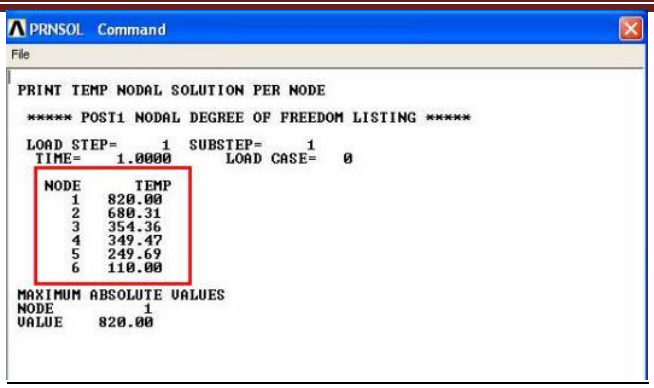
Step 7: Solution

Analysis Type – new Analysis – type of Analysis – steady state - ok

Solve – current LS – ok (Solution is done is displayed) – close.

Step 8: General Post Processor

List Results – Contour Plot - nodel solution – DOF solution – Temperature – ok

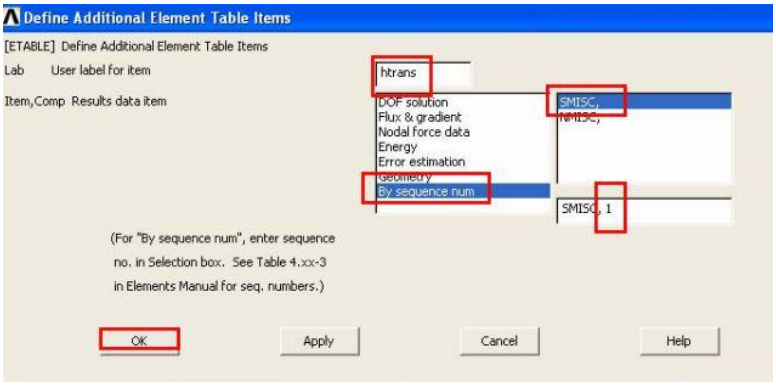


Ansys Utility Menu – plotctrls – Style – Size and Shape – Display of element – ON – Scale - 0.2 – ok

Plot Result – Contour Plot – Nodel solution - DOF solution – Temperature - ok

Step 9: General Post Processor

Element Table – Define Table – Add – user label for item – heat flow – By Sequence num – SMISC – SMISC,1 – ok – close.



Element Table – List Elem Table – select heat flow – ok



COMPARISON of Results

Quantity	ANSYS	Theory
T_2	680.31^{0}C	680.31^{0}C
T_3	354.36^{0}C	354.36^{0}C
T_4	349.47^{0}C	349.47^{0}C
T_5	249.69^{0}C	249.69^{0}C
Heat flow	4889.2W/m^2	4889.21W/m^2

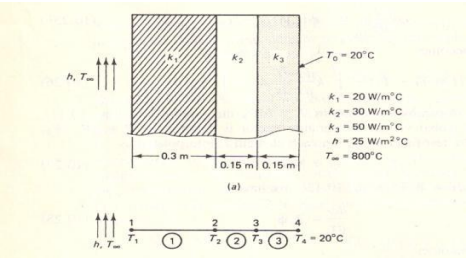
****

PRACTICE PROBLEMS ON THERMAL ANLYSIS

EXERCISE 1: A composite wall consists of three materials as shown. The outer temperature is To=20 deg C. Convection heat transfer takes place on the inner surface of the wall with Tω = 800 deg C and h = 25 W/m^2 .Determine the temperature distribution in the wall.
Find the rate of heat loss from the composite wall and the wall interface temperatures using

a) Theoretical methods

b) Method 1 - Using Ansys Plane77 element (Thermal Solid 8node 77 elements)



·**EXERCISE 2:** The illustration below is a 2 dimensional representation of a 3 dimensional furnace with obvious boundary conditions including constant temperatures. Using symmetry, limit the scope of modeling to only one section of the furnace, as shown.

· Boundary Conditions:

1) The symmetry lines of the corner piece can be assumed to be insulated.
2) The outer most boundaries have a constant temperature of 300K.
3) The inner most boundary has a constant temperature of 1000K.

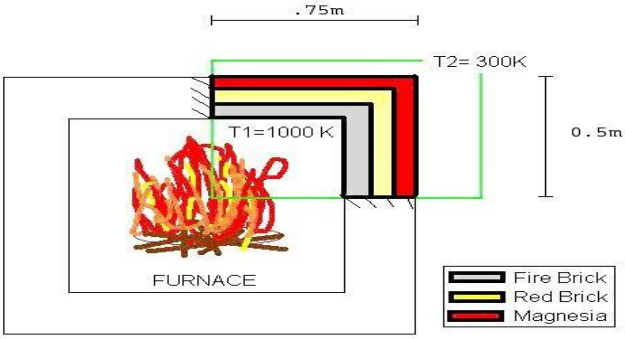
· Material Properties:

Fire Brick(inner layer): K_{FB} = 0.3 W/m*K Red Brick(middle layer): K_{RB} = 0.63 W/m*K Magnesia(outer layer): K_M = 1.41 W/m*K

· Dimensions Length = 1.5 m Width = 1 m

Thickness of each Layer = .05 m

Find the nodal temperature distribution and the rate of heat loss from the furnace.



SELF LEARNING EXERCISE 1: A large industrial furnace is supported on a long column of fireclay brick which is 1 x 1 m on a side .During steady state operation, installation is such that three surfaces of the column are maintained at 600 K while the remaining surface is exposed to an air stream for which Tω = 300 K and h = 12 W/m^{2}K.Determine the temperature distribution in the column and the heat rate to the air stream per unit length of the column. Take K = 1 W/m K

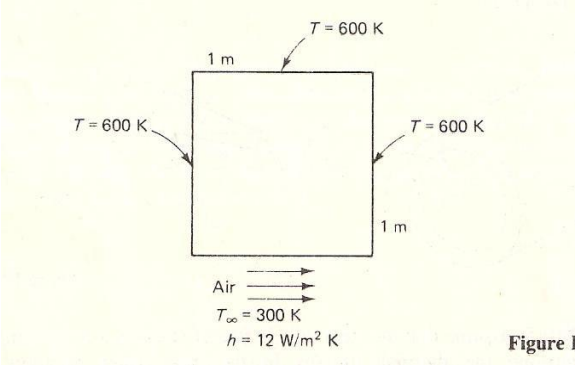


Figure 1

****

Dynamic Analysis

CHAPTER 7

DYNAMIC ANALYSIS

INTRODUCTION

Model Analysis

Definition of Modal Analysis

You use modal analysis to determine the vibration characteristics (natural frequencies and mode shapes) of a structure or a machine component while it is being designed. It also can be a starting point for another, more detailed, dynamic analysis, such as a transient dynamic analysis, a harmonic response analysis, or a spectrum analysis.

Uses for Modal Analysis

You use modal analysis to determine the natural frequencies and mode shapes of a structure. The natural frequencies and mode shapes are important parameters in the design of a structure for dynamic loading conditions. They are also required if you want to do a spectrum analysis or a mode superposition harmonic or transient analysis.

You can do modal analysis on a prestressed structure, such as a spinning turbine blade. Another useful feature is modal cyclic symmetry, which allows you to review the mode shapes of a cyclically symmetric structure by modeling just a sector of it.

Modal analysis in the ANSYS family of products is a linear analysis. Any nonlinearities, such as plasticity and contact (gap) elements, are ignored even if they are defined. You can choose from several mode-extraction methods: Block Lanczos (default), subspace, PowerDynamics, reduced, unsymmetric, damped, and QR damped. The damped and QR damped methods allow you to include damping in the structure. Details about mode-extraction methods are covered later in this section.

7.1 Overview of Steps in a Modal Analysis

The procedure for a modal analysis consists of four main steps:

1. Build the model.
2. Apply loads and obtain the solution.
3. Expand the modes.
4. Review the results.

When building your model, remember these points:

- Only linear behavior is valid in a modal analysis. If you specify nonlinear elements, they are treated as linear. For example, if you include contact elements, their stiffness's are calculated based on their initial status and never change.
- Material properties can be linear, isotropic or orthotropic, and constant or temperature-dependent. You must define both Young's modulus (EX) (or stiffness in some form) and density (DENS) (or mass in some form) for a modal analysis. Nonlinear properties are ignored. If applying element damping, you must define the required real constants for the specific element type (COMBIN7, COMBIN14, COMBIN37, and so on).

Mode-Extraction Method [MODOPT]

Choose one of the extraction methods listed below.

Block Lanczos method (default)

- The Block Lanczos method is used for large symmetric eigenvalue problems. You can use this method for the same types of problems for which you use the subspace method, but you achieve a faster convergence rate.
- Subspace method

The subspace method is used for large symmetric eigenvalue problems. Several solution controls are available to control the subspace iteration process.

When doing a modal analysis with a large number of constraint equations, use the subspace method with the frontal solver instead of the JCG solver, or use the Block Lanczos mode-extraction method.

- PowerDynamics method

The PowerDynamics method is used for very large models (100,000+ DOFs), and is especially useful to obtain a solution for the first several modes to learn how the model will behave.

- Reduced (Householder) method
- The reduced method is faster than the subspace method because it uses reduced (condensed) system matrices to calculate the solution. However, it is less accurate because the reduced mass matrix is approximate.
- Unsymmetric method

The unsymmetric method is used for problems with unsymmetric matrices, such as fluid-structure interaction problems.

- Damped method

The damped method is used for problems where damping cannot be ignored, such as bearing problems.

- QR Damped method

  The QR damped method is faster and achieves better calculation efficiency than the damped method. It uses the reduced modal damped matrix to calculate complex damped frequencies in modal coordinates.

For most applications, you will use the Block Lanczos, subspace, reduced, or Power Dynamics method. The unsymmetric, damped, and QR damped methods are meant for special applications.

When you specify a mode-extraction method, ANSYS automatically chooses the appropriate equation solver.

Prestressed Modal Analysis

Use a pre stressed modal analysis to calculate the frequencies and mode shapes of a pre stressed structure, such as a spinning turbine blade. The procedure to do a pre stressed modal analysis is essentially the same as a regular modal analysis, except that you first need to pre stress the structure by doing a static analysis:

7.2 Harmonic Response Analysis

Definition of Harmonic Response Analysis

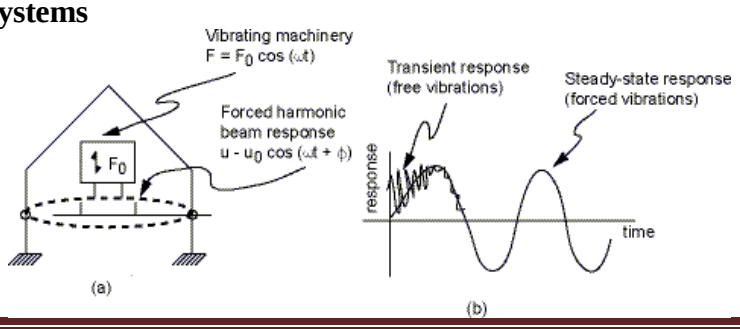
Any sustained cyclic *load* will produce a sustained cyclic *response* (a *harmonic* response) in a structural system. *Harmonic response analysis* gives you the ability to predict the sustained dynamic behavior of your structures, thus enabling you to verify whether or not your designs will successfully overcome resonance, fatigue, and other harmful effects of forced vibrations.

Uses for Harmonic Response Analysis

Harmonic response analysis is a technique used to determine the steady-state response of a linear structure to loads that vary sinusoidally (*harmonically*) with time. The idea is to calculate the structure's response at several frequencies and obtain a graph of some response quantity (usually displacements) versus frequency. "Peak" responses are then identified on the graph and stresses reviewed at those peak frequencies.

This analysis technique calculates only the steady-state, forced vibrations of a structure. The transient vibrations, which occur at the beginning of the excitation, are not accounted for in a harmonic response analysis (see Figure 4.1: "Harmonic Response Systems").

Harmonic Response Systems



The Three Solution Methods

Three harmonic response analysis methods are available: *full*, *reduced*, and *mode superposition.*

How to Do Harmonic Response Analysis

We will first describe how to do a harmonic response analysis using the full method, and then list the steps that are different for the reduced and mode superposition methods.

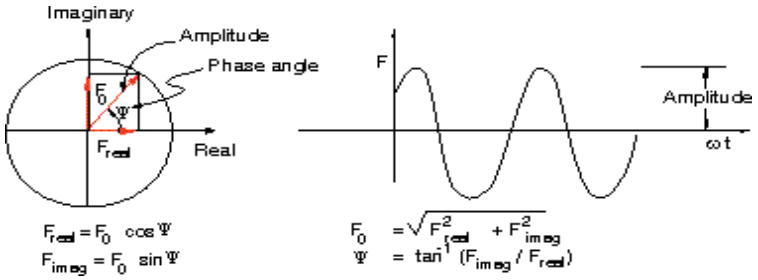
Full Harmonic Response Analysis

The procedure for a full harmonic response analysis consists of three main steps:

1. Build the model.
2. Apply loads and obtain the solution.
3. Review the results.

Apply Loads on the Model

A harmonic analysis, by definition, assumes that any applied load varies harmonically (sinusoidally) with time. To completely specify a harmonic load, three pieces of information are usually required: the *amplitude*, the *phase angle*, and the *forcing frequency range* (see Figure 4.2: "Relationship Between Real/Imaginary Components and Amplitude/Phase Angle").



7.3 Transient Dynamic Analysis

Definition of Transient Dynamic Analysis

Transient dynamic analysis (sometimes called time-history analysis) is a technique used to determine the dynamic response of a structure under the action of any general time-dependent loads. You can use this type of analysis to determine the time-varying displacements, strains, stresses, and forces in a structure as it responds to any combination of static, transient, and harmonic loads. The time scale of the loading is such that the inertia or damping effects are considered to be important.

Preparing for a Transient Dynamic Analysis

A transient dynamic analysis is more involved than a static analysis because it generally requires more computer resources and more of *your* resources, in terms of the "engineering" time involved. You can save a significant amount of these resources by doing some preliminary work to understand the physics of the problem. For example, you can:

1. Analyze a simpler model first. A model of beams, masses, and springs can provide good insight into the problem at minimal cost. This simpler model may be all you need to determine the dynamic response of the structure.
2. If you are including nonlinearities, try to understand how they affect the structure's response by doing a static analysis first. In some cases, nonlinearities need not be included in the dynamic analysis.
3. Understand the dynamics of the problem. By doing a modal analysis, which calculates the natural frequencies and mode shapes, you can learn how the structure responds when those modes are excited. The natural frequencies are also useful for calculating the correct integration time step.

Three Solution Methods

Three methods are available to do a transient dynamic analysis: *full, mode superposition,* and *reduced.* The ANSYS Professional program allows only the mode superposition method.

Performing a Full Transient Dynamic Analysis

We will first describe how to do a transient dynamic analysis using the full method. We will then list the steps that are different for the mode superposition and reduced methods.

The procedure for a full transient dynamic analysis (available in the ANSYS Multiphysics, ANSYS Mechanical, and ANSYS Structural products) consists of these steps:

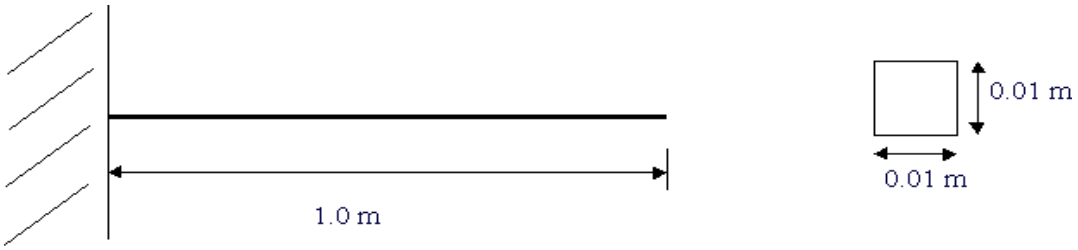
1. Build the Model
2. Establish Initial Conditions
3. Set Solution Controls
4. Set Additional Solution Options
5. Apply the Loads
6. Save the Load Configuration for the Current Load Step
7. Repeat Steps 3-6 for Each Load Step
8. Save a Backup Copy of the Database
9. Start the Transient Solution
10. Exit the Solution Processor
11. Review the Results

Dynamic Analysis
Modal Analysis of a Cantilever Beam

1. For the beam shown determine the following using Subspace method.

i) Determining the mode frequencies and mode shapes of structures.

ii) Animate mode shape.



Modulus of Elasticity (E) = 206800(10^6) N/m^2

Density = 7830 kg/m^3

Step 1: Ansys Utility Menu
File – clear and start new – do not read file – ok – yes.

Step 2: Ansys Main Menu – Preferences
select – STRUCTURAL - ok

Step 3: Preprocessor

Element type – Add/Edit/Delete – Add – **Beam** – **2D elastic 3** – ok – close.

Real constants – Add/Edit/Delete - Add – ok – real constant set no – 1 – c/s area – 0.01*0.01 - moment of inertia – 0.01*0.01**3/12 – total beam height – 0.01 – ok.
Material Properties – material models – Structural – Linear – Elastic – Isotropic – EX – 206800e6 **N/m2** – PRXY – 0.27 – ok - Density – **7830kg/m^3**– ok - close.

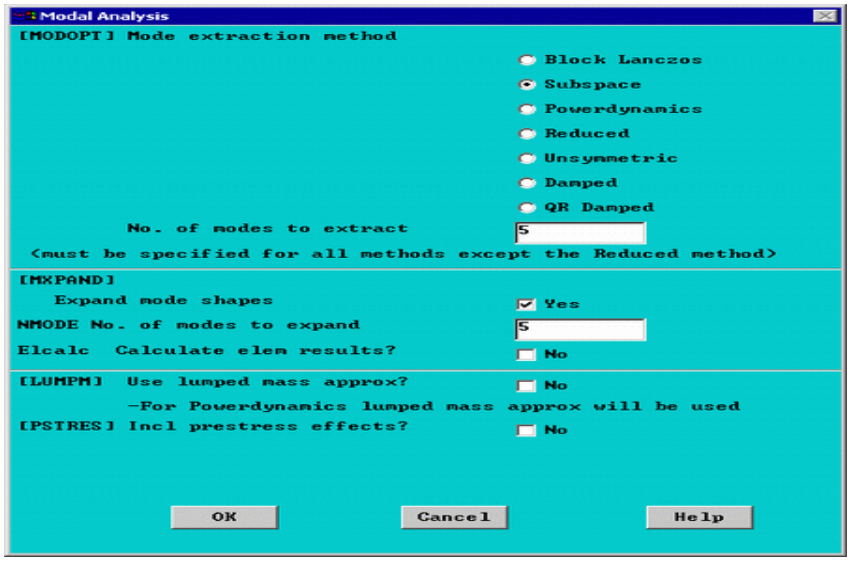
Step 4: Preprocessor

Modeling – Create – Key points – In Active CS – Apply (first Key point is created) – X,Y,Z location in CS – 1mt (X value w.r.t first key point) – ok (second key point is created).
Create – Lines – Straight line – pick 1st key point & 2nd key point – ok (Line created through key point).

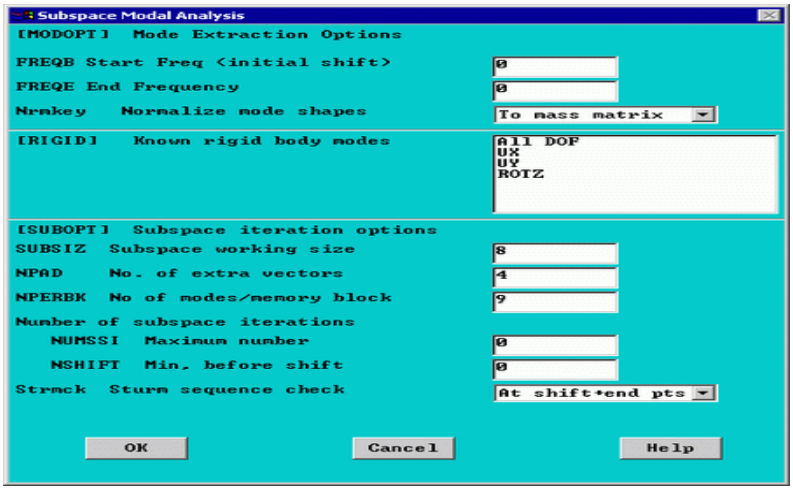
Meshing – Mesh Tool – Size Control – Lines – Set – Pick All – number of element division – 5 - ok - Mesh Tool – Mesh – Lines – Mesh – pick all.

Step 5: Preprocessor

Loads – Analysis Type – new analysis – (type of analysis) Modal – ok.
Analysis Options – (mode extraction method)Subspace – No of modes to extract – 5 – No of modes to expand – 5 – ok



The following window will then appear



For this problem, we will use the default options so click on OK.

Define loads – Apply - Structural – Displacement – on key points - pick key point 1 – apply – DOFs to be constrained – All DOF – ok.

Step 6: Solution

Solve – current LS – ok (Solution is done is displayed) – close.

Step 7: General Post Processor

Result summary – (copy the Frequencies)

```
*****  INDEX OF DATA SETS ON RESULTS FILE  *****

   SET    TIME/FREQ    LOAD STEP   SUBSTEP  CUMULATIVE
     1   8.3018            1         1         1
     2   52.046            1         2         2
     3   146.15            1         3         3
     4   288.64            1         4         4
     5   478.90            1         5         5
```

NOTE: To obtain accurate higher mode frequencies, this mesh would have to be refined even more (i.e. instead of 5 elements, we would have to model the cantilever using 10 or more elements depending upon the highest mode frequency of interest).

Step 8: General Post Processor

View Mode Shapes

- Select **Read Results - First Set**

This selects the results for the first mode shape

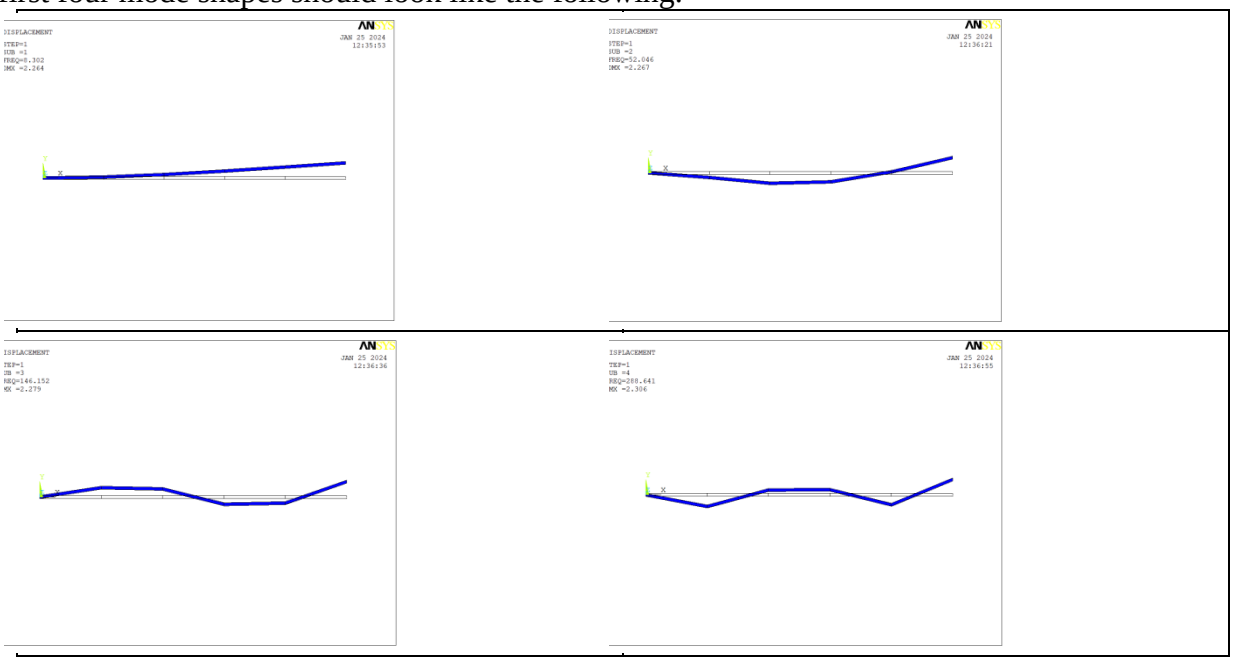
- Select **- Plot Results > Deformed shape** - Select 'Def + undef edge'

The first mode shape will now appear in the graphics window.

To view the next mode shape, select **Read Results > Next Set** .

As above choose **Plot Results > Deformed shape** . Select 'Def + undef edge'.

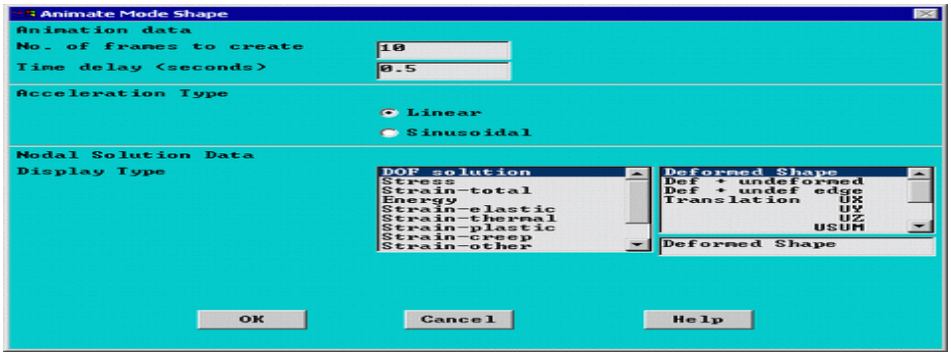
The first four mode shapes should look like the following:



Step 9: Ansys Utility Menu : PlotCtrls – style – size and shape – Display of elements – ON – ok.

Step 10: Ansys Utility Menu : PlotCtrls - Animate – Deformed shape – def+undeformed-ok

The following window will appear

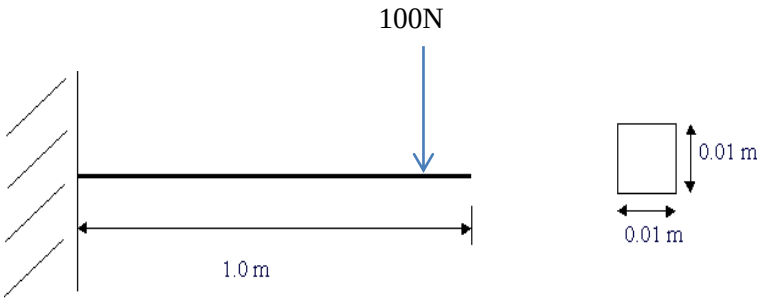


Harmonic Analysis of a Cantilever Beam

For the beam shown perform harmonic response test by applying a cyclic load (harmonic) at the end of the beam. The frequency of the load will be varied from 1 - 100 Hz. Determine the following using Full method,

i) Draw a graph of amplitude Vs frequency and hence determine the first two natural frequencies of the beam.

ii) Draw also the graph of amplitude Vs frequency on a logarithmic scale.



Modulus of Elasticity (E) = 206800(10^6) N/m^2

Density = 7830 kg/m^3

Step 1: Ansys Utility Menu

File – clear and start new – do not read file – ok – yes.

Step 2: Ansys Main Menu – Preferences

select – STRUCTURAL - ok

Step 3: Preprocessor

Element type – Add/Edit/Delete – Add – **Beam** – **2D elastic 3** – ok – close.

Real constants – Add/Edit/Delete - Add – ok – real constant set no – 1 – c/s area – 0.01*0.01 - moment of inertia – 0.01*0.01**3/12 – total beam height – 0.01 – ok.

Material Properties – material models – Structural – Linear – Elastic – Isotropic – EX – 206800e6 **N/m2** – PRXY – 0.27 – ok - Density – **7830kg/m^3**– ok - close.

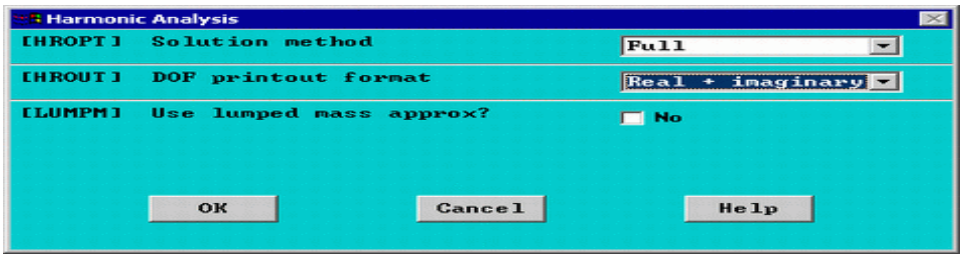
Step 4: Preprocessor

Modeling – Create – Key points – In Active CS – Apply (first Key point is created) – X,Y,Z location in CS – 1mt (X value w.r.t first key point) – ok (second key point is created).
Create – Lines – Straight line – pick 1st key point & 2nd key point – ok (Line created through key point).

Meshing – Mesh Tool – Size Control – Lines – Set – Pick All – number of element division – 5 - ok - Mesh Tool – Mesh – Lines – Mesh – pick all.

Step 5: Preprocessor

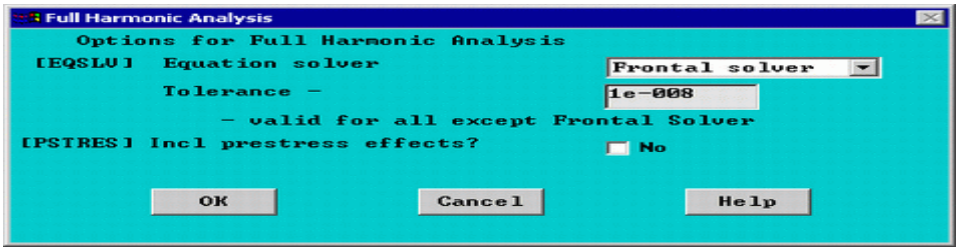
Loads – Analysis Type – new analysis – (type of analysis) Harmonic – ok.
Loads – Analysis Type – Analysis options – (The following window will appear)



select the **Full** Solution method, the **Real + imaginary** DOF printout format and do not use lumped mass approx.

Click 'OK'

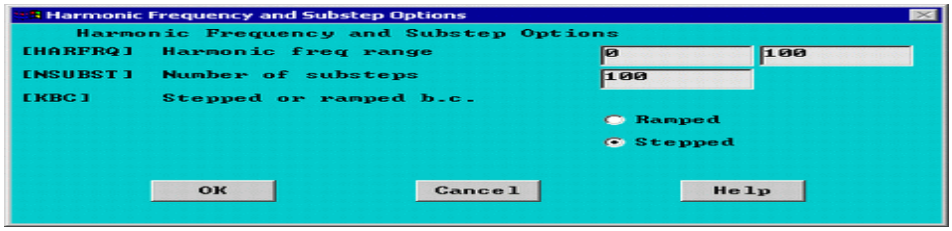
The following window will appear. Use the default settings (shown below).



Define loads – Apply - Structural – Displacement – on key points - pick key point 1 – apply – DOFs to be constrained – All DOF – ok.
Force/moment – on key point – select key pont No 2 – direction of force/moment – FY – force

value – (-100 N) (–ve value) – ok.

Loads – load step options – Time/Frequency – Harmonic Frequency Range – 1Hz To 100Hz – Number of substeps – 100 – select Stepped – ok



By doing this we will be subjecting the beam to loads at 1 Hz, 2 Hz, 3 Hz,. . . . 100 Hz. We will specify a stepped boundary condition as this will ensure that the same load (100 N) which will be applied for each of the frequencies.

Step 6: Solution

Solve – current LS – ok (Solution is done is displayed) – close.

NOTE: We want to observe the response at x=1 (where the load was applied) as a function of frequency. We cannot do this with **General Post Processing (POST1)**, rather we must use **TimeHist Post Processing (POST26)**. POST26 is used to observe certain variables as a function of either time or frequency.

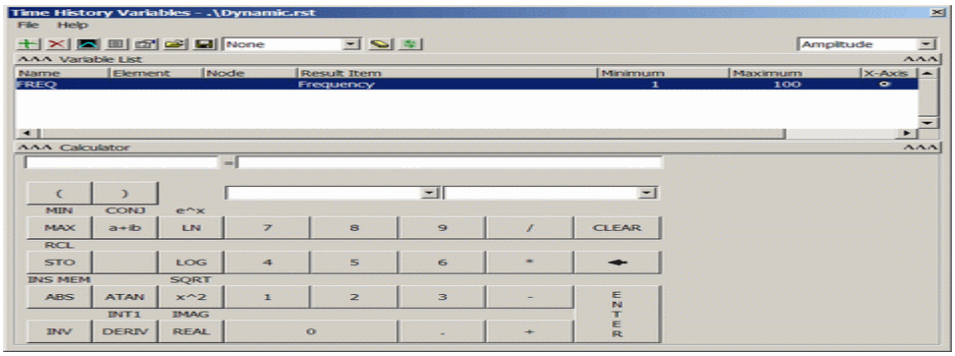
Step 6: TimeHist Processing (POST26) Menu

Select **TimeHist Postpro** from the ANSYS Main Menu.

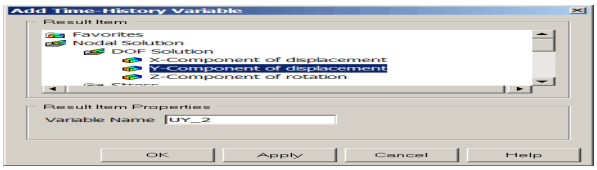
Define Variables

Here we have to define variables that we want to see plotted. By default, **Variable 1** is assigned either **Time** or **Frequency**. In our case it is assigned **Frequency**. We want to see the displacement UY at the node at x=1, which is node #2. (To get a list of nodes and their attributes, select **Utility Menu > List > nodes**).

Select **TimeHist Postpro > Variable Viewer...** and the following window should pop up.

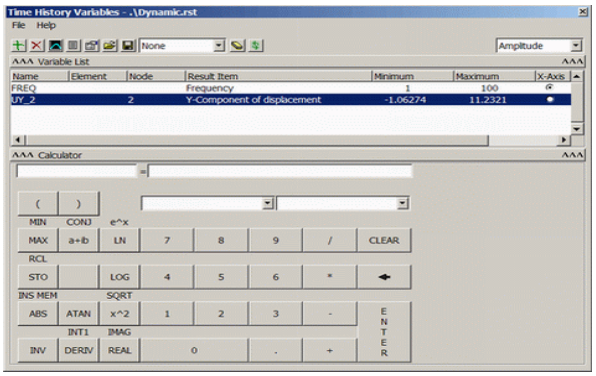


Select `Add` (the green '+' sign in the upper left corner) from this window and the following window should appear.



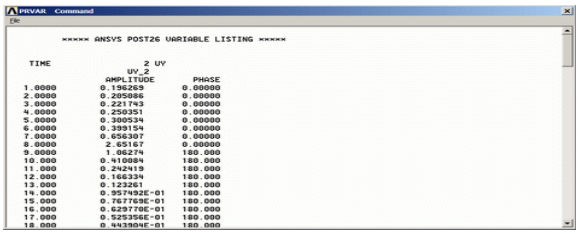
We are interested in the **Nodal Solution > DOF Solution > Y-Component of displacement**. Click OK.

Graphically select node 2 when prompted and click OK. The 'Time History Variables' window should now look as follows



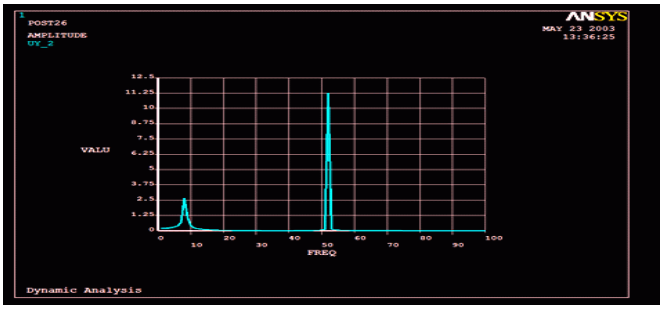
List Stored Variables

In the 'Time History Variables' window click the 'List' button, 3 buttons to the left of 'Add' The following window will appear listing the data:



Plot UY vs. frequency

In the 'Time History Variables' window click the 'Plot' button, 2 buttons to the left of 'Add' The following graph should be plotted in the main ANSYS window.

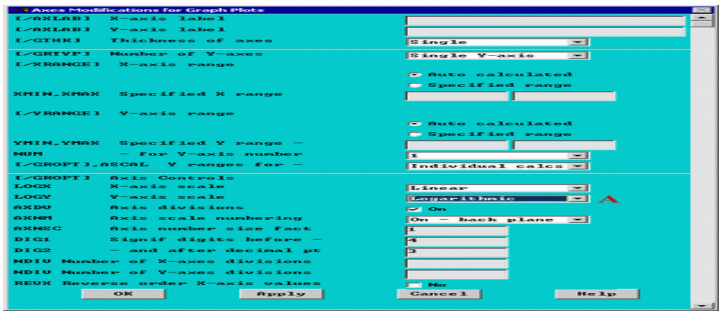


Note that we get peaks at frequencies of approximately 8.3 and 51 Hz which corresponds to first two natural frequencies obtained in modal analysis.

To get a better view of the response, view the log scale of **UY**.

Select Utility Menu > PlotCtrls > Style > Graphs > Modify Axis

The following window will appear

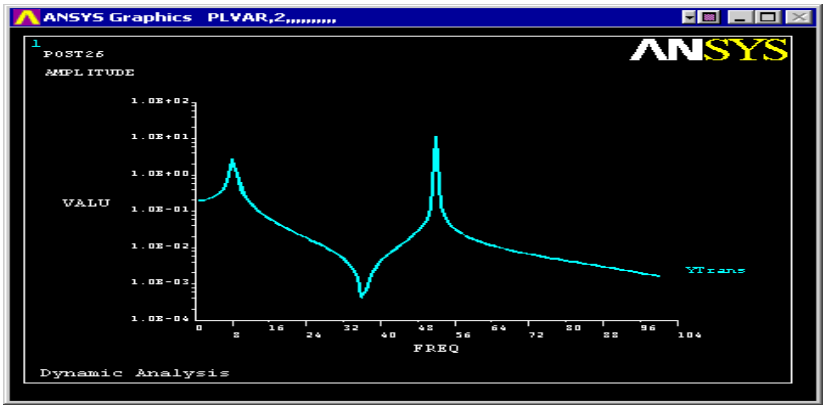


As marked by an 'A' in the above window, change the Y-axis scale to 'Logarithmic'

Select Utility Menu > Plot > Replot

You should now see the following

This is the response at node 2 for the cyclic load applied at this node from 0 - 100 Hz.



PRACTICE PROBLEMS ON DYNAMIC ANALYSIS

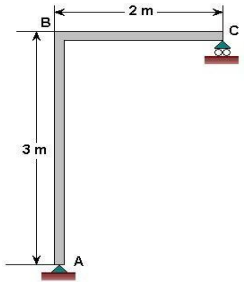
EXERCISE 1: Consider the problem- right-angle frame for vibration analysis. Take the following material properties for the frame.

$E = 200x10^9\ Pa$

$\nu = 0.29$

$\rho = 7860 kg/m^3$

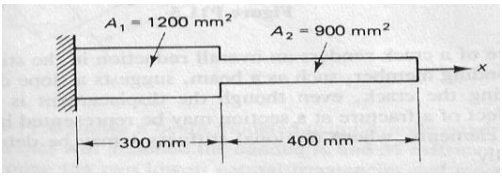
$I = \frac{1x10^{-4}}{12}$



Using a) Subspace method b) Reduced method
i) Determine the mode frequencies and mode shapes of the structure.
ii) Animate mode shape.

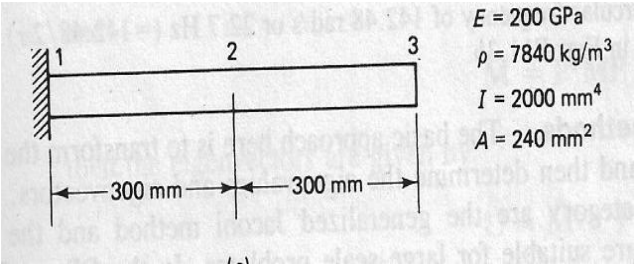
EXERCISE 2: Using a) Subspace method b) Reduced method

i) Determine the mode frequencies and mode shapes of the structure.
ii) Animate mode shape.



EXERCISE 3: For the beam shown perform harmonic response test by applying a cyclic load (harmonic) at the end of the beam. The frequency of the load will be varied from 1 - 100 Hz. Determine the following using Full method

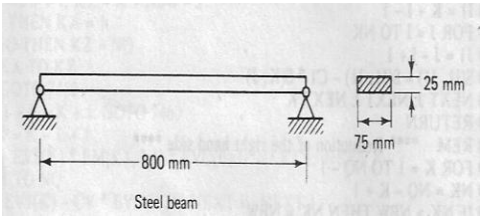
i) Draw a graph of amplitude Vs frequency and hence determine the first two natural frequencies of the beam.
ii) Draw also the graph of amplitude Vs frequency on a logarithmic scale.



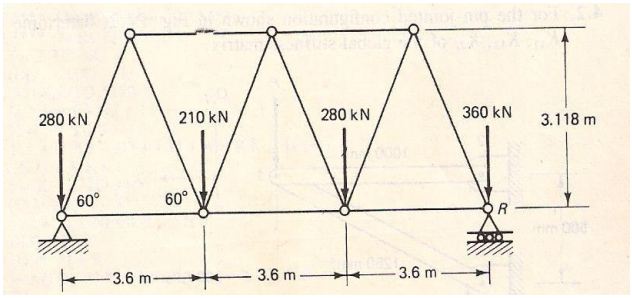
SELF LEARNING EXERCISE 1: For the beam shown perform harmonic response test by applying a cyclic load (harmonic) at the center of the beam. The frequency of the load 200 N will be varied from 1 - 100 Hz. Determine the following using Full method

i) Draw a graph of amplitude Vs frequency and hence determine the first two natural frequencies of the beam.

ii) Draw also the graph of amplitude Vs frequency on a logarithmic scale.



SELF LEARNING EXERCISE 2: Using a) Subspace method b) Reduced method
i) Determine the mode frequencies and mode shapes of the structure.
ii) Animate mode shape.



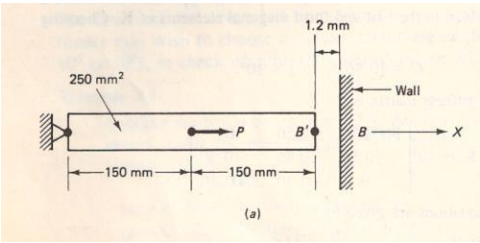
$\rho = 7860 kg/m^3$

Take E = 200 GPa, A=Cross sectional area =3250 mm^2

SAMPLE VIVA QUESTIONS

Computer Aided Modeling And Analysis Lab

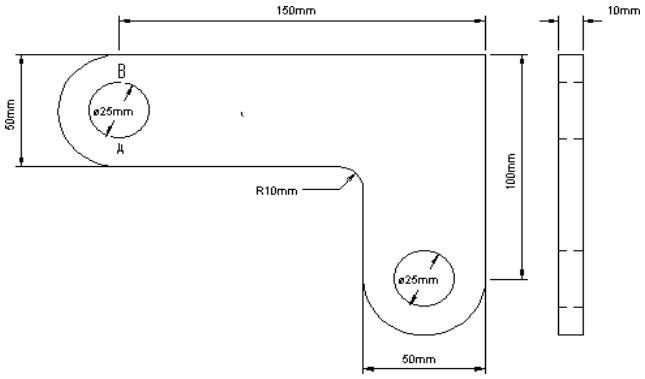
1. What is FEM?
2. State some applications of FEM.
3. What is ANSYS?
4. What is virtual prototyping?
5. What are the three stages involved in finite element analysis?
6. What are the general steps involved in solving any problem in ANSYS?
7. What is Structural analysis?
8. What is Thermal analysis?
9. What is Dynamic analysis?
10. What are "Utility Menu", "Input Line", "Toolbar", "Main Menu", and "Graphics window" in ANSYS environment?
11. What do you understand by the term "FEM Convergence Testing"?
12. What are .db, .dbb, .err, .out, .log files in ANSYS?
13. Which file out of the above you need to save?
14. How do you recall a previously saved job?
15. When do you use "Change Jobname.." and " Change Title…" in ANSYS.?
16. What happens when you click on "Resume Jobname.db…"?
17. What is an "ETABLE" ? What does it contain? Why do you need it?
18. What is "Element output definitions" with ref to an element?
19. What are "Item and sequence numbers" with ref to an element?
20. What is one dimensional problem? Which element supports one dimensional analysis in ANSYS and what are its characteristic features?
21. How do you solve problems of Self Weight in ANSYS?



How do you identify the minimum value of the force "P" for which the gap of 1.2 mm just closes using ANSYS.

22. Which element supports analysis of beams in ANSYS.? What are its capabilities?
23. What is the difference between BEAM 3 and BEAM 188 as applied to solving problems on beams?
24. List the item and sequence numbers for obtaining bending moment, Shear force and bending stress diagrams during analysis of beams using BEAM 3.
25. List the item and sequence numbers for obtaining bending moment, Shear force and bending stress diagrams during analysis of beams using BEAM 188.
26. The BM, SF and BS diagrams will be more accurate if more elements are chosen for a beam. Explain this behavior.

27. When do you use Orientation Key point while analyzing beams using BEAM 188? What is the procedure?
28. A beam 100 mm width 50 mm depth with a span of 2000 mm has a concentrated load of 50000N acting at the center. Does this beam fail? Do you find any design mistake in its orientation and what is the remedy?
29. What are the commonly used elements while analyzing plate problems or two dimensional problems?
30. What is the difference between PLANE 42 and PLANE 82 elements and what are its capabilities?
31. State the difference between Plane stress and Plane strain analysis.
32. What do you understand by triangular and quadrilateral discretization in two dimensional problems? Which element in ANSYS supports triangular discretization and which element supports quadrilateral discretization.
33. How do you plot a stress variation along the curved surface AB using ANSYS. Note that the top left hole is constrained and the bottom right hole is loaded.



34. When do you apply Symmetrical B.C in two dimensional problems?
35. What are the typical thermal quantities of interest after a thermal analysis using ANSYS?
36. Define Conduction and Convection.
37. Define Thermal Conductivity.
38. List all the thermal loads for a thermal analysis problem.
39. When do you use Glue and Overlap in thermal analysis?
40. What are the elements used in thermal analysis?
41. State the difference between Steady state thermal analysis and Transient thermal analysis.
42. What is the material property of steel in a) Structural analysis b) Thermal analysis?
43. How do you interpret the vector plot of Heat Flux in a thermal analysis?
44. What is FLOTRAN CFD Analysis?
45. What are the elements used FLOTRAN CFD for this analysis in ANSYS?
46. What are the different types of FLOTRAN Analyses.?
47. What are the material properties to be defined in FLOTRAN Analyses of air flow between parallel plates?
48. What are the loads to be defined in FLOTRAN Analyses of air flow between parallel plates?
49. Define Modal analysis and Harmonic analysis and explain their significances.
50. Briefly explain how a Modal and Harmonic analyses are done on a Cantilever beam fixed at one end free at the other.